# MATHEMATICS

# STRAIGHT LINE

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**STRAIGHT LINE**

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### COORDINATE SYSTEM, DISTANCE, SECTION FORMULAE

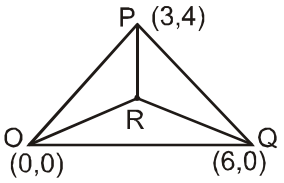
1. Let O(0, 0), P (3, 4), Q(6, 0) be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are :

   माना O(0, 0), P (3, 4) व Q(6, 0) त्रिभुज OPQ के शीर्ष है। त्रिभुज OPQ के अन्दर बिन्दु R इस प्रकार है कि त्रिभुज OPR, PQR व OQR के क्षेत्रफल समान है। तब बिन्दु R के निर्देशांक :

   (A) $\left(\frac{4}{3},3\right)$ (B) $\left(3,\frac{2}{3}\right)$ (C) $\left(3,\frac{4}{3}\right)$ (D) $\left(\frac{4}{3},\frac{2}{3}\right)$

**Ans.** C

**Sol.**




R is centroid hence R $\equiv \left(3, \frac{4}{3}\right)$

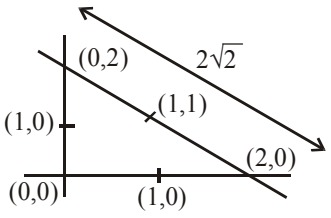
2. The x–coordinate of the incentre of the triangle that has the coordinates of mid point of its sides as (0, 1) (1, 1) and (1, 0) is : **[JEE(Main)2013]**

   एक त्रिभुज जिसकी भुजाओं के मध्य बिन्दुओं के निर्देशांक (0, 1) (1, 1) तथा (1, 0) है, के अन्तः केन्द्र का x निर्देशांक होगा : **[JEE(Main)2013]**

   (A) $2+\sqrt{2}$ (B) $2-\sqrt{2}$ (C) $1+\sqrt{2}$ (D) $1-\sqrt{2}$

**Ans.** B

**Sol.**




$$x=\frac{ax_1+bx_2+cx_3}{a+b+c}$$

$$=\frac{2(0)+2\sqrt{2}(0)+2(2)}{2+2+2\sqrt{2}}$$

$$=\frac{4}{4+2\sqrt{2}}=\frac{2}{2+\sqrt{2}}=2-\sqrt{2}$$

3. Let the orthocentre and centroid of a triangle be A(–3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is : **[JEE(Main)2018]**

माना एक त्रिभुज का लम्ब केन्द्र तथा केन्द्रक क्रमशः A(–3, 5) तथा B(3, 3) है। यदि इस त्रिभुज का परिकेन्द्र C है, तो रेखाखण्ड AC को व्यास मान कर बनाए जाने वाने वृत की त्रिज्या है : **[JEE(Main)2018]**

(A) $3\sqrt{\frac{5}{2}}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\sqrt{10}$ (D) $2\sqrt{10}$

**Ans.** A

**Sol.**




B divides AC in 2 : 1

$AC = \frac{3AB}{2}$; $r = \frac{AC}{2} = \frac{3AB}{4}$

$= \frac{3\sqrt{40}}{4}$; $= 3\sqrt{\left(\frac{5}{2}\right)}$

**AREA OF TRIANGLE**

4. If $x_1, x_2, x_3$ and $y_1, y_2, y_3$ are both in G.P. with the same common ratio, then the points $(x_1\ y_1)$, $(x_2, y_2)$ and $(x_3, y_3)$ :

(A) are vertices of a triangle (B) lie on a straight line

(C) lie on an ellipse (D) lie on a circle

यदि $x_1, x_2, x_3$ तथा $y_1, y_2, y_3$ दोनो गुणौत्तर श्रेढ़ी में है तथा सार्व अनुपात समान है तब बिन्दु $(x_1\ y_1)$, $(x_2, y_2)$ तथा $(x_3, y_3)$ होगे:

(A) त्रिभुज के शीर्ष (B) सरल रेखा पर होंगे

(C) दीर्घवृत्त पर होंगे (D) वृत्त पर होंगे

**Ans.** B

**Sol.** $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_1 r & y_1 r & 1 \\ x_1 r^2 & y_1 r^2 & 1 \end{vmatrix} = \frac{x_1 y_1}{2}\begin{vmatrix} 1 & 1 & 1 \\ r & r & 1 \\ r^2 & r^2 & 1 \end{vmatrix} = 0$

5. Let A (h, k),B(1, 1) and C (2, 1) be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which 'k' can take is given by :

माना किसी समकोण त्रिभुज के शीर्ष A (h, k),B(1, 1) एवं C (2, 1) है जहां AC कर्ण है यदि इस त्रिभुज का क्षेत्रफल एक वर्ग इकाई gS तो 'k' का मान हो सकता है :

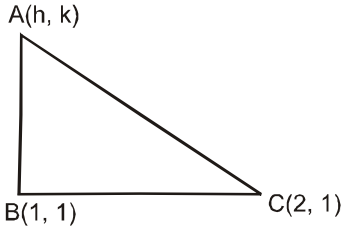
(A) {–1, 3} (B) {–3, – 2) (C) {1, 3} (D) {0, 2}

**Ans.** A

**Sol.** $AB = \sqrt{(h-1)^2+(k-1)^2}$

$BC = 1$

$AC = \sqrt{(h-2)^2+(k-1)^2}$




$AB^2 + BC^2 = AC^2 \Rightarrow (h-1)^2 + (k-1)^2 + 1 = (h-2)^2 + (k-1)^2$

$\Rightarrow 2h = 2 \Rightarrow h = 1$

Area of $\Delta ABC = \frac{1}{2}\sqrt{(h-1)^2+(k-1)^2} \times 1 = 1$

$(K-1)^2 = 4 \Rightarrow k - 1 = \pm 2 \Rightarrow k = 3, -1.$

6. Let k be an integer such that the triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : **[JEE(Main)2017]**

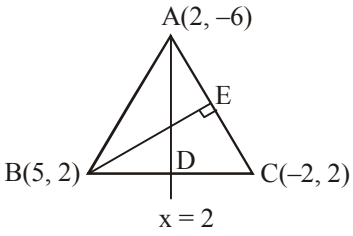
माना k एक ऐसा पूर्णांक है कि त्रिभुज, जिसके शीर्ष (k, –3k), (5, k) तथा (–k, 2) हैं, का क्षेत्रफल 28 वर्ग इकाई है, तो त्रिभुज के y Ec&d sn zfr 1 fcUnq j gSA og gS **[JEE(Main)2017]**

(A) $\left(1, -\frac{3}{4}\right)$ (B) $\left(2, \frac{1}{2}\right)$ (C) $\left(2, -\frac{1}{2}\right)$ (D) $\left(1, \frac{3}{4}\right)$

**Ans.** B

**Sol.** $\begin{vmatrix} k & -3k & 1 \\ 5 & k & 1 \\ -k & 2 & 1 \end{vmatrix} = \pm 56$

$\Rightarrow k = 2$ (k is an integer)




$m_{AC} = \frac{-8}{4} = -2$

$BE \Rightarrow (y-2) = \frac{1}{2}(x-5)$

$x - 2y = 1$

$x = 2$

$y = \frac{1}{2}$

orthocenter $\equiv \left(2, \frac{1}{2}\right)$

**EQUATION OF LINE AND ANGLE BETWEEN LINES**

7. A square of side a lies above the x–axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x–axis. The equation of its diagonal not passing through the origin is :

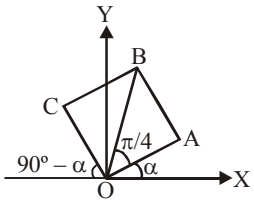
यदि 'a' भुजा का वर्ग x–अक्ष के ऊपर स्थित है उसका एक शीर्ष मूल बिन्दु पर है। मूल बिन्दु से गुजरने वाली भुजा x–अक्ष की धनात्मक fn' kkl $s\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ कोण बनाती है। इसके मूल बिन्दु से न गुजरने वाली विकर्ण की समीकरण है :

(A) $y(\cos\alpha + \sin\alpha) + x(\cos\alpha - \sin\alpha) = a$ (B) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(C) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$ (D) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

**Ans.** A

**Sol.** Co-ordinates of A = $(a\cos\alpha, a\sin\alpha)$




Equation of OB, $y = \tan\left(\frac{\pi}{4} + \alpha\right)x$

$\because$ CA $\perp^r$ to OB

$\therefore$ Slope of CA $= -\cot\left(\frac{\pi}{4} + \alpha\right)$

Equation of CA,

$$y - a\sin\alpha = -\cot\left(\frac{\pi}{4} + \alpha\right)(x - a\cos\alpha)$$

$\Rightarrow y(\sin\alpha + \cos\alpha) + x(\cos\alpha - \sin\alpha) = a$

$\Rightarrow y(\cos\alpha + \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

M – 2 :

$A(a\cos\alpha, a\sin\alpha)$

$C(-a\cos(90^\circ - \alpha), a\sin(90^\circ - \alpha))$

$C(-a\sin\alpha, a\cos\alpha)$

$$AC \Rightarrow (y - a\cos\alpha) = \left(\frac{\cos\alpha - \sin\alpha}{-\sin\alpha - \cos\alpha}\right)(x + a\sin\alpha)$$

$$(\cos\alpha - \sin\alpha)x + (\cos\alpha + \sin\alpha)y = a$$

8. The equation of the straight line passing through the point (4, 3) and making intercepts on the co–ordinate axes whose sum is –1 is :

(A) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$ (B) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$ (D) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

बिन्दु (4, 3) से गुजरने वाली तथा निर्देश अक्षों पर वह अंतःखण्ड बनाने वाली जिनका योग –1 है, रेखा का समीकरण है :

(A) $\frac{x}{2} - \frac{y}{3} = 1$ तथा $\frac{x}{-2} + \frac{y}{1} = 1$ (B) $\frac{x}{2} - \frac{y}{3} = -1$ तथा $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ तथा $\frac{x}{2} + \frac{y}{1} = 1$ (D) $\frac{x}{2} + \frac{y}{3} = -1$ तथा $\frac{x}{-2} + \frac{y}{1} = -1$

**Ans.** A

**Sol.** Here a + b = 1. Required line is

$$\frac{x}{a} - \frac{y}{1+a} = 1 \qquad \text{..... (i)}$$

Since line (i) passes through (4, 3)

$$\therefore \quad \frac{4}{a} - \frac{3}{1+a} = 1 \Rightarrow 4 + 4a - 3a = a + a^2$$

$$\Rightarrow a^2 = 4 \Rightarrow a = \pm 2$$

$\therefore$ Required lines are $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

9. The line parallel to the x–axis and passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0, where (a, b) ≠ (0, 0) is :

(A) below the x–axis at a distance of $\frac{3}{2}$ from it (B) below the x–axis at a distance of $\frac{2}{3}$ from it

(C) above the x–axis at distance of $\frac{3}{2}$ from it (D) above the x–axis at a distance of $\frac{2}{3}$ from it

ग अक्ष के समांतर तथा रेखाओं ax + 2by + 3b = 0 तथा bx – 2ay – 3a = 0 के प्रतिच्छेद बिन्दु से गुजरने वाली रेखा, जहां (a, b) ≠ (0, 0) है :

(A) ग अक्ष के नीचे इससे $\frac{3}{2}$ इकाई दूरी पर (B) ग अक्ष के ऊपर इससे $\frac{2}{3}$ इकाई दूरी पर

(C) ग अक्ष के ऊपर इससे $\frac{3}{2}$ इकाई दूरी पर (D) ग अक्ष के नीचे इससे $\frac{2}{3}$ इकाई दूरी पर

**Ans.** A

**Sol.** M-1 :

Point of intersection of both the lines is $\left(0, -\frac{3}{2}\right)$

M-2 :

The lines passing through the intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$ is

$ax + 2by + 3b + \lambda(bx - 2ay - 3a) = 0$

$\Rightarrow (a + b\lambda)x + (2b - 2a\lambda)y + 3b - 3\lambda a = 0$ ..... (i)

Line (i) is parallel to x-axis

$\therefore\ a + b\lambda = 0 \Rightarrow \lambda = \frac{-a}{b} = 0$

Put the value of $\lambda$ in (i)

$ax + 2by + 3b - \frac{a}{b}(bx - 2ay - 3a) = 0$

$y\left(2b + \frac{2a^2}{b}\right) + 3b + \frac{3a^2}{b} = 0,$

$y\left(\frac{2b^2 + 2a^2}{b}\right) = -\left(\frac{3b^2 + 3a^2}{b}\right)$

$y = \frac{-3(a^2 + b^2)}{2(b^2 + a^2)} = \frac{-3}{2}$, $y = -\frac{3}{2}$

So, it is 3/2 unit below x-axis

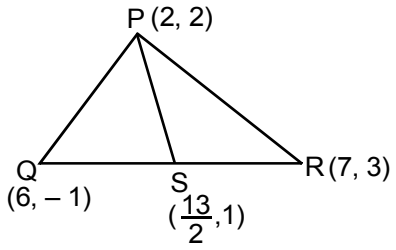
10. Let PS be the median of the triangle with vertices P(2, 2), Q(6, –1) and R(7, 3). The equation of the line passing through (1, –1) and parallel to PS is : **[JEE(Main)2014]**

माना PS एक त्रिभुज की माध्यिका है जिसके शीर्ष P(2, 2), Q(6, –1) तथा R(7, 3) हैं। (1, –1) से होकर जाने वाली रेखा, जो PS के समांतर है, का समीकरण हैं : **[JEE(Main)2014]**

(A) $4x + 7y + 3 = 0$ (B) $2x - 9y - 11 = 0$ (C) $4x - 7y - 11 = 0$ (D) $2x + 9y + 7 = 0$

**Ans.** D

**Sol.**




$$\text{Slope of PS} = \frac{2-1}{2-\frac{13}{2}} = \frac{-2}{9}$$

Hence equation of line through $(1, -1)$ & parallel to PS is:

$$(y+1) = \frac{-2}{9}(x-1)$$

$$9y + 2x + 7 = 0$$

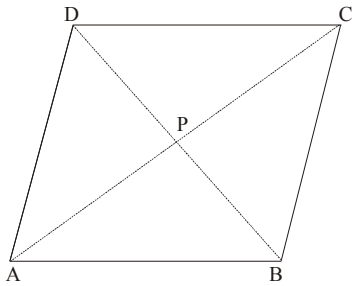
11. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at $(-1, -2)$, then which one of the following is a vertex of this rhombus? **[JEE(Main)2016]**

यदि एक समचतुर्भुज की दो भुजाएँ, रेखाओं $x - y + 1 = 0$ तथा $7x - y - 5 = 0$ की दिशा में हैं तथा इसके विकर्ण बिन्दु $(-1, -2)$ पर i fr PNs d jr sgSr ksbl 1 epr dkZ d kfuEu esa sd ks&d k' kh"kZ gS **[JEE(Main)2016]**

(A) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (B) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (C) $(-3, -9)$ (D) $(-3, -8)$

**Ans.** A

**Sol.**




$AB = 7x - y - 5 = 0$

$AD = x - y + 1 = 0$

$A(+1, 2)$

$P(-1, -2)$

$\Rightarrow C(-3, -6)$

$BC = x - y - 3 = 0$

$CD = 7x - y + 15 = 0$

$\Rightarrow B\left(\frac{1}{3},\frac{-8}{3}\right)$

**PARAMETRIC FORM OF A LINE**

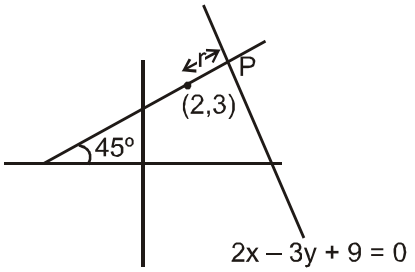
12. The distance of the point (2, 3) from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is :

बिन्दु (2, 3) की रेखा $x - y + 1 = 0$ के अनुदिश रेखा $2x - 3y + 9 = 0$ से दूरी है :

(A) $5\sqrt{3}$ (B) $4\sqrt{2}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

**Ans.** B

**Sol.** Let coordinates of point P by parametric $P(2 + r\cos 45^\circ, 3 + r\sin 45^\circ)$




It satisfies the line $2x - 3y + 9 = 0$ ; $2\left(2+\frac{r}{\sqrt{2}}\right) - 3\left(3+\frac{r}{\sqrt{2}}\right) + 9 = 0 \Rightarrow \quad r = 4\sqrt{2}$

13. P lies on the line $y = x$ and Q lies on $y = 2x$. The equation for the locus of the mid point of PQ, if $|PQ| = 4$, is :

यदि बिन्दु P तथा Q क्रमशः रेखा $y = x$ एवं $y = 2x$ पर है। यदि $|PQ| = 4$ तो P एवं Q के मध्य बिन्दु का बिन्दुपथ होगा :

(A) $25x^2 + 36xy + 13y^2 = 4$ (B) $25x^2 - 36xy + 13y^2 = 4$

(C) $25x^2 - 36xy - 13y^2 = 4$ (D) $25x^2 + 36xy - 13y^2 = 4$

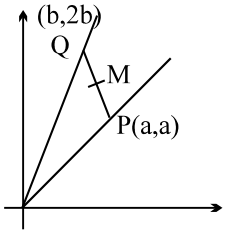
**Ans.** B

**Sol.** $2h = a + b$

$2k = a + 2b$

$\therefore \quad b = 2(k - h)$

$a = 2(2h - k)$ now proceed




**DISTANCE OF A POINT FROM LINE**

14. The line $L_1$ given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point M(13, 32). The line $L_2$ is parallel to $L_1$ and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then the distance between $L_1$ and $L_2$ is :

सरल रेखा $L_1 : \frac{x}{5}+\frac{y}{b}=1$, जो बिन्दु M(13, 32) से गुजरती है तथा रेखा $L_2$, सरल रेखा $L_1$ के समान्तर है तथा उसका समीकरण $\frac{x}{c}+\frac{y}{3}=1$ हो, तो $L_1$ एवं $L_2$ के बीच की दूरी होगी :

(A) $\sqrt{17}$ (B) $\frac{17}{\sqrt{15}}$ (C) $\frac{23}{\sqrt{17}}$ (D) $\frac{23}{\sqrt{15}}$

**Ans.** C

**Sol.** Slope of line $L_1 = \frac{-b}{5}$ and slope of line $L_2 = \frac{-3}{c}$

Line $L_1$ is parallel to line $L_2 \Rightarrow \frac{b}{5}=\frac{3}{c} \Rightarrow bc = 15$

As (13, 32) lies on $L_1$, so $\frac{13}{5}+\frac{32}{b}=1 \Rightarrow \frac{32}{b}=\frac{-8}{5} \Rightarrow b=-20$

So, $c = \frac{-3}{4}$

$\therefore$ Equation of $L_2 : y-4x=3$

Hence distance between $L_1$ and $L_2 = \frac{|52-32+3|}{\sqrt{17}} = \frac{23}{\sqrt{17}}$

**POSITION OF TWO POINTS WITH RESPECT TO A LINE**

15. If $(a, a^2)$ falls inside the angled made by the lines $y=\frac{x}{2}$, $x>0$ and $y=3x$, $x>0$, then a along to :

यदि $(a, a^2)$ रेखाओं $y=\frac{x}{2}$, $x>0$ एवं $y=3x$, $x>0$ के मध्य के कोण के अंदर स्थित है, तब 'a' निम्न में स्थित है :

(A) $\left(0,\frac{1}{2}\right)$ (B) $(3,\infty)$ (C) $\left(\frac{1}{2},3\right)$ (D) $\left(-3,-\frac{1}{2}\right)$

**Ans.** C

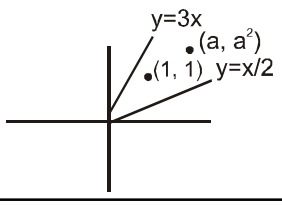
**Sol.** Let $L_1(x-2y)$ & $L_2(3x-y)$

Since (1, 1) and $(a, a^2)$ Both lies same side with respect to both lines

$L_1(1, 1) \times L_1(a, a^2) > 0 \Rightarrow \quad a-2a^2<0 \quad \Rightarrow \quad 2a^2-a>0$

$\Rightarrow \quad a(2a-1)>0$

$a \in (-\infty, 0) \cup \left(\frac{1}{2}, \infty\right)$



similarly $L_2(1, 1) \times L_2(a, a^2) > 0 \Rightarrow 3a - a^2 > 0 \Rightarrow a^2 - 3a < 0 \Rightarrow a \in (0, 3)$

Hence after taking intersection $a \in \left(\frac{1}{2}, 3\right)$

16. A triangle ABC is formed by the lines $2x-3y-6=0$ ; $3x-y+3=0$ and $3x+4y-12=0$. If the points $P(\alpha, 0)$ and $Q(0, \beta)$ always lie on or inside the $\Delta$ ABC, then :

(A) $\alpha \in [-1, 2]$ and $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ and $\beta \in [-2, 4]$

(C) $\alpha \in [-2, 4]$ and $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ and $\beta \in [-2, 3]$

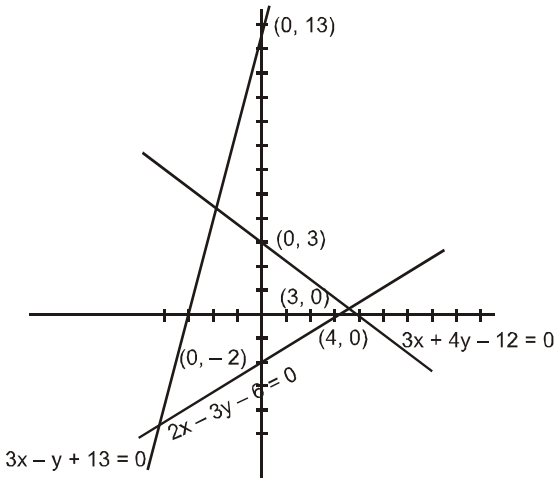
एक त्रिभुज ABC रेखा $2x-3y-6=0$; $3x-y+3=0$ एवं $3x+4y-12=0$ से बन रहा है। यदि बिन्दु $P(\alpha, 0)$ एवं $Q(0, \beta)$ हमेशा त्रिभुज पर या त्रिभुज के भीतर है, तो :

(A) $\alpha \in [-1, 2]$ एवं $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ एवं $\beta \in [-2, 4]$

(C) $\alpha \in [-2, 4]$ एवं $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ एवं $\beta \in [-2, 3]$

**Ans.** D

**Sol.** $L_1 = 2x - 3y - 6 = 0$




$L_2 = 3x - y + 3 = 0 \quad \Rightarrow \quad L_3 = 3x + 4y - 12 = 0$

ऋ मदबम अतः $\alpha \in [-1, 3] \quad \Rightarrow \quad b$ Î $[-2, 3]$

**CONDITION OF CONCURRENCY OF 3 LINES**

17. If the lines

$$\left.\begin{aligned} \lambda x + (\sin\alpha)y + \cos\alpha = 0 \\ x + (\cos\alpha)y + \sin\alpha = 0 \\ x - (\sin\alpha)y + \cos\alpha = 0 \end{aligned}\right]$$ pass through the same point where $\alpha \in R$ then $\lambda$ lies in the interval :

यदि रेखाऐं

$\left.\begin{array}{l}\lambda x+(\sin\alpha)y+\cos\alpha=0\\ x+(\cos\alpha)y+\sin\alpha=0\\ x-(\sin\alpha)y+\cos\alpha=0\end{array}\right]$ एक बिन्दु से गुजरती है। जहाँ $\alpha \in R$ तो $\lambda$ इनमें से किस अन्तराल में होगा :

(A) $[-1, 1]$ (B) $\left[-\sqrt{2},\sqrt{2}\right]$ (C) $[-2, 2]$ (D) $(-\infty, \infty)$

**Ans.** B

**Sol.** $\begin{vmatrix}\lambda & \sin\alpha & \cos\alpha\\ 1 & \cos\alpha & \sin\alpha\\ 1 & -\sin\alpha & \cos\alpha\end{vmatrix}=0$

$\Rightarrow \lambda(1)-1(2\sin\alpha\cos\alpha)+1(\sin^2\alpha-\cos^2\alpha)=0$

$\Rightarrow \lambda=\sin 2\alpha+\cos 2\alpha$

$\left[-\sqrt{2},\sqrt{2}\right]$

**FAMILY OF LINES**

18. The lines $ax + by + c = 0$, where $3a + 2b + 4c = 0$ are concurrent at the point :

यदि $3a + 2b + 4c = 0$ हो, तो $ax + by + c = 0$ से प्रदर्शित सरल रेखाएँ निम्न में से किस बिन्दु पर संगामी हैं :

(A) $\left(\frac{1}{2},\frac{3}{4}\right)$ (B) $(1, 3)$ (C) $(3, 1)$ (D) $\left(\frac{3}{4},\frac{1}{2}\right)$

**Ans.** D

**Sol.** $ax + by + c = 0 \quad \Rightarrow \quad \frac{3a}{4}+\frac{b}{2}+c=0$

compare both $(x, y) \equiv \left(\frac{3}{4},\frac{1}{2}\right)$. Hence given family passes through $\left(\frac{3}{4},\frac{1}{2}\right)$

19. Given the family of lines, $a(2x + y + 4) + b(x - 2y - 3) = 0$. Among the lines of the family, the number of lines situated at a distance of $\sqrt{10}$ from the point $M(2, -3)$ is :

रेखाओं के समूह $a(2x + y + 4) + b(x - 2y - 3) = 0$ में से कितनी रेखाऐं ऐसी है कि जो बिन्दु $M(2, -3)$ से $\sqrt{10}$ की दूरी पर gkstkh:

(A) 0 (B) 1 (C) 2 (D) $\infty$

**Ans.** B

**Sol.** $(2x + y + 4) + \lambda(x - 2y - 3) = 0$

$(2 + \lambda)x + 1(1 - 2\lambda)y + (4 - 3\lambda) = 0$

$m(2, -3)$

$$\frac{4 + 2\lambda + 6\lambda - 3 + 4 - 3\lambda}{\sqrt{(2 \times \lambda)^2 + (1 - 2\lambda)^2}} = \pm\sqrt{10}$$

$$\Rightarrow \frac{(\lambda + 1)^2}{(\lambda^2 + 1)} = 2$$

$$\Rightarrow \lambda^2 + 1 + 2\lambda = 2\lambda^2 + 2$$

$$\Rightarrow \lambda^2 + 1 - 2\lambda = (\lambda - 1)^2 = 0$$

$\lambda = 1$

**CONCEPT OF IMAGE**

20. The equation of image of pair of lines $y = |x - 1|$ with respect to y–axis is :

रेखा युग्म $y = |x - 1|$ का y–अक्ष के सापेक्ष प्रतिबिम्ब का समीकरण होगा :

(A) $x^2 - y^2 - 2x + 1 = 0$ (B) $x^2 - y^2 - 4x + 4 = 0$

(C) $4x^2 - 4x - y^2 + 1 = 0$ (D) $x^2 - y^2 + 2x + 1 = 0$

**Ans.** D

**Sol.** $y = (x - 1)$

$y = x - 1$ $y = 1 - x$

y axis $\rightarrow x \rightarrow -x$

$(x + y + 1) \times (x - y + 1) = 0$

$(x + 1)^2 - y^2 = 0$

$x^2 - y^2 + 2x + 1 = 0$

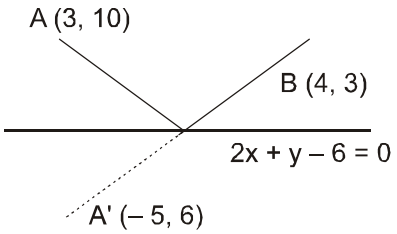
21. A light beam emanating from the point A(3, 10) reflects from the straight line $2x + y - 6 = 0$ and then passes through the point B(4, 3). The equation of the reflected beam is :

बिन्दु A(3, 10) से उत्सर्जित होने वाली एक प्रकाश किरण, सरल रेखा $2x + y - 6 = 0$ से परावर्तित होकर बिन्दु B(4, 3) से गुजरती है, तो परावर्तित प्रकाश किरण का समीकरण होगा :

(A) $3x - y + 1 = 0$ (B) $x + 3y - 13 = 0$ (C) $3x + y - 15 = 0$ (D) $x - 3y + 5 = 0$

**Ans.** B

**Sol.** Image of A(3, 10) in $2x + y - 6 = 0$




$$\frac{x-3}{2} = \frac{y-10}{1} = -2\left(\frac{6+10-6}{2^2+1^2}\right)$$

$$\frac{x-3}{2} = \frac{y-10}{1} = -4$$

$A' = (-5, 6)$

Equation of A'B is $y - 3 = \left(\frac{6-3}{-5-4}\right)(x-4)$ $\Rightarrow$ $y - 3 = -\frac{1}{3}(x-4)$

$3y - 9 = -x + 4$ $\Rightarrow$ $x + 3y - 13 = 0$

22. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x–axis, the equation of the reflected ray is :

**[JEE(Main)2013]**

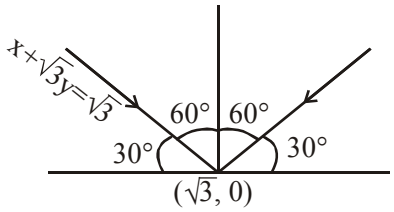
$x + \sqrt{3}y = \sqrt{3}$ की दिशा में जाती है हुई एक प्रकाश की किरण x–अक्ष पर पहुँच कर परावर्तित हो जाती है। इस परावर्तित किरण का l eHkd j .kgS

**[JEE(Main)2013]**

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$ (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

**Ans.** B

**Sol.**




$$y = \frac{1}{\sqrt{3}}(x - \sqrt{3})$$

$$\sqrt{3}y = x - \sqrt{3}$$

**ANGLE BISECTORS**

23. Find the equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$ :

रेखाओं $3x - 4y + 7 = 0$ एवं $12x + 5y - 2 = 0$ के मध्य न्यून कोण के अर्द्धक का समीकरण ज्ञात कीजिए :

(A) $11x - 3y + 9 = 0$ (B) $3x + 11y - 13 = 0$ (C) $3x + 11y - 3 = 0$ (D) $11x - 3y + 2 = 0$

**Ans.** A

**Sol.** $3x - 4y + 7 = 0$ $a_1a_2 + b_1b_2 = -36 + 20$

$-12x - 5x + 2 = 0$ $= -16 < 0$

$$\frac{(3x-4y+7)}{5} = +\frac{(-12x-5y+2)}{13}$$

$39x - 52y + 91 = -60x - 25y + 10$

$99x - 27y + 81 = 0$

$11x - 3y + 9 = 0$

24. The equation of the bisector of the angle between two lines $3x - 4y + 12 = 0$ and $12x - 5y + 7 = 0$ which contains the points $(-1, 4)$ is :

दो रेखाओं $3x - 4y + 12 = 0$ एवं $12x - 5y + 7 = 0$ के बीच बिन्दु $(-1, 4)$ को समाहित करने वाले कोण के अर्द्धक का समीकरण gS

(A) $21x + 27y - 121 = 0$ (B) $21x - 27y + 121 = 0$

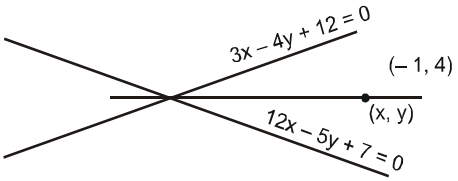
(C) $21x + 27y + 191 = 0$ (D) $\frac{-3x + 4y - 12}{5} = \frac{12x - 5y + 7}{13}$

**Ans.** A

**Sol.** at $(-1, 4)$

$3x - 4y + 12 < 0$ and $12x - 5y + 7 < 0$

$\Rightarrow \quad \frac{3x-4y+12}{12x-5y+12} > 0$ at $(-1, 4)$




So we have to take the bisector with + sign

$$\frac{3x-4y+12}{5} = \frac{12x-5y+7}{13} \quad \Rightarrow \quad 21x + 27y - 121 = 0$$

25. The equation of bisectors of two lines $L_1$ and $L_2$ are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line $L_1$ passes through $(-11, 4)$, the equation of acute angle bisector of $L_1$ and $L_2$ is :

(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) data insufficient (D) none of these

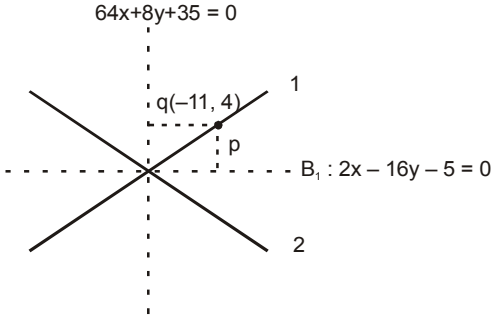
दो रेखाओं $L_1$ एवं $L_2$ के समद्विभाजकों की समीकरण $2x - 16y - 5 = 0$ एवं $64x + 8y + 35 = 0$ हैं। यदि रेखा $L_1$ बिन्दु $(-11, 4)$ से गुजरती है, तो $L_1$ एवं $L_2$ के मध्य न्यून कोण अर्द्धक की समीकरण है :

(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) आँकड़े अपर्याप्त हैं (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $p = \left|\frac{-22-64-5}{\sqrt{2^2+(-16)^2}}\right| = \frac{91}{\sqrt{260}}$

$q = \left|\frac{-64\times11+8\times4+35}{\sqrt{64^2+8^2}}\right| = \frac{637}{\sqrt{4160}} = \frac{159.25}{\sqrt{260}}$



$p < q$ Hence $2x - 16y - 5 =$ - is a cute angle besectory

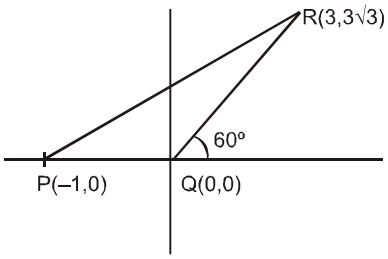
26. Let P = (–1, 0), Q = (0, 0) and R $(3, 3\sqrt{3})$ be three point. The equation of the bisector of the angle PQR is :

माना P = (–1, 0), Q = (0, 0) एवं R $(3, 3\sqrt{3})$ तीन बिन्दु है ∠PQR के समद्विभाजक की समीकरण है :

(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3y} = 0$ (C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

**Ans.** C

**Sol.**




The line segment QR makes an angle 60º with the positive direction of x-axis.
hence bisector of angle PQR will make 120º with +ve direction of x-axis.
$\therefore$ Its equation

$y - 0 = \tan 120^\circ (x - 0) \quad \Rightarrow y = -\sqrt{3}x \Rightarrow x\sqrt{3} + y = 0$

**IMPORTANT CONCEPT RELATED TO OPTIMIZING**

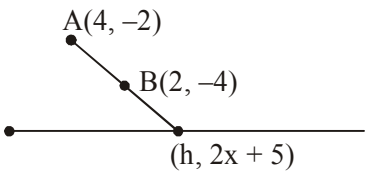
27. The co–ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is $(4, -2)$ and B is $(2, -4)$ will be :

यदि रेखा $2x - y + 5 = 0$ पर मौजूद बिन्दु P इस प्रकार से है कि $|PA - PB|$ अधिकतम है जहाँ A $(4, -2)$ एवं B $(2, -4)$ तो P का निर्देशांक होगा :

(A) $(11, 27)$ (B) $(-11, -17)$ (C) $(-11, 17)$ (D) $(0, 5)$

**Ans.** B

**Sol.**




$L_A, L_B > 0$

$$\begin{vmatrix} h & 4 & 2 & h \\ 2h+5 & -2 & -4 & 2h+5 \end{vmatrix} = 0$$

$-2h - 20 - 16 + 4 + 10 = 0$

$-2h = 22$

$h = -11$

$2h + 5 = -22 + 5 = -17$

**GENERAL LOCUS PROBLEMS**

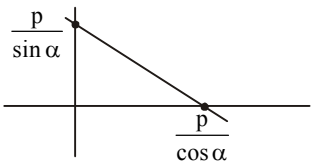
28. Locus of mid point of the portion between the axes of $x\cos\alpha + y\sin\alpha = p$ whre p is constant is :

रेखा $x\cos\alpha + y\sin\alpha = p$ के अक्षों के मध्य काटे गये अन्त खण्ड के मध्य बिन्दु का बिन्दुपथ होगा जहां p अचर है :

(A) $x^2 + y^2 = \frac{4}{p^2}$ (B) $x^2 + y^2 = 4p^2$ (C) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{2}{p^2}$ (D) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

**Ans.** D

**Sol.**




$x\cos\alpha + y\sin\alpha = p$

$x = \frac{p}{2\sin\alpha} \quad \Rightarrow \cos\alpha = \frac{p}{2x}$

$y = \frac{p}{2\sin\alpha} \quad \Rightarrow \sin\alpha = \frac{p}{2y}$

$\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

29. Locus of centroid of the triangle whose vertices are (a cos t, a sin t), (b sin t, – b cos t) and (1, 0) where t is a parameter, is :

एक त्रिभुज जिसके शीर्ष (a cos t, a sin t), (b sin t, – b cos t) तथा (1, 0) है तब त्रिभुज के केन्द्रक का बिन्दुपथ होगा। जहां t परिमाप gS

(A) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$ (B) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$ (D) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

**Ans.** C

**Sol.** (a cot t, a sin t), (b sin t, – b cos t), (1, 0)

$x = \frac{a\cos t + b\sin t + 1}{3}$, $y = \frac{a\sin t - b\cos t}{3}$

$(3x - 1) = a\cos t + b\sin t$ $3y = a\sin t - b\cos t$

$(3x - 1)^2 + (3y)^2 = a^2 + b^2$

30. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is :

माना P बिन्दु (1, 0) तथा Q बिन्दुपथ $y^2 = 8x$ पर स्थित एक बिन्दु है। PQ के मध्य बिन्दु का बिन्दुपथ है :

(A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

**Ans.** A

**Sol.** P(1, 0); Q $y^2 = 8x$

$\left(\frac{\alpha^2}{8}, \alpha\right)$ $x = \frac{\alpha^2}{8} + 1$, $y = \frac{\alpha}{2}$

$\alpha = 2y$

$2x = \frac{y^2}{2} + 1$

$4x = y^2 + 2$

$y^2 - 4x + 2 = 0$

31. The equatons $x = t^3 + 9$ and $y = \frac{3t^3}{4} + 6$ represents a straight line where t is a parameter. Then y–intercept of the line is :

यदि समीकरण $x = t^3 + 9$ तथा $y = \frac{3t^3}{4} + 6$ एक सरल रेखा को निरूपित करता है, जहाँ t एक प्राचल है। तो रेखा का y–काट होगा:

(A) $-\frac{3}{4}$ (B) 9 (C) 6 (D) 1

**Ans.** A

**Sol.** $x = t^3 + 9$

$y = \frac{3t^3}{4} \times 6$

$4y = 3t^3 \times 24$

$\Rightarrow 4y = 3(x - 9) + 24$

$\Rightarrow 4y = 3x - 3$

$\Rightarrow y = \frac{3x}{4} - \frac{3}{4}$

$c = -\frac{3}{4}$

32. Locus of the image of the point (2, 3) in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0$ $k \in R$ is a : **[JEE(Main)-2015]**

(A) straight line parallel to x–axis (B) straight line parallel to y–axis

(C) circle of radius $\sqrt{2}$ (D) circle of radius $\sqrt{3}$

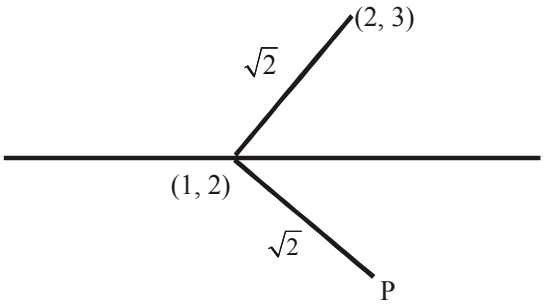
बिन्दु (2, 3) के रेखा $(2x - 3y + 4) + k(x - 2y + 3) = 0$ $k \in R$ से प्रतिबिंब का बिंदुपथ एक : **[JEE(Main)2015]**

(A) ग अक्ष के समांतर रेखा है (B) ल अक्ष के समांतर अक्ष है।

(C) $\sqrt{2}$ त्रिज्या का वृत है (D) $\sqrt{3}$ त्रिज्या का वृत है

**Ans.** C

**Sol.**




$2x - 3y + 4 = 0$

$x - 2y + 3 = 0$

Line passing through (1, 2)

P will at distance of $\sqrt{2}$ from (1, 2)

So locus of P will circle of radius $\sqrt{2}$

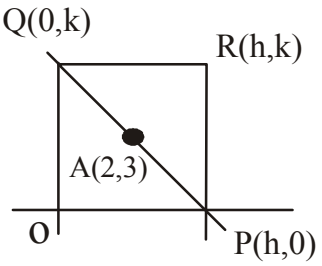
33. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is : **[JEE(Main)2018]**

एक सरल रेखा, जो एक अचर बिन्दु (2, 3) से होकर जाती है, निर्देशांक अक्षों को दो विभिन्न बिन्दुओं P तथा Q पर प्रतिच्छेद करती है। ; fn O मूल बिन्दु है तथा आयत OPRQ को पूरा किया जाता है तो R का बिन्दुपथ है : **[JEE(Main)2018]**

(A) $3x + 2y = xy$ (B) $3x + 2y = 6xy$ (C) $3x + 2y = 6$ (D) $2x + 3y = xy$

**Ans.** A

**Sol.**




P, A, Q will be collinear

$$\begin{vmatrix} h & 0 & 1 \\ 2 & 3 & 1 \\ 0 & k & 1 \end{vmatrix} = 0$$

$3h + 2k = hk$

locus of R (h, k) will be

$3x + 2y = xy$

**PAIR OF STRAIGHT LINES**

34. If the slope of one line of the pair of lines represented by $ax^2 + 10xy + y^2 = 0$ is four times, the slope of the other line, then a equal to :

यदि समीकरण $ax^2 + 10xy + y^2 = 0$ द्वारा निरूपित रेखाओं के युग्म में से एक रेखा की प्रवणता दूसरी रेखा की प्रवणता से चार गुना g§r ksa का मान है :

(A) 1 (B) 2 (C) 4 (D) 16

**Ans.** D

**Sol.** $m_1 + m_2 = -10 \Rightarrow m_1 m_2 = \frac{a}{1}$

given $m_1 = 4m_2 \Rightarrow m_2 = -2, m_1 = -8, a = 16$

35. The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for :

(A) two values of a (B) $\forall$ a $\in$ R (C) for one value of a (D) for no value of a

निम्न में से किसके लिए रेखा युग्म $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ परस्पर लम्बवत् है :

(A) ' के दो मान (B) $\forall$ a $\in$ R (C) ' के एक मान के लिए (D) ' के किसी मान के लिए नही

**Ans.** A

**Sol.** $x^2 - 2pxy - y^2 = 0, \quad x^2 - 2axy - y^2 = 0$

$ca^2 = 2m^2 + 5m + 3a = 0$

$$m_1 m_2 = -1 = \frac{3a}{a^2 - 2} \quad \Rightarrow \quad 2 - a^2 = 3a$$

$$a^2 + 3a - 2 = 0$$

36. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is :

यदि $my^2 + (1 - m^2)xy - mx^2 = 0$ में से सरल रेखाओं $xy = 0$ के मध्य के कोण का समद्विभाजक है, तब m है :

(A) 1 (B) 2 (C) $-\frac{1}{2}$ (D) –2

**Ans.** A

**Sol.** Bisector of $x = 0$ and $y = 0$ is either $y = x$ or $y = -x$
If $y = x$ is Bisector, then
$mx^2 + (1 - m^2)x^2 - mx^2 = 0 \Rightarrow m + 1 - m^2 - m = 0 \Rightarrow m^2 = 1 \Rightarrow m = \pm 1$.

**HOMOGENISATION**

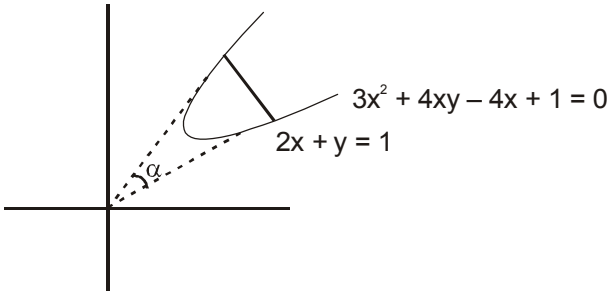
37. The straight lines joining the origin to the points of intersection of the line $2x + y = 1$ and curve $3x^2 + 4xy - 4x + 1 = 0$ include an angle :

सरल रेखा $2x + y = 1$ एवं वक्र $3x^2 + 4xy - 4x + 1 = 0$ के प्रतिच्छेद बिन्दुओं को मूल बिन्दु से मिलाने वाली सरल रेखाओं के e/; dks.kgS

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

**Ans.** A

**Sol.** Homogenize given curve with given line
$3x^2 + 4xy - 4x(2x + y) + 1(2x + y)^2 = 0$
$3x^2 + 4xy - 8x^2 - 4xy + 4x^2 + y^2 + 4xy = 0$
$-x^2 + 4xy + y^2 =$
coeff. $x^2$ + coeff. $y^2 = 0$
Hence angle is 90º




38. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the co–ordinate axes then the value of k :

(A) is equal to 1 (B) is equal to $-1$
(C) is equal to 2 (D) does not exist in the set of real numbers

यदि वक्र $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ तथा $x + ky - 1 = 0$ के प्रतिच्छेद बिन्दु को (0, 0) से मिलाने पर जो रेखा प्राप्त gks hgSog nks.kksa {kks| scjkcj >qdko j[kr hgksa rks k का मान होगा :

(A) 1 के बराबर (B) 1 के बराबर

(C) 2 के बराबर (D) वास्तविक संख्याओं के समुच्चय के लिये अस्तित्व में नहीं

**Ans.** B

**Sol.** Homogenise and put co-efficient of $xy = 0$

39. The angles between the straight lines joining the origin to the points common to $7x^2 + 8y^2 - 4xy + 2x - 4y - 8 = 0$ and $3x - y = 2$ is :

वक्र $7x^2 + 8y^2 - 4xy + 2x - 4y - 8 = 0$ एवं रेखा $3x - y = 2$ के उभयनिष्ट बिन्दु को (0, 0) से मिला दिया जाये तो सरल रेखाओं d scIp d kd ksakgksxk:

(A) $\tan^{-1}\sqrt{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

**Ans.** D

**Sol.** $7x^2 + 8y^2 - 4xy + x(3x - y) - 2y(3x - y) - 2(3x - y^2) = 0$

$\Rightarrow -8x^2 + 8y^2 + 11xy = 0$

$\Rightarrow a + b = 0 \quad Q = \frac{\pi}{2}$

# Exercise J-II

## SUBJECTIVE / INTEGER TYPE QUESTIONS

### COORDINATE SYSTEM, DISTANCE, SECTION FORMULAE

1. Determine the ratio in which the point P(3 , 5) divides the join of A(1, 3) & B(7, 9). Find the harmonic conjugate of P w.r.t. A & B.

वह अनुपात ज्ञात करो जिस अनुपात में बिन्दु P(3 , 5) बिन्दु A(1, 3) एवं B(7, 9) को मिलाने वाले रेखाखण्ड को विभाजित करता है एवं बिन्दु P का A एवं B के सापेक्ष हरात्मक संयुग्मी ज्ञात करो।

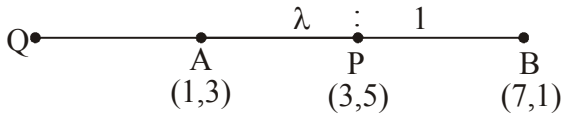
**Ans.** 1 : 2 ; Q(–5, –3)

**Sol.** $\frac{7\lambda+1}{\lambda+1}=3$

$7\lambda + = 3\lambda + 3$

$4\lambda = 2$

$1 : 2$




$x = \frac{2-7}{2-1} = -5$

$y = \frac{6-9}{2-1} = y - 3$

$Q(-5, -3)$

2. Line $\frac{x}{6}+\frac{y}{8}=1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find

(a) the coordinates of the point P and

(b) the radius of the circle escribed opposite to the angle N.

सरल रेखा $\frac{x}{6}+\frac{y}{8}=1$ अक्षों x एवं y को क्रमशः M तथा N बिन्दुओं पर काटती है। यदि बिन्दु P के निर्देशांक (a, b) त्रिभुज OMN के अन्दर इस तरह स्थित हो (जहाँ 'O' मूल बिन्दु है) कि त्रिभुजों POM, PON एवं PMN का क्षेत्रफल बराबर है, ज्ञात कीजिये।

(a) बिन्दु P के निर्देशांक

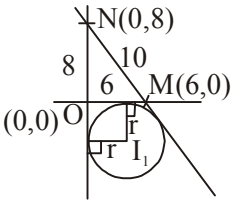
(b) कोण N के विपरीत बाह्य वृत की त्रिज्या

**Ans.** (a) $\left(2, \frac{8}{3}\right)$; (b) 4

**Sol.** $\frac{x}{6}+\frac{y}{8}=1$

(a) $P\left(2,\frac{8}{3}\right)$

(b) $x=\frac{0\times10+6\times8-6\times0}{8-6+10} \quad \Rightarrow \frac{-48}{12}$




$y=-2$

$r=4$

M-2 :

$r_1=\frac{\Delta}{(s-a)}$

3. In an acute triangle ABC, the base BC has the equation $4x-3y+3=0$. If the coordinates of the orthocentre (H) and circumcentre (P) of the triangle are (1, 2) and (2, 3) respectively, then the radius of the circle circumscribing the triangle is $\frac{\sqrt{m}}{n}$ where m and n are relatively prime. Find the value of $(m+n)$.

(You may use the fact that the distance between orthocentre and circumcentre of the triangle is given by $R\sqrt{1-8\cos A\cos B\cos C}$).

एक न्यूकोणीय त्रिभुज ABC में, आधार BC का समीकरण $4x-3y+3=0$ है। यदि त्रिभुज का लम्ब केन्द्र (H) तथा परिकेन्द्र (P) क्रमशः (1, 2) व (2, 3) है, तो त्रिभुज की परित्रिज्या $\frac{\sqrt{m}}{n}$ से व्यक्त की जायेगी। जहाँ m और n का महत्तम समापवर्तक 1 है, तो $(m+n)$ का मान ज्ञात करों।

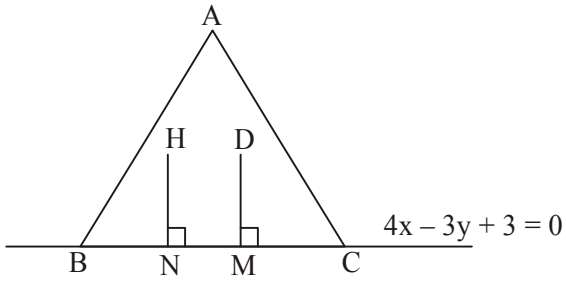
(आप इस तथ्य का इस्तेमाल कर सकते है कि त्रिभुज के लम्ब केन्द्र तथा परिकेन्द्र के बीच की दूरी $R\sqrt{1-8\cos A\cos B\cos C}$ होगी)

**Ans.** 63

**Sol.** $R=\frac{\sqrt{m}}{n}$

$HP=R\sqrt{1-8\cos A\cos B\cos C}$

$R=\frac{\sqrt{2}}{\sqrt{1-8\cos A\cos B\cos C}}$




$HN = 2r \cos B \cos C$

$DM = R \cos A$

$$\cos A \cos B \cos C = \frac{PM \times HN}{2R^2}$$

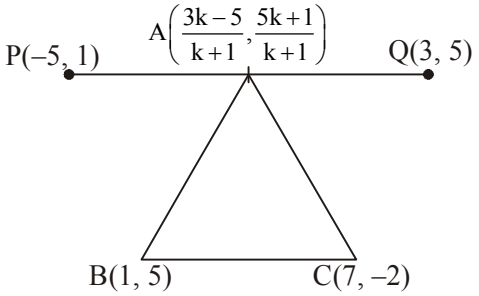
**AREA OF TRIANGLE**

4. The point A divides the join of $P(-5, 1)$ & $Q(3, 5)$ in the ratio $K : 1$. Find the two values of K for which the area of triangle ABC, where B is (1, 5) & C is (7, –2), is equal to 2 units in magnitude.

यदि बिन्दु A, $P(-5, 1)$ एवं $Q(3, 5)$ को मिलाने वाले रेखा खण्ड को $K : 1$ अनुपात में विभाजित करता है, तो K के वह दो मान ज्ञात d j ;ft 1 d sfy , f=kHq ABC के क्षेत्रफल का परिमाण 2 इकाई है। जहाँ बिन्दु B एवं बिन्दु C के निर्देशांक क्रमशः (1, 5) एवं (7, –2) है।

**Ans.** $K = 7$ or $\frac{31}{9}$

**Sol.**




$$\begin{vmatrix} 1 & 7 & \frac{3k-5}{k+1} & 1 \\ 5 & -2 & \frac{5k+1}{k+1} & 5 \end{vmatrix} = \pm 4$$

$$\Rightarrow -2 - 35 + \frac{35k+7}{k+1} - \frac{6k-10}{k+1} + \frac{15k-25}{k+1} - \frac{5k+1}{k-1} = \pm 4$$

$$\Rightarrow -37+\frac{51k-29}{k+1}=\pm 4$$

$$\begin{cases} 10k=70 & k=7 \\ 18k=62 & k=\dfrac{62}{18}=\dfrac{31}{9} \end{cases}$$

5. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$ $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ and $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$ then

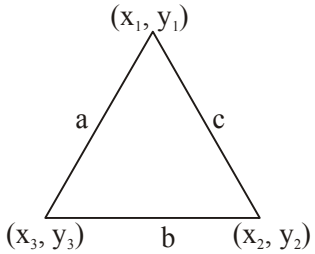
$$\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c).$$ Find the value of $\lambda$.

यदि $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$, $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ एवं $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$ तो

$$\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c).$$ तो $\lambda$ का मान ज्ञात कीजिये।

**Ans.** 4

**Sol.** $A(x_1, y_2)$ $B(x_2, y_2)$, $C(x_3, y_3)$




$$\Delta=\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\Delta=\sqrt{s(s-a)(s-b)(s-c)} \quad \Rightarrow \sqrt{\frac{(a+b+c)}{2}\frac{(b+c-a)}{2}\frac{(a+c-b)}{2}\frac{(a+b-c)}{2}}$$

$$\Delta^2 \Rightarrow \frac{1}{16}(a+b+c)$$

$$(a+b+c)(a+b-c)(a+c-b)(b+c-a)=160^2$$

$\Rightarrow 16 \times \frac{1}{4}$

$\lambda = 4$

6. The area of a triangle is 5. Two of its vertices are (2, 1) & (3, –2). The third vertex lies on y = x + 3. Find the third vertex.

यदि किसी त्रिभुज का क्षेत्रफल 5 ईकाई हो, एवं शीर्ष बिन्दुओं के निर्देशांक (2, 1) एवं (3, –2) हो, तथा तीसरा शीर्ष बिन्दु रेखा y = x + 3 पर स्थित हो तो तीसरे शीर्ष बिन्दु के निर्देशांक ज्ञात करों।

**Ans.** $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

**Sol.** $\Delta = 5$ (2, 1) and (3, –2) and (h, h + 3)

$$\begin{vmatrix} h & 2 & 3 & h \\ h+3 & 1 & -2 & h+3 \end{vmatrix} = \pm 10$$

$h - 2n - 6 - 4 - 3 + 3h + 9 + 2h = \pm 10$

$$4n - 4 = \pm 10 \begin{cases} 4h = 14 & \Rightarrow h = \frac{7}{2} \\ 4n = -6 & h = \frac{-3}{2} \end{cases}$$

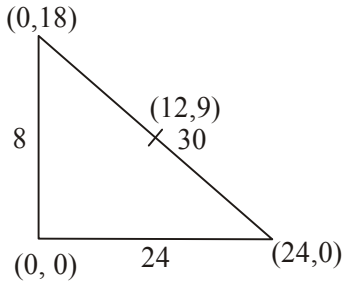
$\left(\frac{7}{2}, \frac{13}{2}\right), \left(\frac{-3}{2}, \frac{3}{2}\right)$

7. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle.

एक त्रिभुज की भुजाओं की लम्बाई 18, 24 एवं 30 है, उस त्रिभुज का क्षेत्रफल ज्ञात कीजिये जिसके शीर्ष बिन्दु दिये गये त्रिभुज के अंतः केन्द्र, परिकेन्द्र एवं केन्द्रक है।

**Ans.** 3 Units

**Sol.**




C(12,9), G(8, 6)

$$I \equiv \left( \frac{0+18\times 24+0}{72}, \frac{0+0+18\times 24}{72} \right)$$

I (6, 6)

$$\Delta CGI = \frac{1}{2}\begin{vmatrix} 12 & 8 & 6 & 2 \\ 9 & 6 & 6 & 9 \end{vmatrix}$$

$$\Rightarrow \frac{1}{2}|72-72+48-36+54-72|$$

$$= \frac{1}{2}\times(6)$$

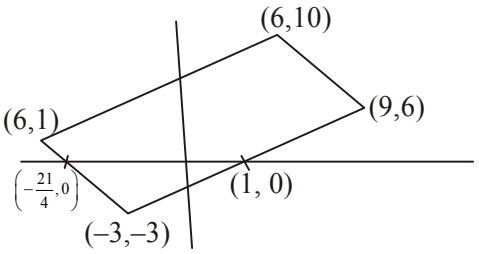
= 3

8. The points (–6, 1), (6, 10), (9, 6) and (–3, –3) are the vertices of a rectangle. If the area of the portion of this rectangle that lies above the x axis is $a/b$, find the value of (a + b), given a and b are coprime.

बिन्दु (–6, 1), (6, 10), (9, 6) एवं (–3, –3) किसी आयत के शीर्ष बिन्दु हो, तथा x अक्ष के उपर स्थित भाग का क्षेत्रफल $a/b$ है तो (a + b) का मान ज्ञात करो यदि a एवं b परस्पर अभाज्य (coprime) है।

**Ans.** 533

**Sol.**




(–6, 1), (6, 10), (9,6) and (–3, –3)

$$\text{Area of rectangle} = \left(\sqrt{81+144}\right)\left(\sqrt{16+9}\right) = 75$$

$$\text{area below x–axis} = \frac{1}{2}\times\frac{25}{4}\times 3 = \frac{75}{8}$$

$$\text{Area above x–axis} = \frac{525}{8}$$

a + b = 533

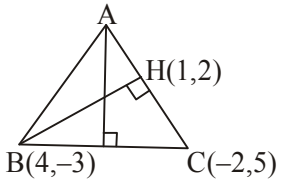
## EQUATION OF LINE AND ANGLE BETWEEN LINES

9. Two vertices of a triangle are (4, –3) & (–2, 5). If the orthocentre of the triangle is at (1, 2), find the coordinates of the third vertex.

यदि त्रिभुज के दो शीर्ष बिन्दुओं के निर्देशांक (4, –3) एवं (–2, 5) है, एवं लम्ब केन्द्र बिन्दु (1, 2) पर स्थित हो, तो तीसरे शीर्ष बिन्दु के निर्देशांक ज्ञात कीजिये।

**Ans.** (33, 26)

**Sol.** orthocentre of ΔBCH will be A. Hence




$$AH \Rightarrow (y-2) = \frac{6}{8}(x-1)$$

$$AC \Rightarrow (y-5) = \frac{3}{5}(x+2)$$

A(33, 26)

10. A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point (1, 5). Obtain the equation.

उस रेखा का समीकरण ज्ञात करे। जिसका सरल रेखाओं $5x - y - 4 = 0$ एवं $3x + 4y - 4 = 0$ के मध्य काटा गया रेखाखण्ड, बिन्दु (1, 5) पर समद्विभाजित होता है।

**Ans.** $83x - 35y + 92 = 0$

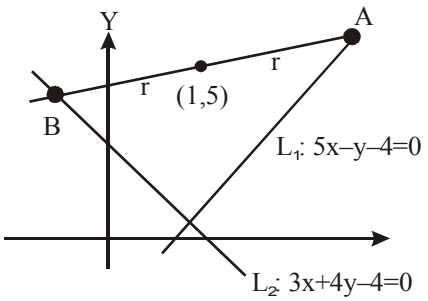
**Sol.** Coordinate of A which is at a distance 'r' from (1, 5) is $(1 + r\cos\theta, 5 + r\sin\theta)$

which satisfies $L_1 = 0$

$5(1 + r\cos\theta) - (5 + r\sin\theta) - 4 = 0$

$r(5\cos\theta - \sin\theta) = 4$

$$r = \frac{4}{5\cos\theta - \sin\theta}$$




Similarly coordinate of point B $(1 - r\cos\theta, 5 - r\sin\theta)$ which lies on $L_2 = 0$

$\therefore\ 3(1 - r\cos\theta) + 4(5 - r\sin\theta) - 4 = 0$

$$r = \frac{19}{3\cos\theta + 4\sin\theta}$$

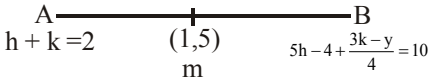
Equating both values of r & get tan θ (= m) and equation of the line passing through (1, 5).

$$m = \tan\theta = \frac{83}{35}$$

**Alternate solution**

$5x - y - 1 = 0$ $\qquad$ $3x + 4y - 4 = 0$

$(4, 5h, -4)$ $\qquad$ $\left(k, \frac{3x-4}{4}\right)$




$20h + 3k = 40 + 20 = 60$

$3h + 3k = 6$

$17h = 54$

$$h = \frac{54}{17}$$

$$k = 2 - \frac{54}{17}$$

$$k = \frac{-20}{17}$$

$$\left(\frac{54}{17}, \frac{270-68}{17}\right)$$

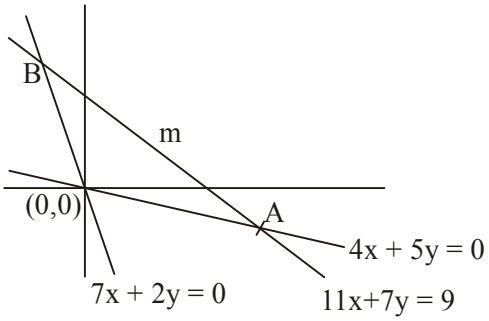
$$\left(\frac{54}{17}, \frac{202}{17}\right)$$

11. Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.

यदि सामान्तर चतुर्भुज की दो संलग्न भुजाओं के समीकरण $4x + 5y = 0$ एवं $7x + 2y = 0$ हो, एवं एक विकर्ण का समीकरण $11x + 7y = 9$, तो दूसरे विकर्ण का समीकरण ज्ञात करो।

**Ans.** $x - y = 0$

**Sol.**



$A\left(\frac{5}{3},-\frac{4}{3}\right)$

$B\left(-\frac{2}{3},\frac{7}{3}\right)$

$M\left(\frac{1}{2},\frac{1}{2}\right)$

Equation of $OM \rightarrow y = x$

12. The line $3x + 2y = 24$ meets the y–axis at A & the x–axis at B. The perpendicular bisector of AB meets the line through $(0, -1)$ parallel to x–axis at C. Find the area of the triangle ABC.

यदि रेखा $3x + 2y = 24$ y–अक्ष को बिन्दु A एवं x अक्ष को बिन्दु B पर मिलती है, एवं रेखाखंड AB का लम्ब समद्विभाजक, रेखा जो कि x अक्ष के समातर एवं बिन्दु $(0, -1)$ से गुजरती है, को बिन्दु C पर प्रतिच्छेद करता है, तो त्रिभुज ABC का क्षेत्रफल ज्ञात करें।

**Ans.** 91 sq.units

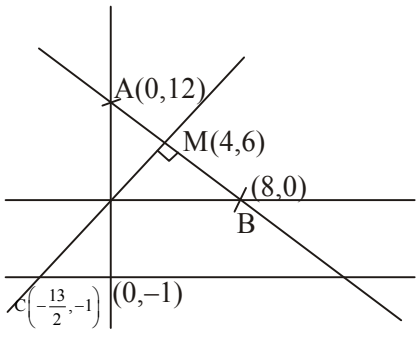
**Sol.** $3x + 2y = 24$

$CM \Rightarrow (y-6) = \frac{2}{3}(x-4)$

$y = -1$ $\qquad C\left(-\frac{13}{2},-1\right)$

$\frac{-7\times 3}{2} + 4 = x \Rightarrow \frac{-21+8}{2} = \frac{-13}{2}$

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} 0 & 8 & -\frac{13}{2} & 0 \\ 12 & 0 & -1 & 12 \end{vmatrix} = \frac{1}{2}[-96-8-78] = \frac{182}{2} = 91$$



13. Find the area of the triangle formed by the straight lines whose equations are $x + 2y - 5 = 0$; $2x + y - 7 = 0$ and $x - y + 1 = 0$. Also compute the tangent of the interior angles of the triangle and hence comment upon the nature of triangle.

रेखाओं, जिनका समीकरण $x + 2y - 5 = 0; 2x + y - 7 = 0$ एवं $x - y + 1 = 0$ है, से बनने वाले त्रिभुज का क्षेत्रफल ज्ञात करो। साथ ही आंतरिक कोणो की स्पर्शज्याऐं ज्ञात कीजिये एवं त्रिभुज की प्रकृति पर टिप्पणी करे।

**Ans.** $\frac{3}{2}$ sq. units, $\left(3, 3, \frac{3}{4}\right)$, isosceles

**Sol.** $x + 2y - 5 = 0$ $\quad m_2 = -\frac{1}{2}$

$2x + y - 7 = 0$ $\quad m_3 = -2$ $\quad m_1 > m_2 > m_3$

$x - y + 1 = 0$ $\quad m_1 = 1$ $\quad 1 \quad -\frac{1}{2} \quad -2$

$$\tan I = \left|\frac{1+\frac{1}{2}}{1+\frac{1}{2}}\right| = 3$$

$$\tan II = \left|\frac{-\frac{1}{2}+2}{1+1}\right| \Rightarrow \frac{\frac{3}{2}}{2} = \frac{3}{4}$$

$$\tan III = \left|\frac{-2-1}{1+2}\right| \Rightarrow 3 \quad \Rightarrow III \Rightarrow \text{Isosceles}$$

$L_2L_3\,(2, 3)$ $\quad 3x - 6 = 0$

$x = 2 \quad y = 3$

$L_3L_1\,(1,2)$ $\quad x + 2y - 5 = 0$

$2x - 2y + 2 = 0$

$3x - 3 = 0$

$x = 1, y = 2$

$L_1L_2\,(3,1)$ $\quad 4x + 2y - 14 = 0$

$x + 2y - 5 = 0$

$3x = 9$

$$x = 3, y = 1$$

$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & 1 & 3 & 2 \\ 3 & 2 & 1 & 3 \end{vmatrix}$$

$$= \frac{1}{2}[4 - 3 + 1 - 6 + 9 - 2]$$

$$\Rightarrow \frac{1}{2}(3) = \frac{3}{2} \text{ sq. units}$$

14. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

बिन्दु (1, 3) एवं (5, 1) किसी आयत के विपरीत शीर्षो के निर्देशांक है एवं अन्य दो शीर्ष बिन्दु रेखा $y = 2x + c$ पर स्थित है, तो c का मान एवं अन्य शीर्ष बिन्दु के निर्देशांक ज्ञात करो।

**Ans.** $c = -4$; B(2 , 0); D(4 , 4)

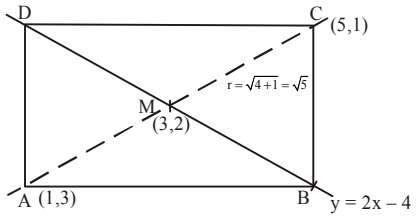
**Sol.** A (1,3) & C(5,1)

M (3,2)

$y = 2x + c \Rightarrow y = 2x - 4$

$2 = 6 + c$

$c = -4$




$$B_1 D(3 \pm r\cos\theta, 2 \pm r\sin\theta)$$

$$B\left(3 + \sqrt{5} \times \frac{1}{\sqrt{5}}, 2 + \sqrt{5} \times \frac{2}{\sqrt{5}}\right) = (4,4)$$

D (3 – 1, 2 – 2) = (2, 0)

15. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

एक सरल रेखा L जो कि रेखा $5x - y = 1$ के लम्बवत् है, यदि रेखा L एवं अक्षो के साथ मिलकर बनने वाले त्रिभुज का क्षेत्रफल 5 हो, तो रेखा का समीकरण ज्ञात करों।

**Ans.** $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

**Sol.** $5x - y = 1$

$\Lambda : \xi + 5\psi = \lambda \qquad a = \lambda,\ b = \frac{\lambda}{5}$

$$\frac{1}{2}\times\lambda\times\frac{\lambda}{5}=5$$

$\lambda^2 = 50$

$\lambda = \pm 5\sqrt{2}$

$x + 5y = 5\sqrt{2}$

$x + 5y - 5\sqrt{2} = 0$

16. The equations of the perpendicular bisectors of the sides AB and AC of a triangle ABC are $x - y + 5 = 0$ and $x + 2y = 0$, respectively. If the point A is $(1, -2)$ find the equation of the line BC.

किसी त्रिभुज ABC की भुजाओं AB एवं AC के लम्ब समद्विभाजको समीकरण क्रमशः $x - y + 5 = 0$ एवं $x + 2y = 0$ हो, एवं बिन्दु A के निर्देशांक $(1, -2)$ हो तो रेखा BC का समीकरण ज्ञात करो।

**Ans.** $14x + 23y = 40$

**Sol.**

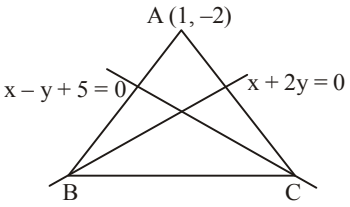



Image of A

B

$$\frac{x-1}{1}=\frac{y+2}{-1}=\frac{-2(1+2+5)}{2}$$

$x = 1 - 8 = -7$ B $(-7, 6)$

$y = -2 + 8 = 6$

C

$$\frac{x-1}{1}=\frac{y+2}{2}=\frac{-2(1-4)}{5}$$

$$x = 1+\frac{6}{5}=\frac{11}{5} \qquad C\left(\frac{11}{5},\frac{2}{5}\right)$$

$$y = -2+\frac{12}{5}=\frac{2}{5}$$

BC :→

$$(y-6)=\left(\frac{6-\frac{2}{5}}{-7-\frac{11}{5}}\right)(x+7)$$

$$(y-6)=\frac{28}{-46}(x+7)$$

$-23y + 138 = 14x + 98$

$14x + 23y - 40 = 0$

17. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the $\Delta$ ABC are G, H and S respectively. $L = 0$ denotes the equation of side BC.

(a) If $L_1 : 2x - y = 0$ and $L_2 : x + y = 3$ and G (2, 3) then find the slope of the line $L = 0$.

(b) If $L_1 : 2x + y = 0$ and $L_2 : x - y + 2 = 0$ and H (2, 3) then find the y-intercept of $L = 0$.

(c) If $L_1 : x + y - 1 = 0$ and $L_2 : 2x - y + 4 = 0$ and S(2, 1) then find the x-intercept of the line $L = 0$.

किसी त्रिभुज ABC जिसकी भुजाओं AB एवं AC के समीकरण यदि $L_1 = 0$ एवं $L_2 = 0$ हो, एवं केन्द्रक, लम्बकेन्द्र एवं परिकेन्द्र क्रमशः G, H एवं S हो, एवं $L = 0$ रेखा BC का समीकरण है।

(a) यदि $L_1 : 2x - y = 0$ एवं $L_2 : x + y = 3$ एवं G (2, 3) हो तो रेखा $L = 0$ की प्रवणता ज्ञात करो।

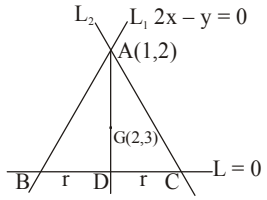
(b) यदि $L_1 : 2x + y = 0$ एवं $L_2 : x - y + 2 = 0$ एवं H (2, 3) तो रेखा $L = 0$ द्वारा y अक्ष पर काटे गये अंतखण्ड की लम्बाई ज्ञात करो।

(c) यदि $L_1 : x + y - 1 = 0$ एवं $L_2 : 2x - y + 4 = 0$ तथा S(2, 1) तो $L = 0$ द्वारा x अक्ष पर काटे गये अंतः खण्ड की लम्बाई ज्ञात करो।

**Ans.** (a) 5 (b) 2 (c) 3/2

**Sol.** (a) $\Delta ABC$, AB & AC $L_1 = 0$ & $L_2 = 0$

BC $\Rightarrow L - 0$ G → Centroid, H → Ortho, S → Circum




D Externally 3,1

$\left(\frac{6-1}{2},\frac{9-2}{2}\right)\equiv\left(\frac{5}{2},\frac{7}{2}\right)$

Let $B\left(\frac{5}{2}+r\cos\theta,\frac{7}{2}+r\sin\theta\right)$

$2x-y=0$

$5+2r\cos\theta-\frac{7}{2}-r\sin\theta=0$

$r\left(2\cos\theta-\sin\theta\right)=-\frac{3}{2}$

$r=\frac{3}{2\left(\sin\theta-2\cos\theta\right)}$

$C\left(\frac{5}{2}-r\cos\theta,\frac{7}{2}-r\sin\theta\right)$

$x+y=3$

$\frac{5}{2}+\frac{7}{2}-3=r\left(\cos\theta-\sin\theta\right)$

$r=\frac{3}{\cos\theta+\sin\theta}$

$2\sin\theta-4\cos\theta=\cos\theta+\sin\theta$

$\sin\theta=5\cos\theta\Rightarrow\tan\theta=5$

(b) $CH\Rightarrow x-2y+4=0$

$AC\Rightarrow x-y+2=0$

C(0,2)

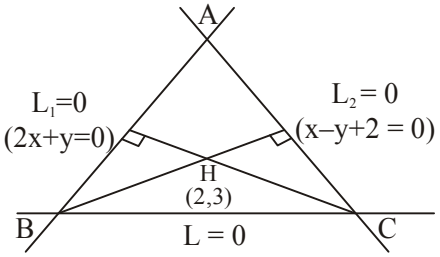
$BH\Rightarrow x+y-5=0$

$AB\Rightarrow 2x+y=0$

B (–5,10)

$BC\Rightarrow 8x+5y=10$

y-intercept = 2




(c) A (–1,2)

$C\rightarrow$ Image of A w.r.t. to SE

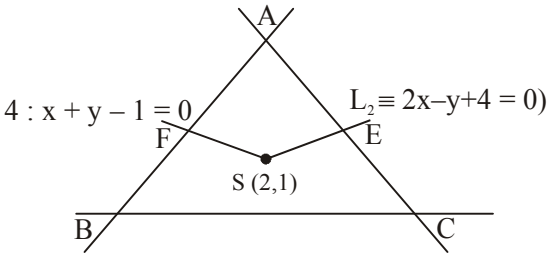
$SE\rightarrow x+2y-4=0$

$C\left(\frac{13}{5},\frac{46}{5}\right)$

$B\rightarrow$ Image of A w.r.t. SF

$SF\rightarrow x-y-1=0$

B(3, –2)

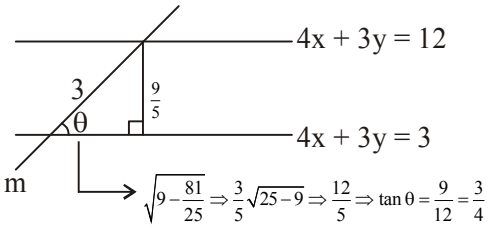


18. Find the equation of the straight lines passing through (–2, –7) & having an intercept of length 3 between the straight lines $4x + 3y = 12$, $4x + 3y = 3$.

बिन्दु (–2, –7) से गुजरने वाली रेखा का समीकरण ज्ञात करो, जो कि रेखाओं $4x + 3y = 12$, $4x + 3y = 3$ के मध्य 3 इकाई का अंतखण्ड काटती है।

**Ans.** $7x + 24y + 182 = 0$ or $x = -2$

**Sol.**




$$\tan\theta = \left|\frac{m + \frac{4}{3}}{1 - \frac{4}{3}m}\right| \Rightarrow \frac{3m+4}{3-4m} = \pm\frac{3}{4}$$

$\frac{3m+4}{3-4m} = \frac{3}{4}$ $\qquad$ $\frac{3m+4}{3-4m} = -\frac{3}{4}$

$12m + 16 = 9 - 12m$ $\qquad$ $12m + 16 = -9 + 12m$

$24m = -7$ $\qquad$ $\mu = \infty$

$$m = -\frac{7}{24}$$

$$(y + 7) = m(x + 2)$$

$24y + 168 = -7x - 14$ $\qquad$ $x = -2$

$7x + 24y + 182 = 0$

19. Consider a $\Delta$ ABC whose sides AB, BC and CA are represented by the straight lines $2x + y = 0$, $x + py = q$ and $x - y = 3$ respectively. The point P is (2, 3).

(a) If P is the centroid, then find the value of $(p + q)$.

(b) If P is the orthocentre, then find the value of $(p + q)$.

(c) If P is the circumcentre, then find the value of $(p + q)$.

किसी त्रिभुज ABC , जिसकी भुजाऐं AB, BC एवं CA क्रमशः रेखाओं के समीकरण $2x + y = 0$, $x + py = q$ तथा $x - y = 3$ से i nf' kZ gSA; fn P के निर्देशांक (2, 3)

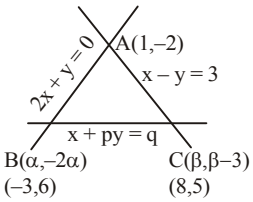
(a) यदि P केन्द्रक है, तो $(p + q)$ का मान ज्ञात करो

(b) यदि P लम्बकेन्द्र है, तो $(p + q)$ का मान ज्ञात करो

(c) यदि P परिकेन्द्र है, तो $(p + q)$ का मान ज्ञात करो

**Ans.** (a) 74 (b) 50 (c) 47

**Sol.**




(a) centroid (2, 3)

$1 + \alpha + \beta = 6$ ...(1)

$-2 - 2\alpha + \beta - 3 = 9$ ...(2)

$\alpha + \beta = 5$

$2\alpha - \beta = -14$

$3\alpha = -9$

$\alpha = -3,\ \beta = 8$

$x + py = q$ $\quad 8 + 5p = q$

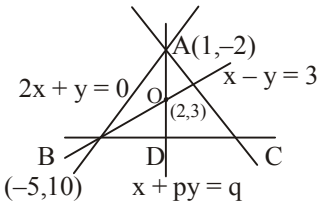
$-3 + p6 = q$

$-3 + 6p = 8 + 5p \Rightarrow p = 11$ & $q = 63$

$x + 11y = 63$

$p + q = 11 + 63 = 74$

(b) orthocentre (2, 3)




$$m_{AD} = \frac{3+2}{2-1} = 5$$

$$m_{BC} = -\frac{1}{5} = -\frac{1}{p}$$

$p = 5$

$BE \rightarrow x + y = \lambda$

(2, 3)

$x + y = 5$

$2x + y = 0$

$-x = 5$

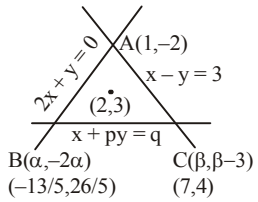
$x = -5$

$y = 10$

$-5 + 50 = q$

$q = 45$

$p + q = 45 + 5 = 50$

(C) 


(2,3) circumcentre

$R = \sqrt{1+25} = \sqrt{26}$

$(\alpha-2)^2 + (3+2\alpha)^2 = 26$

$\Rightarrow 5\alpha^2 - 4\alpha + 4 + 12\alpha + 9 = 26$

$\Rightarrow 5\alpha^2 + 8\alpha - 13 = 0 \rightarrow 1, \alpha$

$\alpha = -\frac{13}{5}$

$\alpha(\beta-2)^2 + (\beta-6)^2 = 26$

$\Rightarrow 2\beta^2 - 16\beta + 40 - 26 = 0$

$\Rightarrow 2\beta^2 - 16\beta + 14 = 0 \rightarrow 1, \beta$

$\beta = 7$

$7 + 4p = q$

$-13 + 26\,p = 5q$

$\underline{35 + 20\,p = 5q}$

$-48 + 6p = 0$

$p = 8$

$q = 7 + 32 = 39$

$\Rightarrow p + q = 8 + 39 = 47$

**PARAMETRIC FORM OF A LINE**

20. A line through the point P(2, –3) meets the lines $x - 2y + 7 = 0$ and $x + 3y - 3 = 0$ at the points A and B respectively. If P divides AB externally in the ratio 3 : 2 then find the equation of the line AB.

यदि बिन्दु P(2, –3) से गुजरने वाली रेखा, सरल रेखाओं $x - 2y + 7 = 0$ एवं $x + 3y - 3 = 0$ को क्रमशः बिन्दुओं A एवं B पर मिलती है। यदि बिन्दु P रेखाखण्ड AB को अनुपात 3 : 2 में बाह्यविभाजित करे तो AB का समीकरण ज्ञात कीजिये।

**Ans.** $2x + y - 1 = 0$

**Sol.** P(2, –3) $x - 2yb + 7 = 0$ & $x + 3y - 3 = 0$

$2 + 3k\cos\theta + 6 - 6k\sin\theta + 7 = 0$

$6k\sin\theta - 3k\cos\theta = 15$

$2k\sin\theta - k\cos\theta = 5$ ...(1)

$2 + 2k\cos\theta - 9 + 6k\sin\theta - 3 = 0$

$2k\cos\theta + 6k\sin\theta = 10$

$k\cos\theta + 3k\sin\theta = 5$ ...(2)

(1)/(2)

$$\frac{2s - c}{c + 3s} = 1$$

$$\Rightarrow \frac{2t - 1}{3t +} = 1$$

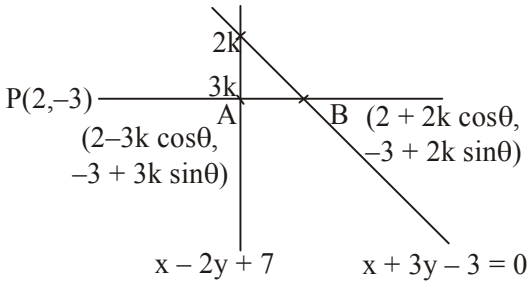
$\Rightarrow 2t - 1 = 3t + 1$

$\Rightarrow t = -2$

$(y + 3) = -2(x - 2)$

$y + 3 = -2x + y$

$2x + y = 1$




21. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ.

बिन्दु $P(\sqrt{3}, 2)$ से गुजरने वाली एवं x अक्ष से $\frac{\pi}{6}$ कोण बनाने वाली रेखा, किसी अन्य रेखा $\sqrt{3}x - 4y + 8 = 0$ को बिन्दु Q पर मिलती है, तो रेखाखण्ड PQ की लम्बाई ज्ञात करो।

**Ans.** 6 units

**Sol.** $P\left(\sqrt{3},2\right)$ $\quad Q\left(\sqrt{3}+\frac{r\sqrt{3}}{2},2+\frac{r}{2}\right)$

$\sqrt{3}x-4y+8=0$

$\Rightarrow 3+\frac{3r}{2}-8-2r+8=0$

$\Rightarrow \frac{r}{2}=3\Rightarrow r=6$

**FAMILY OF LINES**

22. Consider the family of lines $(x-y-6)+\lambda(2x+y+3)=0$ and $(x+2y-4)+\mu(3x-2y-4)=0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

रेखाओं का निकाय $(x-y-6)+\lambda(2x+y+3)=0$ एवं $(x+2y-4)+\mu(3x-2y-4)=0$. यदि दोनों निकायो की रेखायें एक दूसरे से समकोण बनाती है तो इनके प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात करो।

**Ans.** $x^2+y^2-3x+4y-3=0$

**Sol.** $(x-y-6)+l(2x+y+3)=0$

$2x+y+3=0$

$3x=3$

$x=1 \quad y=-5 \qquad (1,-5)$

$x+2y-4=0 \qquad 2+2y-4=0$

$3x-2y-4=0 \qquad 1+y-2=0$

$4x=8 \qquad y=1 \qquad (2,1)$

$x=2$

$(x-1)(x-2)+(y+5)(y-1)=0$

$x^2+y^2-3x+4y-3=0$

**CONCEPT OF IMAGE**

23. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

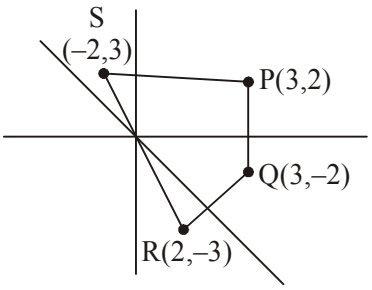
बिन्दु P (3, 2), जहाँ बिन्दु Q, बिन्दु P का, x अक्ष के सापेक्ष एवं बिन्दु R, बिन्दु Q का, रेखा $y = -x$ के सापेक्ष प्रतिबिम्ब बिन्दु है। R का मूल बिन्दु के सापेक्ष प्रतिबिम्ब S है तथा PQRS एक उत्तल चतुर्भुज है तो PQRS का क्षेत्रफल ज्ञात करो।

**Ans.** 15

**Sol.** P (3, 2)

Q (3, –2)

$$\Delta = \frac{1}{2}\begin{vmatrix} 3 & 3 & 2 & -2 & 3 \\ 2 & -2 & -3 & 3 & 2 \end{vmatrix}$$




$$= \frac{1}{2}[-6-6-9+4+6-6-4-9]$$

$$= \frac{1}{2}[30] = 15$$

**ANGLE BISECTORS**

24. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point $(1, -10)$. Determine the equation of the third side.

यदि समद्विबाहु त्रिभुज की दो समान भुजाओं के समीकरण $7x - y + 3 = 0$ एवं $x + y - 3 = 0$ हो, एवं तीसरी भुजा बिन्दु $(1, -10)$ से गुजरती हो, तो तीसरी भुजा का समीकरण ज्ञात करो।

**Ans.** $x - 3y - 31 = 0$ or $3x + y + 7 = 0$

**Sol.** $7x - y + 3 = 0$ $\qquad \frac{7x - y + 3}{x\sqrt{5}} = \infty$ $\qquad \pm\frac{x + y - 3}{\sqrt{2}}$

$x + y - 3 = 0$

$7x - y + 3 = 5x + 5y - 15$ $\qquad$ $7x - y + 3 = -5x - 5y + 15$

$2x - 6y + 18 = 0$ $\qquad$ $12x + 4y - 12 = 0$

$x - 3y + 9 = 0$ $\qquad$ $3x + y - 3 = 0$

$m = \frac{1}{3}$ $\qquad$ $m = -3$

$(y+10)=\frac{1}{3}(x-1)$ $\qquad$ $(y+10)=-3(x-1)$

$3y+30=x-1$ $\qquad$ $y+10=-3x+3$

$x-3y-31=0$ $\qquad$ $3x+y+7=0$

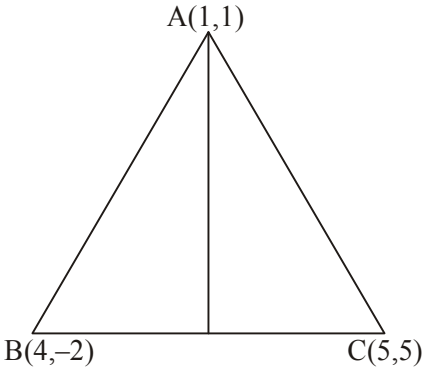
25. Given vertices A(1, 1), B(4, –2) and C(5, 5) of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

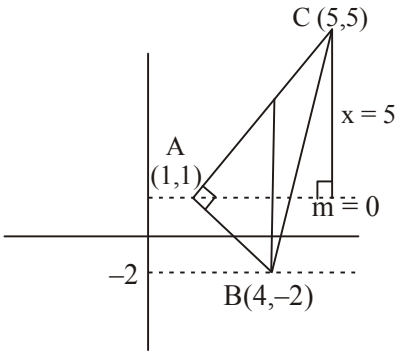
। (1, 1), B (4, –2) एवं C (5, 5) त्रिभुज के शीर्ष बिन्दु हो, तो बिन्दु C से कोण A के आन्तरिक कोणार्धक पर डाले गये लम्ब का समीकरण ज्ञात करो।

**Ans.** $x-5=0$

**Sol.** A(1,1), B(4, –2) & C(5, 5)

$$\left.\begin{matrix} m_{AC}=1 \\ m_{AB}=-1 \end{matrix}\right] \text{bisector} \qquad m=0$$




**GENERAL LOCUS PROBLEMS**

26. A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a}+\frac{y}{b}=1$ & $\frac{x}{b}+\frac{y}{a}=1$, meets the coordinate axes in A and B. Find the locus of the mid point of AB.

रेखाओं $\frac{x}{a}+\frac{y}{b}=1$ एवं $\frac{x}{b}+\frac{y}{a}=1$ के प्रतिच्छेद बिन्दुओं से गुजरने वाली चर रेखा, निर्देशांक अक्षों को क्रमशः A एवं B पर मिलती है, तो रेखाखंड AB के मध्य बिन्दु के बिन्दुपथ (locus) का समीकरण ज्ञात करो।

**Ans.** $2xy(a+b)=ab(x+y)$

**Sol.** By family of lines $\left(\frac{x}{a}+\frac{y}{b}-1\right)+\lambda\left(\frac{x}{b}+\frac{y}{a}-1\right)=0$ it cuts x-axis at $A\left(\frac{\lambda+1}{\frac{1}{a}+\frac{\lambda}{b}},\ 0\right)$

y - axis at $B\left(0,\ \frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}\right)$. Let mid point of AB is P(h, k) $\Rightarrow$ $2h=\frac{\lambda+1}{\frac{1}{a}+\frac{\lambda}{b}},\ 2k=\frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}$

eliminate $\lambda$ from both equations we get $2hk(a+b)=ab(h+k)$

Hence locus of P(h, k) is $2xy(a+b)=ab(x+y)$

27. In the xy plane, the line '$l_1$' passes through the point (1, 1) and the line '$l_2$' passes through the point (–1, 1). If the difference of the slopes of the lines is 2. Find the locus of the point of intersection of the lines $l_1$ and $l_2$

xy समतल में, यदि सरल रेखा line '$l_1$' बिन्दु (1, 1) एवं सरल रेखा '$l_2$' बिन्दु (–1, 1) से गुजरती है, एवं उसकी प्रवणताओं का अंतर 2 है तो सरल रेखाओं $l_1$ एवं $l_2$ के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात करो।

**Ans.** $y=x^2$ and $y=2-x^2$

**Sol.**

| | | |
|---|---|---|
| $m_1-m_2=2$ | or | $m_1-m_2=-2$ |
| $\frac{k-1}{h-1}-\frac{k-1}{h+1}=2$ | | $\frac{k-1}{h-1}-\frac{k-1}{h+1}=-2$ |
| ..... | | ..... |
| $x^2=y$ | | $y=2-x^2$ |

28. The area of the triangle formed by the intersection of a line parallel to x–axis and passing through P (h, k) with the lines $y=x$ and $x+y=2$ is $4h^2$. Find the locus of the point P.

ग अक्ष के समान्तर एवं P (h, k) से गुजरने वाली रेखा तथा रेखाओं $y=x$ एवं $x+y=2$ द्वारा निर्मित त्रिभुज का क्षेत्रफल $4h^2$ है तो बिन्दु P का बिन्दु पथ ज्ञात कीजिये।

**Ans.** $y=2x+1,\ y=-2x+1$

**Sol.** A line passing through P(h, k) and parallel to x-axis is $y=k$. .....(i)

The other lines given are $y=x$ .....(ii)

and $y+x=2$ .....(iii)

Let ABC be the $\Delta$ formed by the points of intersection of the lines (i), (ii) and (iii).

$\therefore$ A(k, k) , B(1, 1), C (2–k, k)

$$\therefore \quad \text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} k & k & 1 \\ 1 & 1 & 1 \\ 2-k & k & 1 \end{vmatrix} = 4h^2$$

$$C_1 \to C_1 - C_2 \quad \frac{1}{2}\begin{vmatrix} 0 & k & 1 \\ 0 & 1 & 1 \\ 2-2k & k & 1 \end{vmatrix} = 4h^2$$

$$\Rightarrow \quad \frac{1}{2}|(2-2k)(k-1)| = 4h^2 \quad \Rightarrow \quad (k-1)^2 = 4h^2$$

$$\Rightarrow \quad k-1 = 2h, \quad k-1 = -2h \quad \Rightarrow \quad k = 2h+1 \quad k = -2h+1$$

$\therefore$ locus of (h, k) is $y = 2x + 1$ $y = -2x + 1$

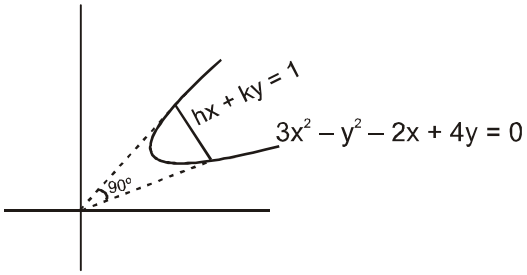
**HOMOGENISATION**

29. Show that all the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrency & if not, give reasons.

दशाओं कि किसी वक्र $3x^2 - y^2 - 2x + 4y = 0$ की सभी जीवाऐं, जो कि मूल बिन्दु पर समकोण बनाती है, संगामी होती है। क्या उपरोक्त कथन वक्र $3x^2 + 3y^2 - 2x + 4y = 0$ के लिये भी सत्य है, यदि हाँ, तो संगमन बिन्दु ज्ञात करो यदि नही, तो कारण बताओं।

**Ans.** $(1, -2)$, yes $\left(\frac{1}{3}, -\frac{2}{3}\right)$

**Sol.** Let equation of chords $hx + ky = 1$




By homogenisation

$3x^2 - y^2 - 2x(hx + ky) + 4y(hx + ky) = 0$

$\therefore$ it makes 90°. Hence

coeff. $x^2$ + coeff. $y^2 = 0$

$3 - 2h - 1 + 4k = 0 \Rightarrow h - 2k = 1$

Hence all chords are concurrent at $(1, -2)$

$3x^2 + 3y^2 - 2x + 4y = 0$

$$x^2 + y^2 - \frac{2}{3}x + \frac{y}{3}y = 0$$

circle centre at $\left(\frac{1}{3},-\frac{2}{3}\right)$

Passing through origin chords subtend 90º at origin will be diameters P.O. concurrents

$\left(\frac{1}{3},-\frac{2}{3}\right)$

30. A straight line is drawn from the point (1, 0) to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

यदि बिन्दु (1, 0) से वक्र $x^2 + y^2 + 6x - 10y + 1 = 0$ पर रेखा इस तरह खीची जाए कि रेखा द्वारा वक्र पर काटे जाने वाला अन्तः खण्ड मूल बिन्दु पर समकोण बनाये तो रेखा का समीकरण ज्ञात करो।

**Ans.** $x + y = 1;\ x + 9y = 1$

**Sol.** (1,0) $\qquad x^2 + y^2 + 6x - 10y + 1 = 0$

$x + py = 1 \qquad x^2 + y^2 + 6x(x + py) - 10y(x + py) + (x + py)^2 = 0$

$\Rightarrow x^2 + y^2 + 6x^2 + 6pxy - 10xy - 10py^2 + x^2 + p^2y^2 + 2pxy = 0$

$\Rightarrow 8x^2 + (1 - 10p)y^2 + (8p - 10)xy = 0$

Perpendicular if $a + b = 0 \Rightarrow p^2 - 10p + 9 = 0$

$\Rightarrow p = 9, 1$

$x + y = 1$

$x + 9y = 1$

31. If the straight line joining the origin to the points of intersection of $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ and $2x + 3y = k$ are at right angles, then find the value of $5k - 6k^2$.

यदि वक्र $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ तथा रेखा $2x + 3y = k$ की उभयनिष्ट निर्देशांको को मूल बिन्दु से मिलाकर रेखा बनायी जाये तथा वो दोनों रेखा के बीच 90° का कोण बनता है, तो $5k - 6k^2$ का मान बताओ।

**Ans.** 52

**Sol.** Curve : $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0 \qquad .....(1)$

Line : $2x + 3y = k \quad \Rightarrow \quad \frac{2x + 3y}{k} = 1 \quad ....(2)$

homogenise equation (1) with the help of equation (2), combined equation of line pair is

$$(3x^2 - xy + 3y^2) + (2x - 3y)\left(\frac{2x + 3y}{k}\right) + 4\left(\frac{2x + 3y}{k}\right)^2 = 0$$

given angle between the lines is 90° $\quad \Rightarrow \quad$ coefficient of $x^2$ + coefficient of $y^2 = 0$

$$\left(3+\frac{4}{k}+\frac{16}{k^2}\right)+\left(3-\frac{9}{k}+\frac{66}{k^2}\right)=0$$

$$6-\frac{5}{k}+\frac{52}{k^2}=0 \quad \Rightarrow \quad 6k^2-5k+52=0$$

$\therefore \quad 5k-6k^2=52$

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question Nos. 32 to 34**

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the points E(3, 4) and F(1, 2) respectively and centroid of $\Delta$ ABC is G(3, 2), then answer the following questions:

मानाकि ABC एक न्यूनकोण त्रिभुज है तथा AD, BE तथा CF इसकी माध्यिकाएँ है जहाँ E तथा F क्रमशः बिन्दु E(3, 4) और F(1, 2) है तथा त्रिभुज ABC का केन्द्रक G(3, 2) है, तो निम्न प्रश्नों के उत्तर दीजिए।

32. The equation of side AB is :

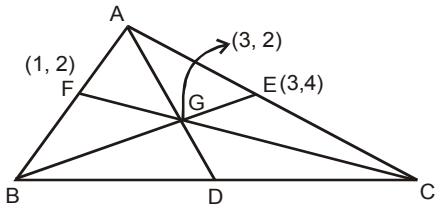
(A) $2x+y=4$ (B) $x+y-3=0$ (C) $4x-2y=0$ (D) none of these

भुजा AB का समीकरण है :

(A) $2x+y=4$ (B) $x+y-3=0$ (C) $4x-2y=0$ (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** Taking $A(x_1, y_1)$ , $B(x_2, y_2)$ and $C(x_3, y_3)$




then by $\frac{x_1+x_2}{2}=1$, $\frac{x_2+x_3}{2}=5$, $\frac{x_3+x_1}{2}=3$ and $\frac{y_1+y_2}{2}=2$, $\frac{y_2+y_3}{2}=0$, $\frac{y_1+y_3}{2}=4$

we get A(–1, 6) , B(3, –2) , C(7, 2) equation of AB is $2x+y=4$

33. Coordinate of D are :

D के निर्देशांक है :

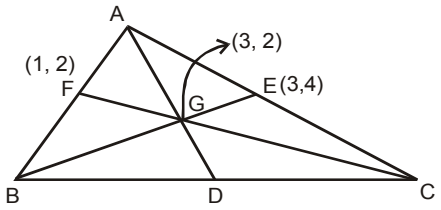
(A) (7, –4) (B) (5, 0) (C) (7, 4) (D) (– 3, 0)

**Ans.** B

**Sol.** Let the co-ordinates of D$(\alpha, \beta)$ then $\frac{\alpha+1+3}{3}=3 \quad \Rightarrow \quad a = 5$

and $\frac{\beta+2+4}{3}=2 \quad \Rightarrow \quad b = 0$

$\therefore \quad D(5, 0)$



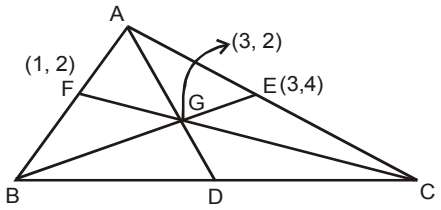

34. Height of altitude (in units) drawn from point A is :

बिन्दु A से डाले गये शीर्षाभिलम्ब की ऊँचाई (इकाई में) है :

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$ (C) $6\sqrt{2}$ (D) $2\sqrt{3}$

**Ans.** C

**Sol.**




Height of altitude from A is $= \frac{2 \times \text{area}\,(\Delta ABC)}{BC} = 6\sqrt{2}$

**Paragraph for Question Nos. 35 to 36**

Consider a triangle ABC with vertex A(2, –4). The internal bisectors of the angles B and C are $x + y = 2$ and $x – 3y = 6$ respectively. Let the two bisectors meet at I.

माना एक त्रिभुज ABC जिसका शीर्ष A(2, –4) है। कोण B तथा C के अन्तःकोण समद्विभाजकों का समीकरण क्रमशः $x + y = 2$ तथा $x – 3y = 6$ है। माना दो समद्विभाजक I पर मिलते है।

35. If (a, b) is incentre of the triangle ABC then (a + b) has the value equal to :

यदि (a, b) त्रिभुज ABC का अन्तः केन्द्र है, तो (a + b) का मान बराबर होगा :

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** B

**Sol.** $x – 3y = 6$

$x + y = 2$

$I\,(3, -1) \equiv (a, b)$

$a + b = 2$

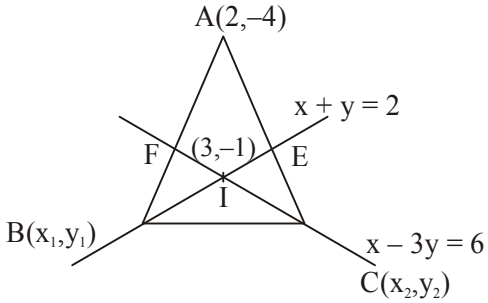
36. If $(x_1, y_1)$ and $(x_2, y_2)$ are the co–ordinates of the point B and C respectively, then the value of $(x_1 x_2 + y_1 y_2)$ is equal to :

यदि $(x_1, y_1)$ तथा $(x_2, y_2)$ क्रमशः बिन्दु B तथा C के निर्देशांक है, तो $(x_1 x_2 + y_1 y_2)$ का मान होगा :

(A) 4 (B) 5 (C) 6 (D) 8

**Ans.** D

**Sol.** P = image of A (2, – 4) in $x - 3y = 6$




$P\left(\frac{2}{5}, \frac{4}{5}\right)$

Q = Image of A (2,–4) in $x + y = 2$ Q (6,0)

Equation of BX = Equation of PQ $\Rightarrow$

$BC \Rightarrow x + 7y = 6$

$C \rightarrow x + 7y = 6$ $B \rightarrow x + 7y = 6$

$x + y = 2$ $x - 3y = 6$

$C\left(\frac{4}{3}, \frac{2}{3}\right)$ B (6,0)

$x_1x_2 + y_1y_2 = 8$

## ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

37. The area of triangle ABC is 20 $cm^2$ . The co–ordinates of vertex A are (–5, 0) and B are (3, 0). The vertex C lies on the line, $x - y = 2$ . The co–ordinates of C are :

$\Delta$ ABC का क्षेत्रफल 20 $cm^2$ है। शीर्ष A (–5, 0) तथा B (3, 0) है। शीर्ष C रेखा $x - y = 2$ पर स्थित है। C के निर्देशांक है :

(A) (5, 3) (B) (–3, –5) (C) (–5, –7) (D) (7, 5)

**Ans.** BD

**Sol.** Let any point on the line $x - y = 2$ be $C \equiv (h, h-2)$. Given area of $\Delta ABC$ is

$$\left|\frac{1}{2}\begin{vmatrix} h & h-2 & 1 \\ -5 & 0 & 1 \\ 3 & 0 & 1 \end{vmatrix}\right| = 20$$

$$\Rightarrow |8(h-2)| = 40$$

$$\Rightarrow h - 2 = \pm 5$$

$$h = 7, -3$$

Hence, the points are (7,5) and (–3,–5)

38. Three vertices of a triangle are A(4, 3) ; B(1, – 1) and C(7, k) . Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the $\Delta$ ABC lie on the same straight line is/are :

(A) 7 (B) – 1 (C) – 19/8 (D) none

त्रिभुज के तीन शीर्ष A(4, 3) ; B(1, – 1) तथा C(7, k) है। k के मान जिनके लिये केन्द्रक, लम्बकेन्द्र, अन्तःकेन्द्र तथा परिकेन्द्र एक ghl jy j§ ki j gS

(A) 7 (B) – 1 (C) – 19/8 (D) कोई नहीं

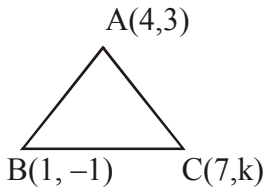
**Ans.** BC

**Sol.** AB = AC

$\Rightarrow 9 + 16 = 9 + (k-3)^2$

$\Rightarrow k = 3 \pm 4 \Rightarrow k = 7, -1$

for k = 7 C(7,7), B(1,–1), A(4,3) are collinear.




so, $k \neq 7$

AB = BC

$\Rightarrow 9 + 16 = 36 + (k+1)^2$

Not possible

AC = BC

$\Rightarrow 36 + (k+1)^2 = 9 + (k-3)^2$

$27 + k^2 + +2k + 1 = k^2 - 6k + 9$

$\Rightarrow 8k = -19 \Rightarrow k = -\frac{19}{8}$

39. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :

यदि $a^2 + 9b^2 - 4c^2 = 6ab$ तो रेखाओं का समूह $ax + by + c = 0$ किस बिन्दु पर संगामी होगा :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, –3/2)

**Ans.** CD

**Sol.** Given $a^2 + 9b^2 - 4c^2 = 6ab$

$\Rightarrow (a-3b)^2 = (2c)^2$

$\Rightarrow (a-3b) = \pm c$

$\therefore a - 3b + 2c = 0$ $a - 3b = 2c$

$c = \frac{3b-a}{2}$ $c = \frac{a-3b}{2}$

family of lines is $ax + by + c = 0$

$ax + by + \frac{3b-a}{2} = 0$ $ax + by + \frac{a-3b}{2} = 0$

if these two lines are concurrent then

$3b - a = 0$

now, $ax + by + \frac{3b-a}{2} = 0$

$3bx + by = 0$ $y = 3x$

$\therefore$ The points will be $\left(-\frac{1}{2}, \frac{3}{2}\right)$ and $\left(\frac{1}{2}, \frac{-3}{2}\right)$.

40. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is :

(A) isosceles (B) right angled (C) obtuse angled (D) equilateral

रेखाओं $x + y = 0$, $3x + y - 4 = 0$ एवं $x + 3y - 4 = 0$ से बनने का वाला त्रिभुज होगा :

(A) समद्विबाहु (B) समकोणीय (C) अधिक कोणीय (D) समबाहु

**Ans.** AC

**Sol.** $x + y = 0$ _____(i)

$3x + y - 4 = 0$ _____(ii)

$x + 3y - 4 = 0$ _____(iii)

Solving lines (i) and (ii), we get

$-x = -3x + 4 \Rightarrow x = 2 \Rightarrow y = -2$

$\therefore$ (i) and (ii) intersect at $A = (2, -2)$

solving lines (ii) and (iii), we get

$$-3x + 4 = \frac{4 - x}{3} \Rightarrow -9x + 12 = 4 - x \Rightarrow x = 1 \Rightarrow y = 1$$

$\therefore$ (ii) and (iii) intersect at $B = (1,1)$

Solving lines (i) and (iii), we get

$$-x = \frac{4 - x}{3} \Rightarrow -3x = 4 - x \Rightarrow x = -2 \Rightarrow y = 2$$

$\therefore$ (i) and (iii) intersect at $C = (-2,2)$

So, AB, BC, AC form a triangle ABC

Now, $AB = \sqrt{(1-2)^2 + (1+2)^2} = \sqrt{10}$

$$BC = \sqrt{(-2-1)^2 + (2-1)^2} = \sqrt{10}$$

$$AC = \sqrt{(-2-2)^2 + (2+2)^2} = \sqrt{32}$$

$\therefore AB = BC$

Since, two sides of a triangle are equal then the triangle formed by A, B, C is isosceles triangle.

41. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point $(1, -10)$. The equation of the third side can be :

किसी समद्विबाहु त्रिभुज की दो समान भुजा क्रमशः $7x - y + 3 = 0$ एवं $x + y - 3 = 0$ है तथा इसकी तीसरी भुजा यदि $(1, -10)$ से गुजरती हो, तो तीसरी भुजा का समीकरण होगा :

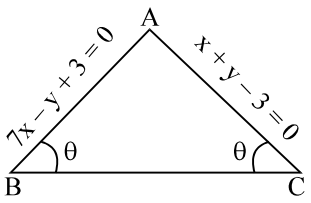
(A) $x + 3y + 29 = 0$ (B) $x - 3y = 31$ (C) $3x - y = 13$ (D) $3x + y + 7 = 0$

**Ans.** BD

**Sol.** $m_{AB} = 7$ and $m_{AC} = -1$

Let slope of BC (third side) be m.

Now $\left|\frac{7-m}{1+7m}\right| = \left|\frac{-1-m}{1-m}\right| \Rightarrow \left(\frac{7-m}{1+7m}\right) = \pm\left(\frac{-1-m}{1-m}\right)$




$\therefore$ Taking '+' sign, we get

$m^2 + 1 = 0$ (no real values of m possible)

And taking '–' sign, we get

$3m^2 + 8m - 3 = 0 \Rightarrow (3m-1)(m+3) = 0 \Rightarrow m = \frac{1}{3}, -3$

$\therefore$ Required equation of third side can be $x - 3y - 31 = 0$ or $3x = y + 7 = 0$

42. Straight lines $2x + y = 5$ and $x - 2y = 3$ intersect at the point A. Points B and C are chosen on these two lines such that $AB = AC$. Then the equation of a line BC passing through the point $(2, 3)$ is :

रेखा $2x + y = 5$ एवं $x - 2y = 3$ एक दूसरे को बिन्दु A पर प्रतिच्छेदित करती है। बिन्दु B एवं C क्रमशः इन दोनों रेखाओं पर bl i dkj l sgSSd AB = AC। यदि BC रेखा बिन्दु (2, 3) से गुजरती हो, तो BC का समीकरण होगा :

(A) $3x - y - 3 = 0$ (B) $x + 3y - 11 = 0$ (C) $3x + y - 9 = 0$ (D) $x - 3y + 7 = 0$

**Ans.** AB

**Sol.** Note that the lines are perpendicular . Find the equation of the lines through (2, 3) and parallel to the bisectors of the given lines, the slopes of the bisectors being $-1/3$ and 3

43. The lines $L_1$ and $L_2$ denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients $m_1$ and $m_2$ respectively. The acute angles between them is θ. Which of the following relations hold good?

(A) $m_1 + m_2 = 5/4$

(B) $m_1 m_2 = 3/8$

(C) acute angle between $L_1$ and $L_2$ is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$

(D) sum of the abscissa and ordinate of the point P is – 1

रेखा युग्म $L_1$ एवं $L_2$ समीकरण $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ से व्यक्त होता हैं तथा P पर प्रतिच्छेदित करता है तथा ft ud hi ø.k dk $m_1$ एवं $m_2$ है। यदि उनके बीच का न्यून कोण θ हो, तो इनमें से कौनसा सम्बन्ध सत्य होगा?

(A) $m_1 + m_2 = 5/4$

(B) $m_1 m_2 = 3/8$

(C) $L_1$ एवं $L_2$ का बीच का न्यून कोण $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$ होगा

(D) बिन्दु P की भुजा एवं कोटि का योग – 1 होगा

**Ans.** BCD

**Sol.** $f(x, y) = (x + 2y + 3)(3x + 4y + 5)$

$m_1 + m_2 = -\frac{2h}{b} = -\frac{10}{8} = -\frac{5}{4}$

$m_1 m_2 = \frac{a}{b} = \frac{3}{8} \Rightarrow$ (B) is correct

also $L_1$ and $L_2$ intersect at $(1, -2)$

$\therefore$ sum of abscissa and ordinate $= -1 \Rightarrow$ (D) is correct

$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b} = \frac{2\sqrt{25-24}}{11} = \frac{2}{11}$

$\sin\theta = \frac{2}{5\sqrt{5}} \Rightarrow q = \sin^{-1}\left(\frac{2}{5\sqrt{5}}\right) \Rightarrow$ (C) is correct

44. Let B $(1, -3)$ and D $(0, 4)$ represent two vertices of rhombus ABCD in (x, y) plane, then coordinates of vertex A, if $\angle BAD = 60°$, can be equal to :

यदि B $(1, -3)$ एवं D $(0, 4)$ किसी समचतुर्भुज के शीर्ष है यदि $\angle BAD = 60°$ तो शीर्ष A का निर्देशांक होगा :

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

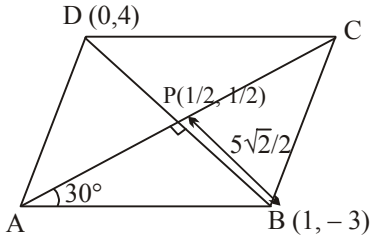
(C) $\left(\frac{1-14\sqrt{3}}{2}, \frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2}, \frac{1+2\sqrt{3}}{2}\right)$

**Ans.** AB

**Sol.** Let inclination of AC is $\theta$

$\tan\theta = \frac{1}{7}$ $\sin\theta = \frac{1}{5\sqrt{2}}$, $\cos\theta = \frac{7}{5\sqrt{2}}$

$AP = \frac{5}{2}\sqrt{2}\cot 30° = \frac{5\sqrt{2}}{2}\sqrt{3}$




$A = \left(\frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$

or $\left(\frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$

$\Rightarrow \quad A = \left(\frac{1}{2} - \frac{7\sqrt{3}}{2}, \frac{1}{2} - \frac{\sqrt{3}}{2}\right)$ or $\left(\frac{1}{2} + \frac{7\sqrt{3}}{2}\right), \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)$

45. The equation of the bisectors of the angle between the two intersecting lines :

$\frac{x-3}{\cos\theta} = \frac{y+5}{\sin\theta}$ and $\frac{x-3}{\cos\varphi} = \frac{y+5}{\sin\varphi}$ are $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$ and $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$ then :

दो प्रतिच्छेदी रेखाओं $\frac{x-3}{\cos\theta} = \frac{y+5}{\sin\theta}$ एवं $\frac{x-3}{\cos\varphi} = \frac{y+5}{\sin\varphi}$ के मध्य कोणों के अर्द्धकों की समीकरण $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$ एवं $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$ हो, तो :

(A) $\alpha = \frac{\theta+\phi}{2}$ (B) $\beta = -\sin\alpha$ (C) $\gamma = \cos\alpha$ (D) $\beta = \sin\alpha$

**Ans.** ABC

**Sol.** one bisector makes an angle $\left(\frac{\theta+\phi}{2}\right)$ with x-axis, then other bisector makes an angle $90^\circ + \left(\frac{\theta+\phi}{2}\right)$ with x-axis

Equation of bisector are

$$\frac{x-3}{\cos\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\theta+\phi}{2}\right)} \qquad ...(1)$$

and $\frac{x-3}{\cos\left(\frac{\pi}{2}+\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\pi}{2}+\frac{\theta+\phi}{2}\right)}$

$$\frac{x-3}{-\sin\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\cos\left(\frac{\theta+\phi}{2}\right)} \qquad ...(2)$$

But given bisector are $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$

$\therefore \alpha = \frac{\theta+\phi}{2}$

and $\frac{x-3}{\beta}=\frac{y+5}{\gamma}$

$\therefore \beta=-\sin\left(\frac{\theta+\phi}{2}\right)=-\sin\alpha$

and $y=\cos\left(\frac{\theta+\phi}{2}\right)=\cos\alpha$

Hence, option (A), (B), (C) is correct answer.

46. Equation of a straight line passing through the point (4, 5) and equally inclined to the lines,$3x = 4y + 7$ and $5y = 12x + 6$ is :

बिन्दु (4, 5) से गुजरने वाली एवं रेखाओं $3x = 4y + 7$ तथा $5y = 12x + 6$ से समान कोण पर झुकी रेखा की समीकरण है :

(A) $9x - 7y = 1$ (B) $9x + 7y = 71$ (C) $7x + 9y = 73$ (D) $7x - 9y + 17 = 0$

**Ans.** AC

**Sol.** The lines will pass through (4, 5) & parallel to the bisectors between them $\frac{3x-4y-7}{5}=\pm\frac{12x-5y+6}{13}$

by taking + sign, we get $21x + 27y + 121 = 0$. Now by taking – sign, we get $99x - 77y - 61 = 0$

so slopes of bisectors are $-\frac{7}{9}, \frac{9}{7}$

Equation of lines are $y-5=\frac{-7}{9}(x-4)$ and $y-5=\frac{9}{7}(x-4)$

$\Rightarrow$ $7x + 9y = 73$ and $9x - 7y = 1$

47. One side of a rectangle lies along the line $4x + 7y + 5 = 0$. Two of its vertices are (–3, 1) and (1, 1). Then the equations of other sides are :

एक आयत की एक भुजा रेखा $4x + 7y + 5 = 0$ के अनुदिश है। इसके दो शीर्ष (–3, 1) एवं (1, 1) हैं। तो इसकी अन्य भुजाओं के l eld j . kgS

(A) $7x - 4y + 25 = 0$ (B) $7x + 4y + 25 = 0$ (C) $7x - 4y - 3 = 0$ (D) $4x + 7y = 11$

**Ans.** ACD

**Sol.** Line $\perp$ to $4x + 7y + 5 = 0$ is $7x - 4y + \lambda = 0$

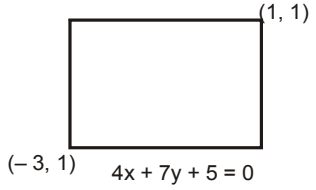
It passes through (– 3, 1) and (1, 1)

$-21 - 4 + \lambda = 0 \quad \Rightarrow \lambda = 25$

$7 - 4 + \lambda = 0 \quad \Rightarrow \lambda = -3$

Hence lines are $7x - 4y + 25 = 0$, $7x - 4y - 3 = 0$

line || to $4x + 7y + 5 = 0$ passing through (1, 1) is $4x + 7y + \lambda = 0$



48. Suppose ABCD is a quadrilateral such that the coordinates of A, B and C are (1, 3), (–2, 6) and (5, – 8) respectively. For what choice of coordinates of D will make ABCD a trapezium :

माना ABCD एक चतुर्भुज है, तथा बिन्दु A, B तथा C के निर्देशांक क्रमशः (1, 3), (–2, 6) तथा (5, – 8) है। ABCD एक समलम्ब pr dhZ gksus ds fy, fn; sx; sfod Yi ksa ds lEHkkfor funsZ'kkad gksxs :

(A) (3, – 6) (B) (6, –9) (C) (0, 5) (D) (3, –1)

**Ans.** ABCD

**Sol.** One pair of parallel line should have

49. The points A(0, 0), B(cosα, sinα) and C(cosβ, sinβ) are the vertices of a right angled triangle if :

बिन्दु A(0, 0), B(cosα, sinα) तथा C(cosβ, sinβ) एक समकोण त्रिभुज के शीर्ष है, यदि :

(A) $\sin\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (B) $\cos\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

(C) $\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (D) $\sin\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

**Ans.** ABCD

**Sol.** OA = 1 and OB = 1 $\quad\therefore AB=\sqrt{2}$

$$\therefore(\cos\beta-\cos\alpha)^2+(\sin\beta-\sin\alpha)^2=2$$

$$\Rightarrow\cos^2\alpha+\sin^2\alpha+\cos^2\beta+\sin^2\beta-2\sin\beta\sin\alpha-2\cos\beta\cos\alpha=2$$

$$\Rightarrow 2\sin\beta\sin\alpha+2\cos\beta\cos\alpha=0$$

$$\therefore\cos(\alpha-\beta)=0$$

$$\Rightarrow\sin\left(\frac{\alpha-\beta}{2}\right)=\pm\frac{1}{\sqrt{2}}$$

$$\rightarrow\cos\left(\frac{\alpha-\beta}{2}\right)=\pm\frac{1}{\sqrt{2}}$$

50. Two vertices of the Δ ABC are at the points A(– 1, – 1) and B(4, 5) and the third vertex lines on the straight line $y=5(x-3)$. If the area of the Δ is 19/2 then the possible co–ordinates of the vertex C are :

ΔABC का शीर्ष A(– 1, – 1) तथा B(4, 5) है। यदि तीसरा शीर्ष रेखा $y=5(x-3)$ पर है तथा त्रिभुज का क्षेत्रफल 19/2 है, तो बिन्दु C का समीकरण निर्देशांक होगा :

(A) (5, 10) (B) (3, 0) (C) (2, –5) (D) (5, 4)

**Ans.** AB

**Sol.** $A(-1,-1)$, $B(4,5)$, $C(\alpha, 5\alpha-15)$

$$\begin{vmatrix} -1 & -1 & 1 \\ 4 & 5 & 1 \\ \alpha & 5\alpha-15 & 1 \end{vmatrix} = \pm 19 \Rightarrow \alpha = 5, 3$$

51. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x - \sqrt{3}\, y = 0$ then the co–ordinates of the third vertex are :

यदि a भुजा की लम्बाई वाले समबाहु त्रिभुज की एक शीर्ष (0, 0) पर है तथा दूसरी रेखा $x - \sqrt{3}\, y = 0$ पर हो, तो तीसरे शीर्ष का funkZkd gksxk:

(A) (0, a) (B) $\left(\frac{\sqrt{3}a}{2}, -\frac{a}{2}\right)$ (C) (0, – a) (D) $\left(-\frac{\sqrt{3}a}{2}, \frac{a}{2}\right)$

**Ans.** ABCD

**Sol.** Make a figure and interpret

52. The sides of a triangle are the straight lines $x + y = 1$; $7y = x$ and $\sqrt{3}\, y + x = 0$. Then which of the following is an interior point of the triangle?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

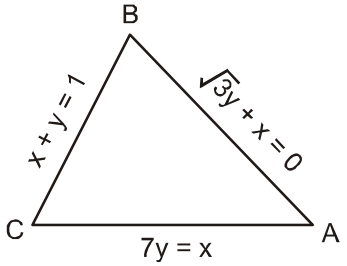
यदि त्रिभुज ABC की भुजाओं का समीकरण $x + y = 1$ ; $7y = x$ और $\sqrt{3}\, y + x = 0$ हैं, तो इनमें से कौनसा बिन्दु त्रिभुज के भीतर gksxk?

(A) परिकेन्द्र (B) केन्द्रक (C) अन्तः केन्द्र (D) लम्बकेन्द्र

**Ans.** BC

**Sol.** Let slope of given lines $m_1 = \frac{1}{7}$, $m_2 = \frac{-1}{\sqrt{3}}$, $m_3 = -1$

Hence interior angle of triangle




$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\frac{1}{7} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{7\sqrt{3}}} = \frac{\sqrt{3} + 7}{7\sqrt{3} - 1} > 0$$

$$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{-\frac{1}{\sqrt{3}} + 1}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} > 0 \quad \Rightarrow \quad \tan C = \frac{m_3 - m_1}{1 + m_2 m_1} = \frac{-1 - \frac{1}{7}}{1 - \frac{1}{7}} = \frac{-8}{6} < 0$$

Hence angle C is obtuse therefore circumcentre and orthocentre lies outside the triangle.

## MATCH THE COLUMN TYPE QUESTION(S)

53. Match the following column :

| **Column –I** | **Column – II** |
|---|---|
| (A) Slope of line bisecting co–ordinate axis, is | (P) 3 |
| (B) Area of $\Delta$ formed by line $3x + 4y + 12 = 0$, with co–ordinate axis is | (Q) 1 |
| (C) If the equation $2x^2 - 2xy - y^2 - 6x + 6y + c = 0$ represents a pair of lines then 'C' is | (R) 6 |
| (D) If distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $25\sqrt{2}$, then 'a/5 is equal to | (S) – 1 |

निम्न कॉलम को सुमेलित कीजिए :

| **Column –I** | **Column – II** |
|---|---|
| (A) निर्देशांक अक्षों के मध्य के कोण को समद्विभाजित करने वाली रेखा की प्रवणता होगी | (P) 3 |
| (B) रेखा $3x + 4y + 12 = 0$ तथा निर्देशांक अक्षों से बने त्रिभुज का क्षेत्रफल होगा | (Q) 1 |
| (C) समीकरण $2x^2 - 2xy - y^2 - 6x + 6y + c = 0$ एक रेखा युग्म को निरूपित करती है तो 'C' का मान होगा | (R) 6 |
| (D) यदि समान्तर रेखाओं $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ के मध्य की दूरी $25\sqrt{2}$, है तो 'a/5 का मान होगा | (S) – 1 |

**Ans.** A – QS, B – R, C – P, D – QS

**Sol.** (A) Equation of bisector

$y = mx$

$m = \tan 45^\circ = 1$

or $m = \tan 135^\circ = -1$

$\therefore\ y = \pm x$

slope $= m = \pm 1$

(B) Equation of x-axis is $y = 0$ _____(1)

Equation of y-axis is $x = 0$ _____(2)

Let given equation $3x + 4y + 12 = 0$ meets x-axis at point A and y-axis at B.

At A we have $y = 0 \Rightarrow 3x = -12$

$\Rightarrow x = -4$

$\therefore$ A is $(-4,0)$

$\Rightarrow OA = \sqrt{(-4)^2 + 0} = 4$ units

At B we have $x = 0 \Rightarrow 4y = -12$

$\Rightarrow y = -3$

$\therefore$ B is $(0, -3)$

$\Rightarrow OB = \sqrt{0 + (-3)^2} = 3$ *units*

$$\text{Area of triangle} = \frac{1}{2} \times OA \times OB = \frac{1}{2} \times 4 \times 3 = 6 \text{ sq units}$$

(C) $\Delta = 0 \Rightarrow C = 3$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow -2c + 18 - 18 + 9 - c = 0$

$\Rightarrow -3c = -9 \Rightarrow c = 3$

(D) $(x + y)^2 - 8a(x + y) - 9a^2 = 0$

$(x + y - 9a)(x + y + a) = 0$

$$25\sqrt{2} = \left|\frac{10a}{\sqrt{2}}\right|$$

$|a| = 5 \Rightarrow a = \pm 5$

54. Match the following column :

| **Column–I** | **Column–II** |
|---|---|
| (A) If a, b, c are in A. P., then lines $ax + by + c = 0$ are concurent at | (P) $(-4, -7)$ |
| (B) A point on the line $x + y = 4$ which lies at a unit distance from the line $4x + 3y = 10$ is | (Q) $(-7, 11)$ |
| (C) Orthocentre of trianlge made by lines $x + y = 1$, $x - y + 3 = 0$, $2x + y = 7$ is | (R) $(1, -2)$ |
| (D) Two vertice of a triangle are $(5, -1)$ and $(-2, 3)$. If orthcentre is the origin then coordinated of the third vertex are | (S) $(-1, 2)$<br>(T) $(0, 0)$ |

निम्न कॉलम को सुमेलित कीजिए :

| **Column–I** | **Column–II** |
|---|---|
| (A) यदि a, b, c समान्तर श्रेणी में है, तो रेखाऐं $ax+by+c= 0$ किस बिन्दु पर संगामी है | (P) $(-4, -7)$ |
| (B) एक बिन्दु रेखा $x + y = 4$ पर स्थित है, रेखा $4x + 3y = 10$ से इकाई दूरी पर स्थित है, के निर्देशांक होंगे | (Q) $(-7, 11)$ |
| (C) रेखाओं $x + y = 1$, $x - y + 3 = 0$, $2x + y = 7$ से बने त्रिभुज का लम्बकेन्द्र होगा | (R) $(1, -2)$ |
| (D) यदि त्रिभुज के दो शीर्ष $(5, -1)$ तथा $(-2, 3)$ है। यदि इसका लम्बकेन्द्र मूल बिन्दु पर स्थित है तो त्रिभुज का तीसरा शीर्ष होगा | (S) $(-1, 2)$ |

**Ans.** A – R, B – Q, C – S, D – P

**Sol.** (A) $a + c = 2b \Rightarrow a + c - 2b = 0$

$\Rightarrow ax + by + c = 0$ satisfies $(1, -2)$

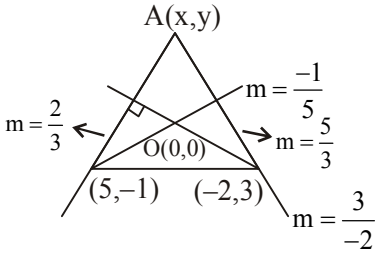
(B) Perpendicular distance of $P(\lambda, 4-\lambda)$ from $4x + 3y = 10$

$$\left|\frac{4\lambda + 3(4-\lambda) - 10}{5}\right| = 1$$

$$\Rightarrow |\lambda + 2| = 5 \Rightarrow \lambda - 2 \pm 5 = 3, -7$$

P(3,1), (–7,+11)

(C) Point B = (–1,2)




(D) $\frac{y}{x} = \frac{7}{4}$

$$\frac{4+1}{y-5} = \frac{2}{3}, \frac{21}{4}x = 2x - 13$$

$x = -4$, $y = -7$

55. Consider the lines given by

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

Match the statements / Expression in Column–I with the statements / Expressions in Column–II and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in OMR :

| **Column–I** | **Column–II** |
|---|---|
| (A) $L_1, L_2, L_3$ are concurrent, if | (P) $k = -9$ |
| (B) One of $L_1, L_2, L_3$ is parallel to at least one of the other two, if | (Q) $k = -\frac{6}{5}$ |
| (C) $L_1, L_2, L_3$ form a triangle, if | (R) $k = \frac{5}{6}$ |
| (D) $L_1, L_2, L_3$ do not form a triangle, if | (S) $k = 5$ |

माना रेखाओं के समीकरण

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

कॉलम–I से कॉलम–II का मिलान कीजिये तथा अपने उत्तर को 4 × 4 के आव्यूह के रूप में उत्तर पुस्तिका पर अंकित कीजिये :

| **Column–I** | **Column–II** |
|---|---|
| (A) स$_1$, $L_2$, $L_3$ संगामी है, यदि | (P) $k = -9$ |
| (B) स$_1$, $L_2$, $L_3$ मे से कोई एक शेष दो में से एक के समान्तर है, यदि | (Q) $k = -\frac{6}{5}$ |
| (C) स$_1$, $L_2$, $L_3$ एक त्रिभुज बनाती है, यदि | (R) $k = \frac{5}{6}$ |
| (D) स$_1$, $L_2$, $L_3$ एक त्रिभुज नही बनाती है, यदि | (S) $k = 5$ |

**Ans.** (A) S (B) P,Q (C) R (D) P,Q,S

**Sol.** (A) $L_1$, $L_2$ and $L_3$, intersect at (2,1).

$L_1$, $L_2$, $L_3$ are concurrent if $L_2$, passes through (2,1) that is if $k = 5$.

(B) As $L_1$ and $L_3$ intersect at (2,1) they are not parallel.

So $L_2$ is parallel to $L_1$ if $\frac{3}{k} = -3 \Rightarrow k = -9$

and $L_2$ is parallel to $L_3$ if $\frac{3}{k} = -\frac{5}{2} \Rightarrow k = -\frac{6}{5}$.

(C) The lines $L_1$, $L_2$, $L_3$ will form a triangle, fi no two of them are parallel and they are not concurrent. Thsu $k = \frac{5}{6}$.

(D) Lines $L_1$, $L_2$, $L_3$ will not form a triangle, if two of them are parallel or all of them are concurrent.

**Alter :**

$x + 3y - 5 = 0$, $3x - xy - 1 = 0$, $5x + 2y - 12 = 0$

(A) For concurrency

$$\begin{vmatrix} 1 & 3 & -5 \\ 3 & -k & -1 \\ 5 & 2 & -12 \end{vmatrix} = 0$$

$\Rightarrow$ $(12k + 2) - 3(-36 + 5) - 5(6 + 5k) = 0$

$\therefore$ $k = 5$

(B) For parallel

either $-\frac{1}{3} = \frac{3}{k}$ $\therefore k = -9$

or $-\frac{5}{2} = \frac{3}{k}$ $\therefore k = \frac{-6}{5}$

(C) They will form triangle

when $k \neq 5, -9, \frac{-6}{5}$

(D) They will not form triangle

when $k = 5, -9, \frac{-6}{5}$

56. Let ABC be a triangle such that the coordinates of A are (– 3, 1). Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column–I with their corresponding correct entries of column–II :

| **Column–I** | **Column–II** |
|---|---|
| (A) Equation of the line AB is | (P) $2x + y - 3 = 0$ |
| (B) Equation of the line BC is | (Q) $2x - 3y + 9 = 0$ |
| (C) Equation of CA is | (R) $4x + 7y + 5 = 0$ |
| | (S) $18x - y - 49 = 0$ |

यदि त्रिभुज ABC जिसके बिन्दु A के निर्देशांक (– 3, 1) है, बिन्दु B से गुजरने वाली माध्यिका का समीकरण $2x + y - 3 = 0$ एवं कोण C के कोर्णाधक का समीकरण $7x - 4y - 1 = 0$ है तो कॉलम–I एवं कॉलम–II का मिलान कीजिए :

| **Column–I** | **Column–II** |
|---|---|
| (A) रेखा AB का समीकरण | (P) $2x + y - 3 = 0$ |
| (B) रेखा BC का समीकरण | (Q) $2x - 3y + 9 = 0$ |
| (C) रेखा CA का समीकरण | (R) $4x + 7y + 5 = 0$ |
| | (S) $18x - y - 49 = 0$ |

**Ans.** (A) R; (B) S ; (C) Q

**Sol.** $D\left(\frac{\alpha-3}{2}, \frac{7\alpha+3}{8}\right)$ In $2x + y - 3 = 0$

$$\alpha - 3 + \frac{7\alpha+3}{8} - 3 = 0$$

$$\Rightarrow 8\alpha - 24 + 7\alpha + 3 - 24 = 0$$

$$\Rightarrow 15\alpha = 45$$

$$(\alpha - 3)$$

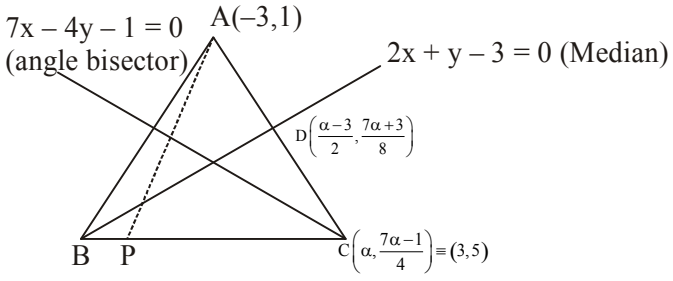
P is image of A(–3,1) in $7x - 4y - 1 = 0$

$$\frac{x+3}{7} = \frac{y-1}{-4} = -\frac{2(-21-4-1)}{49+16} = \frac{2\times 26}{65} = \frac{4}{5}$$

$$x = \frac{28}{5} - 3 = \frac{13}{5},\ y = -\frac{16}{5} + 1 = -\frac{11}{5}$$

$$P\left(\frac{13}{5}, -\frac{11}{5}\right)$$

$$\text{Equation of BC} = (y-1) = \frac{-\frac{11}{5} - 1}{\frac{13}{5} + 3}(x+3)$$




$$(y-1) - \frac{16}{28}(x+3)$$

$$\Rightarrow 7y - 4 = -4x - 12$$

$$\Rightarrow 4x + 7y + 8 = 0 \qquad \text{BC}$$

$$2x + y - 3 = 0 \qquad \text{Median}$$

$$B\left(\frac{13}{5}, -\frac{11}{5}\right)$$

(A) AB $\Rightarrow 4x + 7y + 5 = 0$

(B) BC $\Rightarrow 18x - y - 49 = 0$

(C) CA $\Rightarrow 2x - 3y + 9 = 0$

## Exercise J-M

1. Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is : **[JEE(Main)July2021]**

माना रेखाओं के एक युग्म $y = px$ तथा $y = qx$ का समीकरण $(y - px)(y - qx) = 0$, द्वारा लिखा जाता है, तो रेखाओं $x^2 - 4xy - 5y^2 = 0$ के बीच कोणों को समद्विभाजित करती हुई रेखाओं का समीकरण है : **[JEE(Main)July2021]**

(A) $x^2 - 3xy + y^2 = 0$ (B) $x^2 + 3xy - y^2 = 0$ (C) $x^2 + 4xy - y^2 = 0$ (D) $x^2 - 3xy - y^2 = 0$

Ans. B

Sol. $x^2 - 4xy - 5y^2 = 0$

$$\frac{x^2 - y^2}{1 - (-5)} = \frac{xy}{-2}$$

$$\frac{x^2 - y^2}{6} = \frac{xy}{-2}$$

$$-x^2 + y^2 = 3xy$$

$$x^2 - y^2 + 3xy = 0$$

2. The point P (a, b) undergoes the following three transformations successively : **[JEE(Main)July2021]**

(a) reflection about the line $y = x$.

(b) translation through 2 units along the positive direction of x-axis.

(c) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$, then the value of $2a + b$ is equal to :

बिन्दु P (a, b) में क्रम से निम्न तीन रूपांतरण होते हैं : **[JEE(Main)July2021]**

(a) रेखा $y = x$ के सापेक्ष परावर्तन

(b) x-अक्ष की धनात्मक दिशा के अनुदिश 2 इकाई स्थानांतरण

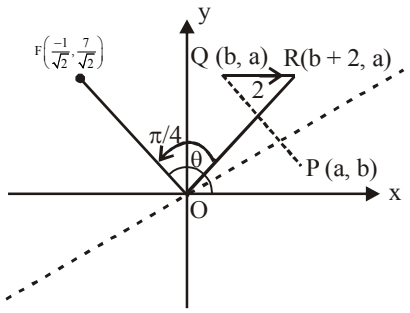
(c) मूलबिन्दु के सापेक्ष वामावृत दिशा में कोण $\frac{\pi}{4}$ का आवर्तन

; fn P की अंतिम स्थिति के निर्देशांक $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ हैं, तो $2a + b$ का मान बराबर है :

(A) 9 (B) 13 (C) 7 (D) 5

Ans. A

Sol.




$$OF = r = \sqrt{\frac{1}{2}+\frac{49}{2}} = 5 = OR$$

$$\tan\theta = -7$$

$$\tan\left(\theta-\frac{\pi}{4}\right)=\frac{4}{3}$$

$$\cos\left(\theta-\frac{\pi}{4}\right)=\frac{3}{5}, \qquad \sin\left(\theta-\frac{\pi}{4}\right)=\frac{4}{5}$$

$$R\left(b+2, a\right) = R\left(r\cos\left(\theta-\frac{\pi}{4}\right), r\sin\left(\theta-\frac{\pi}{4}\right)\right)$$

$$b+2=3 \Rightarrow b=1, \qquad a=4$$

$2a + b = 9$

3. Two sides of a parallelogram are along the lines $4x + 5y = 0$ and $7x + 2y = 0$. If the equation of one of the diagonals of the parallelogram is $11x + 7y = 9$, then other diagonal passes through the point :

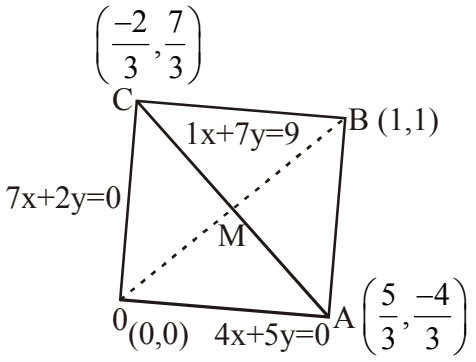
**[JEE(Main)July2021]**

एक समांतर चतुर्भुज की दो भुजाएँ रेखाओं $4x + 5y = 0$ तथा $7x + 2y = 0$ के अनुदिश हैं। इस समांतर चतुर्भुज के एक विकर्ण का l ehdj.k $11x + 7y = 9$ हैं, तो दूसरा विकर्ण निम्न में से किस बिन्दु से होकर जाता है? **[JEE(Main)July2021]**

(A) (2, 2) (B) (1, 3) (C) (1, 2) (D) (2, 1)

Ans. A

Sol.



Mid-point of AC $= \left(\frac{\frac{5}{3}-\frac{2}{3}}{2}, \frac{\frac{-4}{3}+\frac{7}{3}}{2}\right)$

$= \left(\frac{1}{2}, \frac{1}{2}\right)$

Equation of OB (other diagonal) is :

$y - 0 = \frac{1-0}{1-0}(x-0)$

$y = x$

$\because (2,2)$ lies on it

Ans: (2, 2)

4. Consider a triangle having vertices A(–2, 3), B(1, 9) and C(3, 8). If a line L passing through the circum-center of triangle ABC, bisects line BC, and intersects y-axis at point $\left(0, \frac{\alpha}{2}\right)$, then the value of real number $\alpha$ is _____.

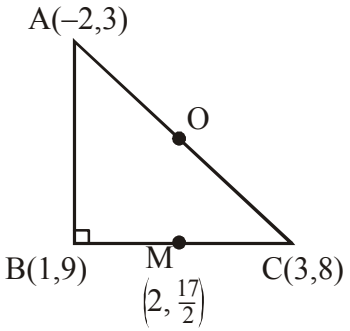
**[JEE(Main)July2021]**

एक त्रिभुज जिसके शीर्ष बिन्दु A(–2, 3), B(1, 9) तथा C(3, 8) हैं, पर विचार कीजिए। यदि एक रेखा L, जो त्रिभुज ABC के ι φδ ΫΖ.σ κδ φτ ιρη ΣβρκBC को समद्विभाजित करती है तथा y-अक्ष को बिन्दु $\left(0, \frac{\alpha}{2}\right)$ पर काटती है, तो वास्तविक संख्या $\alpha$ का मान है _____

**[JEE(Main)July2021]**

Ans. 9

Sol.




$AC = 5\sqrt{2}$

$AB = \sqrt{45}$

$BC = \sqrt{5}$

Circumcentre is $\left(\frac{1}{2}, \frac{11}{2}\right)$ $\quad M_{OM} = \frac{\frac{17}{2}-\frac{11}{2}}{2-\frac{1}{2}}$

equation of OM

$$y-\frac{17}{2}=2\,(x-2)$$

$2y-17=4x-8$

$4y-2y+9=0$

put $\left(0,\frac{\alpha}{2}\right)$

$-\alpha+9=0$

$\alpha=9$

5. Let ABC be a triangle with A (– 3, 1) and $\angle ACB=\theta, 0<\theta<\frac{\pi}{2}$. If the equation of the median through B is $2x+y-3=0$ and the equation of angle bisector of C is $7x-4y-1=0$, then $\tan\theta$ is equal to :

**[JEE(Main)Aug.2021]**

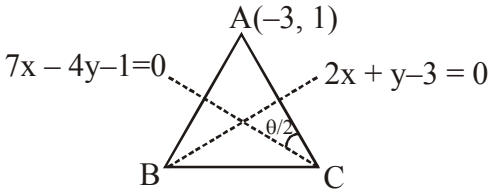
माना ABC एक त्रिभुज है जिसमें A (– 3, 1) तथा $\angle ACB=\theta, 0<\theta<\frac{\pi}{2}$ है। यदि B से माध्यिका रेखा का समीकरण $2x+y-3=0$ है तथा कोण C की समद्विभाजक रेखा का समीकरण $7x-4y-1=0$ है, तो $\tan\theta$ बराबर है :

**[JEE(Main)Aug.2021]**

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{4}{3}$ (D) 2

Ans. C

Sol.




$C\left(h,\frac{7h-1}{4}\right)$

Mid point of AC will lie on median.

$=$ Mid point $\left(\frac{h-3}{2},\frac{7h+3}{8}\right)$

$2\left(\frac{h-3}{2}\right)+\frac{7h+3}{8}-3=0$

$8h-24+7h+3-24=0$

$15h=45 \Rightarrow h=3$

$\Rightarrow C(3,5)$

$$m_{AC} = \frac{5-1}{3+3} = \frac{2}{3}$$

$$\tan\frac{\theta}{2} = \left|\frac{\frac{7}{4}-\frac{2}{3}}{1+\frac{7}{6}}\right| = \frac{1}{2}$$

$$\tan\theta = \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}} = \frac{2\left(\frac{1}{2}\right)}{1-\frac{1}{4}} = \frac{4}{3}$$

6. Let A be a fixed point (0, 6) and B be a moving point (2t, 0). Let M be the mid-point of AB and the perpendicular bisector of AB meets the y-axis at C. The locus of the mid-point P of MC is : **[JEE(Main)Aug.2021]**

माना A एक स्थिर बिन्दु (0, 6) है तथा B एक चर बिन्दु (2t, 0) है। माना AB का मध्य बिन्दु है तथा AB का लंबद्विभाजक y-अक्ष को बिन्दु C पर मिलता है तो MC के मध्य बिन्दु P का बिन्दुपथ है : **[JEE(Main)Aug.2021]**

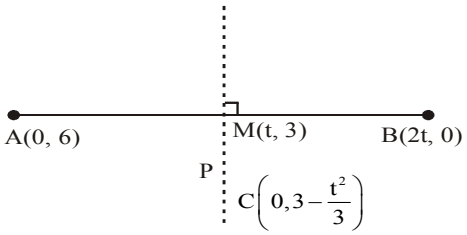
(A) $3x^2 - 2y - 6 = 0$ (B) $2x^2 - 3y + 9 = 0$ (C) $3x^2 + 2y - 6 = 0$ (D) $2x^2 + 3y - 9 = 0$

Ans. D

Sol. $m_{AB} = \frac{-3}{t}$

$$m_{MC} = \frac{t}{3}$$

Eq. of MC $\Rightarrow (y-3) = \frac{t}{3}(x-t)$




$$x = 0, y = 3 - \frac{t^2}{3} \quad \Rightarrow C\left(0, 3-\frac{t^2}{3}\right)$$

$$P = \left(\frac{t}{2}, 3-\frac{t^2}{6}\right) \equiv (h, k)$$

$2h = t \qquad \Rightarrow 4h^2 = t^2$

$$3 - \frac{t^2}{6} = k \Rightarrow 3 - \frac{4h^2}{6} = k$$

$\Rightarrow 2h^2 + 3k - 9 = 0$

$\Rightarrow 2x^2 + 3y - 9 = 0$

7. Let A(a, 0), B(b, 2b + 1) and c(0, b) $b \neq 0, |b| \neq 1$, be points such that the area of triangle ABC is 1 sq. unit, then the sum of all possible values of a is : **[JEE(Main)Aug.2021]**

माना बिन्दुओं A(a, 0), B(b, 2b + 1) तथा c(0, b) $b \neq 0$, $|b| \neq 1$, से बने त्रिभुज ABC का क्षेत्रफल 1 वर्ग इकाई है, तो a के सभी सम्भव मानों का योग है : **[JEE(Main)Aug.2021]**

(A) $\frac{-2b^2}{b+1}$ (B) $\frac{-2b}{b+1}$ (C) $\frac{2b^2}{b+1}$ (D) $\frac{2b}{b+1}$

Ans. A

Sol. $\frac{1}{2}\begin{vmatrix} a & 0 & 1 \\ b & 2b+1 & 1 \\ 0 & b & 1 \end{vmatrix} = \pm 1$

$$\begin{vmatrix} a & 0 & 1 \\ b & 2b+1 & 1 \\ 0 & b & 1 \end{vmatrix} = \pm 2$$

$$a(2b+1-b)+1(b^2) = \pm 2$$

$$a(b+1)+b^2 = \pm 2$$

Case - I Case - II

$$a = \frac{2-b^2}{b+1} \qquad a = \frac{-2-b^2}{b+1}$$

$$\text{Sum} = \frac{-2b^2}{b+1}$$

8. If p and q are the lengths of perpendiculars from the origin on the lines, x cosec $\alpha$ – y sec $\alpha$ = k cot $2\alpha$ and x sin $\alpha$ + y cos $\alpha$ = k sin $2\alpha$ respectively, then $k^2$ is equal to : **[JEE(Main)Aug.2021]**

यदि रेखाओं x cosec $\alpha$ – y sec $\alpha$ = k cot $2\alpha$ तथा x sin $\alpha$ + y cos $\alpha$ = k sin $2\alpha$ पर मूल बिन्दु से डाले गए लम्बों की लम्बाईयाँ Øe' lh/; तथा q है, तो $k^2$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $p^2 + 4q^2$ (B) $4p^2 + q^2$ (C) $p^2 + 2q^2$ (D) $2p^2 + q^2$

Ans. B

Sol. $L_1 : x\cos ec\alpha - y\sec\alpha = k\cot 2\alpha$

$L_2 : x\sin\alpha - y\cos\alpha = k\sin 2\alpha$

$$P = \left|\frac{0-0-k\cot 2\alpha}{\sqrt{\cos ec^2\alpha + \sec^2\alpha}}\right|$$

$$P^2 = \frac{k^2\cot^2 2\alpha}{\cos ec^2\alpha + \sec^2\alpha} = k^2\cot^2 2\alpha.\sin^2\alpha\cos^2\alpha$$

$$P^2 = \frac{k^2}{4}\cos^2 2\alpha$$

$$q = \left|\frac{0 - 0 - k\sin^2\alpha}{\sqrt{\sin^2\alpha + \cos^2\alpha}}\right|$$

$$q^2 = k^2 \sin^2 2\alpha$$

$$4p^2 + q^2 = k^2\cos^2 2\alpha + k^2 \sin^2 2\alpha$$

$$k^2 = 4p^2 + q^2$$

9. Let A be the set of all points (α, β) such that the area of triangle by the points (5, 6), (3, 2) and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A, is :

**[JEE(Main)Aug.2021]**

माना A उन सभी बिन्दुओं (α, β) जिनके लिए बिन्दुओं (5, 6), (3, 2) तथा (α, β) द्वारा बनाए गए त्रिभुज का क्षेत्रफल 12 वर्ग इकाई gS rks A esa fLFkr fdlh fcUnq ls ewy fcUnq dks feykus okys js[kk[k.M dh fuEure lEHko yEckbZ gS : **[JEE(Main)Aug.2021]**

(A) $\frac{8}{\sqrt{5}}$ (B) $\frac{4}{\sqrt{5}}$ (C) $\frac{12}{\sqrt{5}}$ (D) $\frac{16}{\sqrt{5}}$

Ans. A

Sol. $$\left|\frac{1}{2}\begin{vmatrix}\alpha & \beta & 1\\ 5 & 6 & 1\\ 3 & 2 & 1\end{vmatrix}\right| = 12$$

$$|\alpha(4) - \beta(2) + 1(-8)| = 24$$

$$|4\alpha - 2\beta - 8| = 24$$

$$|2\alpha - \beta - 4| = 12$$

$$2\alpha - \beta - 4 = \pm 12$$

$2\alpha - \beta - 4 = 12$ $\qquad$ $2\alpha - \beta - 4 = -12$

$2\alpha - \beta = 16$ $\qquad$ $2\alpha - \beta = -8$

$2x - y = 16$ $\qquad$ $2x - y = -8$

$d_1 = \left|\frac{-16}{\sqrt{5}}\right|$ $\qquad$ $d_2 = \left|\frac{8}{\sqrt{5}}\right|$

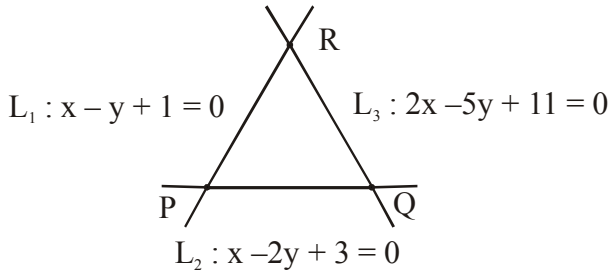
$d_1 = \frac{16}{\sqrt{5}}$ $\qquad$ $d_2 = \frac{8}{\sqrt{5}}$

10. Let the points of intersections of the lines $x - y + 1 = 0$, $x - 2y + 3 = 0$ and $2x - 5y + 11 = 0$ are the mid points of the sides of a triangle ABC. Then the area of the triangle ABC is_____. **[JEE(Main)Aug.2021]**

माना रेखाओं $x - y + 1 = 0$, $x - 2y + 3 = 0$ तथा $2x - 5y + 11 = 0$ के प्रतिच्छेदन बिन्दु एक त्रिभुज ABC की भुजाओं के मध्य fcUnqvksa ls cuus okys f=kHkqt ABC का क्षेत्रफल है _____ **[JEE(Main)Aug.2021]**

Ans. 6

Sol.




$P = (1, 2)$ , $Q = (7, 5)$ , $R = (2, 3)$

$\because$ Since P, Q, R are the mid–points of the sides of the $\Delta ABC$. Hence

Area of the $\Delta ABC = 4 \times$ (area of $\Delta PQR$)

$$= 4\left|\frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 7 & 5 & 1 \\ 2 & 3 & 1 \end{vmatrix}\right|$$

$$= 2\left|1(5-3) - 2(7-2) + 1(21-10)\right|$$

$$= 2\left|2 - 10 + 11\right|$$

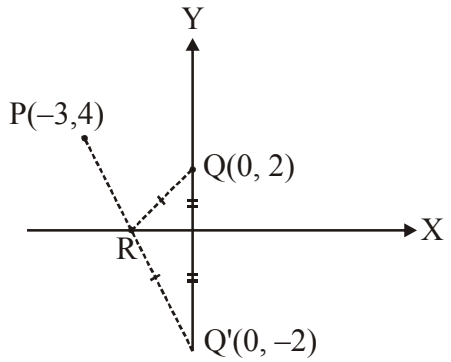
$$= 6.$$

11. A man starts walking from the point P(–3, 4), touches the x-axis at R, and then turns to reach at the point Q(0, 2). The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50((PR)^2 + (RQ)^2)$ is equal to_____ . **[JEE(Main)Aug.2021]**

एक व्यक्ति बिन्दु P(–3, 4) से चलना शुरू करता है, तथा x-अक्ष को R पर छूता है और तब मुड़कर बिन्दु Q(0, 2) पर पहुँच जाता है। ;fn ,d fLFkj pky l spy jgkgSA; fn ,d fLFkj U;wure le; esa Q बिन्दु पर पहुँचता है तब $50((PR)^2 + (RQ)^2)$ बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 1250

Sol.



Since man is walking at constant speed and the man reaches the point Q in the minimum time hence the distance (PR + RQ) is minimum.

$\because$ for (PR + RQ) min , P, R, Q' are collinear

where Q' is the image of Q.

Q' = (0, –2)

RQ = RQ'

Equation of PQ' $y-4=\frac{6}{-3}(x+3)$

$y-4=-2x-6$

$2x+y+2=0$

for R, $y=0, \quad x=-1$

$R=(-1,0)$

$PR^2=2^2+4^2=20$

$RQ^2=1^2+2^2=5$

$50[(PR)^2+(RQ)^2]=50[20+5]=1250$

12. Let the system of linear equations $x+y+\alpha z=2$, $3x+y+z=4$, $x+2z=1$ have a unique solution $(x*, y*, z*)$. If $(\alpha, x*)$, $(y*, \alpha)$ and $(x*, -y*)$ are collinear points, then the sum of absolute values of all possible values of $\alpha$ is : **[JEE(Main)June2022]**

माना रैखिक समीकरण निकाय $x+y+\alpha z=2$, $3x+y+z=4$, $x+2z=1$ का केवल एक हल $(x*, y*, z*)$ है। यदि $(\alpha, x*)$, $(y*, \alpha)$ तथा $(x*, -y*)$ संरेख हैं, है तो $\alpha$ के सभी संभव मानों के निरपेक्ष मानों d k; kx Qy gS: **[JEE(Main)June2022]**

(A) 4 (B) 3 (C) 2 (D) 1

Ans. C

Sol. $\Delta=-(\alpha+3)$

$Dx=-(3+\alpha)$

$Dy=-(\alpha+3)$

$Dz=0$

for unique solutions $\alpha\neq-3$

Now

$x*=1$

$y*=1$

$z*=0$

Now points are collinear

$$\begin{vmatrix} 1 & \alpha & 1 \\ 1 & 1 & \alpha \\ 1 & 1 & -1 \end{vmatrix} = 0$$

$1(-1-\alpha) - \alpha(-1-\alpha) + 1(1-1) = 0$

$-1 + \alpha^2 = 0$

$\alpha^2 = 1$

$\alpha = \pm 1$

sum of absolute values $= 1 + 1 = 2$

13. Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the points (α, –α), (–α, α) and ($\alpha^2$, β) are collinear, then β is equal to : **[JEE(Main)June2022]**

माना शीर्ष A(1, α), B(α, 0) तथा C(0, α) के त्रिभुज का क्षेत्रफल 4 वर्ग इकाई है। यदि बिन्दु (α, –α), (–α, α) तथा ($\alpha^2$, β) संरेख हैं, तो β बराबर है : **[JEE(Main)June2022]**

(A) 64 (B) –8 (C) –64 (D) 512

Ans. C

Sol. Area of triangle $\left|\frac{1}{2}\begin{vmatrix} 1 & 1 & \alpha \\ 1 & \alpha & 0 \\ 1 & 0 & \alpha \end{vmatrix}\right| = 4$

$\alpha = \pm 8$

Now given points are collinear then

$$\begin{vmatrix} \alpha & -\alpha & 1 \\ -\alpha & \alpha & 1 \\ \alpha^2 & \beta & 1 \end{vmatrix} = 0$$

If put $\alpha = +8$ or $\alpha = -8$ in above determinant we get

$\beta = -64$

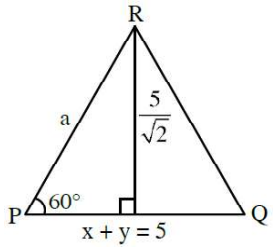
14. Let R be the point (3, 7) and let P and Q be two points on the line $x + y = 5$ such that PQR is an equilateral triangle. Then the area of ΔPQR is : **[JEE(Main)June2022]**

माना बिन्दु (3, 7) है तथा माना रेखा $x + y = 5$ पर दो बिन्दु P तथा Q इस प्रकार हैं कि PQR एक समबाहु त्रिभुज है। ΔPQR एक l eckgqf=Hkq gS **[JEE(Main)June2022]**

(A) $\frac{25}{4\sqrt{3}}$ (B) $\frac{25\sqrt{3}}{2}$ (C) $\frac{25}{\sqrt{3}}$ (D) $\frac{25}{2\sqrt{3}}$

Ans. D

Sol.




$$\sin 60^o = \frac{\frac{5}{\sqrt{2}}}{a}$$

$$a = \frac{5\sqrt{2}}{3}$$

$$\text{Area of } \Delta PQR = \frac{\sqrt{3}}{4}a^2 = \frac{25}{2\sqrt{3}}$$

15. In an isosceles triangle ABC, the vertex A is (6, 1) and the equation of the base BC is $2x + y = 4$. Let the point B lie on the line $x + 3y = 7$. If $(\alpha,\beta)$ is the centroid of $\Delta ABC$, then $15(\alpha + \beta)$ is equal to :

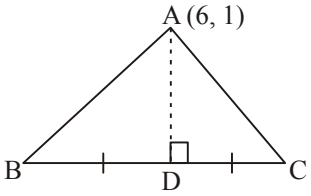
**[JEE(Main)June2022]**

एक समद्विबाहु त्रिभुज ABC का शीर्ष A बिन्दु (6, 1) पर है तथा आधार BC का समीकरण $2x + y = 4$ है। माना बिन्दु B, रेख $x + 3y = 7$ पर स्थित है। यदि $\Delta ABC$ का केन्द्रक $(\alpha, \beta)$ है, तो $15(\alpha + \beta)$ बराबर है : **[JEE(Main)June2022]**

(A) 39 (B) 41 (C) 51 (D) 63

Ans. C

Sol. Equation of BC $\Rightarrow 2x + y = 4$ ———(1)




B lie on line $x + 3y = 7$ ———(2)

from (1) & (2) B(1, 2)

Foot of perpendicular A on line BC

$$\frac{x-6}{2} = \frac{y-1}{1} = -\left(\frac{12+1-4}{5}\right)$$

$$\frac{x-6}{2} = \frac{y-1}{1} = \frac{-9}{5}$$

$$x = \frac{12}{5} \qquad y = \frac{-4}{5}$$

B(1, 2) $D\left(\frac{12}{5}, \frac{-4}{5}\right)$ C(x, y)

$$\frac{x+1}{2}=\frac{12}{5} \qquad \frac{y+2}{2}=\frac{-4}{5}$$

$$x=\frac{19}{5} \qquad y=\frac{-18}{5}$$

Centroid $\qquad \alpha=\dfrac{6+1+\frac{19}{5}}{3}=\dfrac{54}{15}$

$$\beta=\frac{1+2-\frac{18}{5}}{3}=\frac{-3}{15}$$

$$15|\alpha+\beta|=51$$

16. Let a triangle be bounded by the lines $L_1 : 2x + 5y = 10$; $L_2 : -4x + 3y = 12$ and the line $L_3$, which passes through the point P(2, 3), intersects $L_2$ at A and $L_1$ at B. If the point P divides the line-segmetn AB, internally in the ratio 1 : 3, then the area of the triangle is equal to : **[JEE(Main)June2022]**

माना एक त्रिभुज रेखाओं $L_1 : 2x + 5y = 10$; $L_2 : -4x + 3y = 12$ तथा $L_3$ से घिरा है। रेखा $L_3$ बिन्दु P(2, 3) से होकर जाती है, $L_2$ को बिन्दु A पर काटती है तथा $L_1$ को बिन्दु B पर काटती है। यदि बिन्दु P, रेखा–खण्ड AB को अंतः 1 : 3 के अनुपात में fo Hkkft r djrk gS rks f=Hkqt dk {ks=Qy cjkcj gS: **[JEE(Main)June2022]**

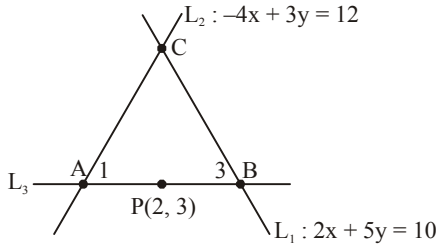
(A) $\frac{110}{13}$ (B) $\frac{132}{13}$ (C) $\frac{142}{13}$ (D) $\frac{151}{13}$

Ans. B

Sol. $L_1 : 2x + 5y = 10$ _____(1)

$L_2 : -4x + 3y = 12$ _____(2)

Solving eq. (1) and eq. (2) we get




$$C=\left(\frac{-15}{13},\frac{32}{13}\right)$$

Now

let $B\left(x_1,\frac{1}{5}(10-2x_1)\right);A=\left(x_2,\frac{1}{3}12+4x_2\right)$

from section formula $2=\dfrac{x_1+3x_2}{4}$

$$\Rightarrow x_1 + 3x_2 = 8 \qquad \text{_____(3)}$$

and

$$3 = \frac{12 + 4x_1 + \frac{10 - 2x_1}{5}}{4}$$

$$\Rightarrow 10x_2 - x_1 = -5 \qquad \text{_____(4)}$$

from eq. (3) and (4) $x_1 = \frac{95}{13}, x_2 = \frac{3}{13}$

Point $A\left(\frac{3}{13}, \frac{56}{13}\right)$, $B\left(\frac{95}{13}, \frac{-12}{13}\right)$

Now area of $\Delta ABC$

$$\text{Point area of } \Delta ABC = \left|\frac{1}{2}\left\{\frac{3}{13} \times \left(\frac{-44}{13}\right) + \frac{95}{13} \times \frac{-24}{13} + \frac{-15}{13} \times \frac{68}{13}\right\}\right|$$

$$= \frac{132}{13} \text{ sq. units}$$

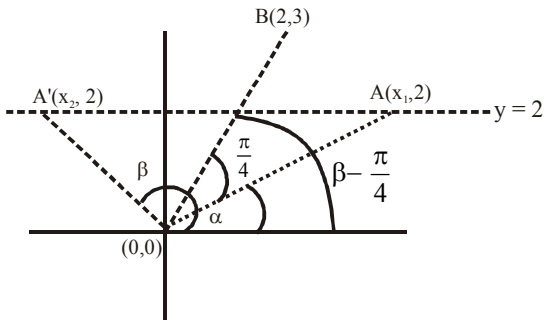
17. The distance between the two points A and A′ which lie on $y = 2$ such that both the line segments AB and A′B (where B is the point (2, 3)) subtent angle $\frac{\pi}{4}$ at the origin, is equal to : **[JEE(Main)June2022]**

माना रेखा $y = 2$ पर दो बिन्दु A तथा A′ इस प्रकार हैं कि दोनों रेखा खंड AB तथा A′B (जहाँ B, बिन्दु (2, 3) है), मूल बिन्दु पर $\frac{\pi}{4}$ का कोण बनाते हैं। तो बिन्दुओं A तथा A′ के बीच की दूरी है : **[JEE(Main)June2022]**

(A) 10 (B) $\frac{48}{5}$ (C) $\frac{52}{5}$ (D) 3

Ans. C

Sol.




$$\tan\left(\frac{\pi}{4} + \alpha\right) = \frac{3}{2}, \tan\alpha = \frac{2}{x_1}$$

$$\frac{1+\frac{2}{x_1}}{1-\frac{2}{x_1}}=\frac{3}{2}\Rightarrow x_1=10$$

And

$$\begin{cases}\tan\left(\beta-\frac{\pi}{4}\right)=\frac{3}{2}\\ \tan\beta=\frac{2}{x_2}\end{cases}\quad\rightarrow\text{solving}$$

$$\Rightarrow\ x_2=-\frac{2}{5}$$

$$AA^1=|x_2-x_1|\ =\frac{52}{5}$$

18. Let $A\left(\frac{3}{\sqrt{a}},\sqrt{a}\right)$, $a>0$, be a fixed point in the xy-plane. The image of A in y-axis be B and the image of B in x-axis be C. If $D(3\cos\theta,a\sin\theta)$ is a point in the fourth quadrant such that the maximum area of $\Delta ACD$ is 12 square units, then a is equal to _____. **[JEE(Main)June2022]**

माना xy-समतल में एक निश्चित बिन्दु $A\left(\frac{3}{\sqrt{a}},\sqrt{a}\right)$, $a>0$ है। y-अक्ष में A का प्रतिबिंब B तथा x-axis में B का प्रतिबिंब C है। यदि चतुर्थ चतुर्थांश में एक बिंदु $D(3\cos\theta,a\sin\theta)$ के लिए $\Delta ACD$ का अधिकतम क्षेत्रफल 12 वर्ग इकाई है, तो a बराबर है _____ **[JEE(Main)June2022]**

Ans. 8

Sol. Clearly B is $\left(-\frac{3}{\sqrt{a}},+\sqrt{a}\right)$ and C is $\left(-\frac{3}{\sqrt{a}},-\sqrt{a}\right)$

$$\text{Area of }\Delta ACD=\frac{1}{2}\begin{vmatrix}\frac{3}{\sqrt{a}} & \sqrt{a} & 1\\ -\frac{3}{\sqrt{a}} & -\sqrt{a} & 1\\ 3\cos\theta & a\sin\theta & 1\end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1 \\ -\frac{3}{\sqrt{a}} & -\sqrt{a} & 1 \\ 3\cos\theta & a\sin\theta & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \left|-3\sqrt{a}\sin\theta + 3\sqrt{a}\cos\theta\right| = 3\sqrt{a}\left|\cos\theta - \sin\theta\right|$$

$$\Rightarrow \Delta_{max} = 3\sqrt{a}.\sqrt{2} = 12 \Rightarrow a = \left(2\sqrt{2}\right)^2 = 8$$

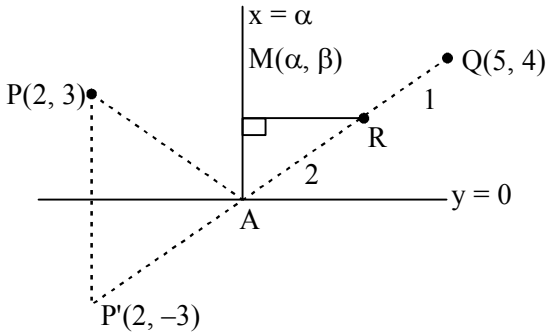
19. A ray of light passing through the point P(2, 3) reflects on the x-axis at point A and the reflected ray passes through the point Q(5, 4). Let R be the point that divides the line segment AQ internally into the ratio 2 : 1. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be$(\alpha, \beta)$. Then, the value of $7\alpha + 3\beta$ is equal to _____. **[JEE(Main)June2022]**

बिन्दु P(2, 3) से होकर जाने वाली रोशनी की एक किरण x-अक्ष पर बिन्दु A से परावर्तित होती है तथा परावर्तित किरण बिन्दु Q(5, 4) से होकर जाती है। माना बिन्दु R रेखाखण्ड AQ को 2 : 1 के अनुपात में अंतः विभाजित करता है। माना कोण PAQ के l ef} Hkt d ij l sR लंब के पाद M के निर्देषांक $(\alpha, \beta)$ हैं। तो $7\alpha + 3\beta$ का मान बराबर है _____.

**[JEE(Main)June2022]**

Ans. 31

Sol.




By observation we see that $A(\alpha, 0)$.

And $\beta$ = y-cordinate of R

$$= \frac{2\times 4 + 1\times 0}{2+1} = \frac{8}{3} \qquad \text{_____(1)}$$

Now P′ is image of P in y = 0 which will be P'(2,-3)

$\therefore$ Equation of P′Q is $\left(y+3\right) = \frac{4+3}{5-2}\left(x-2\right)$

i.e. $3y + 9 = 7x - 14$

$A \equiv \left(\frac{23}{7}, 0\right)$ by solving with $y = 0$

$\therefore\ \alpha = \frac{23}{7}$ _____(2)

By (1), (2)

$7\alpha + 3\beta = 23+8 = 31$

20. A line, with the slope greater than one, passes through the point A(4, 3) and intersects the line $x - y - 2 = 0$ at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line : **[JEE(Main)July2022]**

एक रेखा, जिसकी प्रवणता एक से अधिक है, बिन्दु A(4, 3) से होकर जाती है तथा रेखा $x - y - 2 = 0$ को बिन्दु B पर काटती g8A; fn j§kf[kd ka MAB की लंबाई $\frac{\sqrt{29}}{3}$ है, तो बिन्दु निम्न में से किस रेखा पर भी स्थित है : **[JEE(Main)July2022]**

(A) $2x + y = 9$ (B) $3x - 2y = 7$ (C) $x + 2y = 6$ (D) $2x - 3y = 3$

Ans. C

Sol. Let $B(x_1, x_1 - 2)$

$$\sqrt{(x_1 - 4)^2 + (x_1 - 2 - 3)^2} = \frac{\sqrt{29}}{3}$$

Squaring on both side

$18x_1^2 - 162x_1 + 340 = 0$

$x_1 = \frac{51}{9}$ or $x_1 = \frac{10}{3}$

$y_1 = \frac{33}{9}$ or $y_1 = \frac{4}{3}$

Option (C) will satisfy $\left(\frac{10}{3}, \frac{4}{3}\right)$

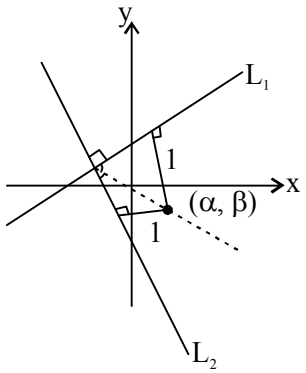
21. Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below $L_1$ and above $L_2$, then $100(\alpha + \beta)$ is equal to : **[JEE(Main)July2022]**

माना बिन्दु $P(\alpha, \beta)$ दो रेखाओं $L_1 : 3x - 4y + 12 = 0$, तथा $L_2 : 8x + 6y + 11 = 0$ में से प्रत्येक से इकाई दूरी पर है। यदि P, $L_1$ के नीचे तथा $L_2$ के ऊपर स्थित है, तो $100(\alpha + \beta)$ बराबर है : **[JEE(Main)July2022]**

(A) –14 (B) 42 (C) –22 (D) 14

Ans. D

Sol.




$L_1$: $3x - 4y + 12 = 0$

$L_2$: $8x + 6y + 11 = 0$

Equation of angle bisector of $L_1$ and $L_2$ of angle containing origin

$2(3x - 4y + 12) = 8x + 6y + 11$

$2x + 14y - 13 = 0$ _____(i)

$$\frac{3\alpha - 4\beta + 12}{5} = 1$$

$\Rightarrow 3\alpha - 4\beta + 7 = 0$ _____(ii)

Solution of $2x + 14y - 13 = 0$ and $3x - 4y + 7 = 0$ gives the required point $P(\alpha, \beta), \alpha = \frac{-23}{25}, \beta = \frac{53}{50}$

$100(\alpha + \beta) = 14$

22. A point P moves so that the sum of squares of its distances from the points (1, 2) and (–2, 1) is 14. Let f(x, y) = 0 be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the points C, D. Then the area of the quadrilateral ACBD is equal to : **[JEE(Main)July2022]**

एक बिन्दु P इस प्रकार चलायमान है कि इसकी बिन्दुओं (1, 2) तथा (–2, 1) से दूरियों के वर्गों का योग 14 है। माना P का बिन्दुपथ f(x, y) = 0 है, जो x-अक्ष को बिन्दुओं A, B पर तथा y-अक्ष को बिन्दुओं C, D पर काटता है। तो चतुर्भुज ACBD का क्षेत्रफल बराबर gS **[JEE(Main)July2022]**

(A) $\frac{9}{2}$ (B) $\frac{3\sqrt{17}}{2}$ (C) $\frac{3\sqrt{17}}{4}$ (D) 9

Ans. B

Sol. $(x-1)^2 + (y-2)^2 + (x+2)^2 + (y-1)^2 = 14$

$\Rightarrow x^2 + y^2 + x - 3y - 2 = 0$

Put $x = 0$

$\Rightarrow y^2 - 3y - 2 = 0$

$\Rightarrow y = \dfrac{3 \pm \sqrt{17}}{2}$

Put $y = 0$

$\Rightarrow x^2 + x - 2 = 0$

$(x + 2)(x - 1) = 0$

$\therefore A(-2, 0), B(1, 0), C\left(0, \dfrac{3+\sqrt{17}}{2}\right), D\left(0, \dfrac{3-\sqrt{17}}{2}\right)$

$\text{Area} = \dfrac{1}{2} \cdot 3 \cdot \sqrt{17} = \dfrac{3\sqrt{17}}{2}$

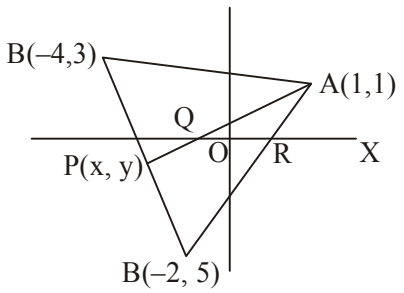
23. Let A(1, 1), B(–4, 3), C(–2, –5) be vertices of a triangle ABC, P be a point on side BC, and $\Delta_1$ and $\Delta_2$ be the areas of triangle APB and ABC, respectively. If $\Delta_1 : \Delta_2 = 4 : 7$ then the area enclosed by the lines AP, AC and the x-axis is : **[JEE(Main)July2022]**

माना एक त्रिभुज ABC के शीर्ष A(1, 1), B(–4, 3), C(–2, –5) हैं, भुजा BC पर एक बिन्दु P है तथा त्रिभुजों APB तथा ABC के {ksk Qy Øe' k% $\Delta_1$ तथा $\Delta_2$ हैं। यदि $\Delta_1 : \Delta_2 = 4 : 7$ हैं, तो रेखाओं AP, AC और x -अक्ष से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\dfrac{1}{4}$ (B) $\dfrac{4}{3}$ (C) $\dfrac{1}{2}$ (D) 1

Ans. C

Sol.




$\text{Given } \Delta_1 = \dfrac{1}{2}\begin{vmatrix} x & y & 1 \\ 1 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$

& $\Delta_2 = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ -4 & 3 & 1 \\ -2 & -5 & 1 \end{vmatrix}$

Given $\frac{\Delta_1}{\Delta_2} = \frac{4}{7} \Rightarrow \frac{-2x-5y+7}{36} = \frac{4}{7}$

$\Rightarrow 14x + 35y = -95$ ..(1)

Equation of BC is $4x + y = -13$ ..(2)

Solve equation (1) & (2)

Point $P\left(\frac{-20}{7}, \frac{-11}{7}\right)$

Here point $Q\left(\frac{-1}{2}, 0\right)$ & $R\left(\frac{1}{2}, 0\right)$

So Area of triangle $AQR = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

24. The equation of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 39$ and $x - y = 3$ respectively and P(2, 3) is its circumcentre. Then which of the following is NOT true? **[JEE(Main)July2022]**

(A) $(AC)^2 = 9p$ (B) $(AC)^2 + p^2 = 136$

(C) $32 < \text{area } (\Delta ABC) < 36$ (D) $34 < \text{area } (\Delta ABC) < 38$

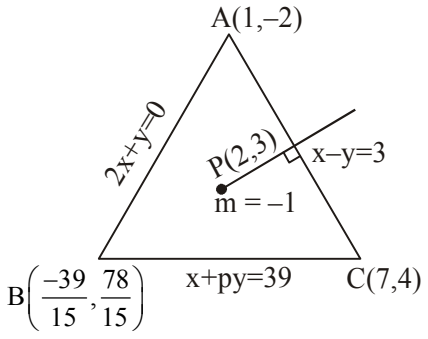
त्रिभुज ABC की भुजाओं AB, BC तथा CA के समीकरण क्रमशः $2x + y = 0$, $x + py = 39$ तथा $x - y = 3$ हैं तथा इसका परिकेन्द्र P(2, 3) है। तो निम्न में से कौनसा सत्य नहीं है? **[JEE(Main)July2022]**

(A) $(AC)^2 = 9p$ (B) $(AC)^2 + p^2 = 136$

(C) $32 <$ क्षेत्रफल $(\Delta ABC) < 36$ (D) $34 <$ क्षेत्रफल $(\Delta ABC) < 38$

Ans. D

Sol.



Perpendicular bisector of AB

$x + y = 5$

Take image of A

$$\frac{x-1}{1} = \frac{y+2}{1} = \frac{-2(-6)}{2} = 6$$

(7, 4)

$7 + 4ap = 39$

$p = 8$

solving $x + 8y = 39$ and $y = -2x$

$$x = \frac{-39}{15} \qquad y = \frac{78}{15}$$

$AC^2 = 72 = 9p$

$AC^2 + p^2 = 72 + 64 = 136$

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} 1 & -2 & 1 \\ 7 & 4 & 1 \\ \frac{-39}{15} & \frac{78}{15} & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[4 - \frac{78}{15} + 2\left(7 + \frac{39}{15}\right) + 7\left(\frac{78}{15}\right) + \frac{4 \times 39}{15}\right]$$

$$= \frac{1}{2}\left[18 + 18 \times \frac{13}{5}\right]$$

$$= 9\left[\frac{18}{5}\right] = \frac{162}{5} = 32.4$$

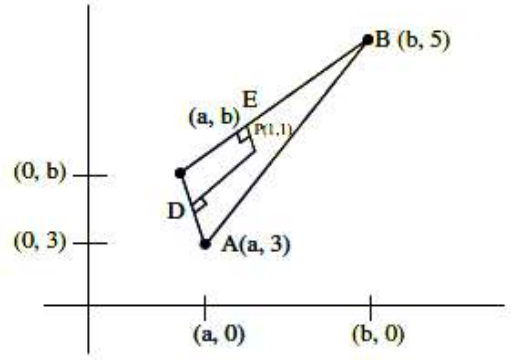
25. Let the circumcentre of a triangle with vertices A(a, 3), B(b, 5) and C(a, b), $ab > 0$ be P(1, 1). If the line AP intersects the line BC at the point $Q(k_1, k_2)$, then $k_1 + k_2$ is equal to : **[JEE(Main)July2022]**

माना एक त्रिभुज, जिसके शीर्ष A(a, 3), B(b, 5) तथा C(a, b), $ab > 0$ हैं, का परिकेन्द्र P(1, 1) है। यदि रेखा AP, रेखा BC को बिन्दु $Q(k_1, k_2)$ पर काटती है, तो $k_1 + k_2$ बराबर है : **[JEE(Main)July2022]**

(A) 2 (B) $\frac{4}{7}$ (C) $\frac{2}{7}$ (D) 4

Ans. B

Sol.




$m_{AC} \to \infty$

$m_{PD} = 0$

$D\left(\frac{a+a}{2}, \frac{b+3}{2}\right)$

$D\left(a, \frac{b+3}{2}\right)$

$m_{PD} = 0$

$\frac{b+3}{2} - 1 = 0$

$b + 3 - 2 = 0$

$b = -1$

$E\left(\frac{b+a}{2}, \frac{5+b}{2}\right) = \left(\frac{af}{2}, 2\right)$

$m_{CB} \cdot m_{EP} = -1$

$\left(\frac{5-b}{b-a}\right) = \left(\frac{2}{a-3}\right) = -1$

$\left(\frac{6}{-1-a}\right) = \left(\frac{2}{a-3}\right) = -1$

$12 = (1 + a)(a - 3)$

$12 = a^2 - 3a + a - 3$

$\Rightarrow a^2 - 2a - 15 = 0$

$(a - 5)(a + 3) = 0$

$a = 5$ or $a = -3$

Given $ab > 0$

$a(-1) > 0$

$-a > 0$

$a < 0$

a = –3 Accept

AP line A (–3, 3) P(1, 1)

$$y-1=\left(\frac{3-1}{-3-1}\right)(x-1)$$

$\Rightarrow x + 2y = 3$ Appling .....(1)

Line BC B(–1, 5)

C(–3, –1)

$$(y-5)=\frac{6}{2}(x+1)$$

$y – 5 = 3x + 3$

$y = 3x + 8$ .....(2)

Solving (1) & (2)

$x + 2 (3x + 8) = 3$

$\Rightarrow 7x + 16 = 3$

$7x = –13$

$$x=-\frac{13}{7}$$

$$y=3\left(-\frac{13}{7}\right)+8$$

$$=\frac{-39+56}{7}$$

$$y=\frac{17}{7}$$

$$x+y=\frac{-13+17}{7}=\frac{4}{7}$$

26. Let $m_1$, $m_2$ be the slopes of two adjacent sides of a square of side a such that $a^2+11a+3\left(m_1^2+m_2^2\right)=220$. If one vertex of the square is $(10(\cos\alpha-\sin\alpha), 10(\sin\alpha+\cos\alpha))$, where $\alpha\in\left(0,\frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha-\sin\alpha)x+(\sin\alpha+\cos\alpha)y=10$, then $72(\sin^4\alpha+\cos^4\alpha)+a^2-3a+13$ is equal to :

**[JEE(Main)July2022]**

माना भुजा a के एक वर्ग की संलग्न भुजाओं की प्रवणताऐं $m_1$, $m_2$ हैं तथा $a^2+11a+3\left(m_1^2+m_2^2\right)=220$ हैं। यदि वर्ग का एक शीर्ष $(10(\cos\alpha-\sin\alpha), 10(\sin\alpha+\cos\alpha))$, $\alpha\in\left(0,\frac{\pi}{2}\right)$ है तथा एक विकर्ण का समीकरण $(\cos\alpha-\sin\alpha)x+(\sin\alpha+\cos\alpha)y=10$ है, तो $72(\sin^4\alpha+\cos^4\alpha)+a^2-3a+13$ बराबर है :

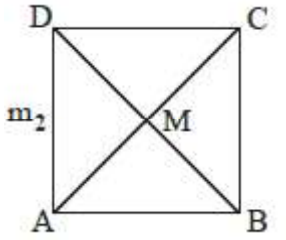
**[JEE(Main)July2022]**

(A) 119 (B) 128 (C) 145 (D) 155

Ans. B

Sol. $m_1 m_2 = -1$

$$a^2 + 11a + 3\left(m_1^2 + \frac{1}{m_1^2}\right) = 220$$




Eq. of AC

$$AC = (\cos\alpha - \sin\alpha) + (\sin\alpha + \cos\alpha)y = 10$$

$$BD = (\sin\alpha - \cos\alpha)x + (\sin\alpha - \cos\alpha)y = 0$$

$$(10(\cos\alpha - \sin\alpha), 10(\sin\alpha - \cos\alpha))$$

$$\text{Slope of AC} = \left(\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right) = \tan\theta = M$$

Eq. of line making an angle $\pi_4$ with AC

$$m_1, m_2 = \frac{m \pm \tan\frac{\pi}{4}}{1 \pm m\tan\frac{\pi}{4}}$$

$$= \frac{m+1}{1-m} \text{ or } \frac{m-1}{1+m}$$

$$\frac{\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} + 1}{1 - \left(\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right)}, \frac{\sin\alpha - \cos\alpha}{1 + \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}}$$

$$m_1, m_2 = \tan\alpha, \cot\alpha$$

mid point of AC & BD

$$= M(5(\cos\alpha - \sin\alpha), 5(\cos\alpha + \sin\alpha))$$

$$B(10(\cos\alpha - \sin\alpha), 10(\cos\alpha + \sin\alpha))$$

$$a = AB = \sqrt{2}BM = \sqrt{2}(5\sqrt{2}) = 10$$

$$a = 10$$

$\because a^2+11a+3\left(m_1^2+\frac{1}{m_1 2}\right)=220$

$100+110+3\left(\tan^2\alpha+\cot^2\alpha\right)=220$

Hence $\tan^2\alpha=3, \tan^2\alpha=\frac{1}{3}\Rightarrow\alpha=\frac{\pi}{3}$ or $\frac{\pi}{6}$

Now $72\left(\sin^4\alpha+\cos^4\alpha\right)+a^2-3a+13$

$=72\left(\frac{9}{16}+\frac{1}{16}\right)+100-30+13$

$=72\left(\frac{5}{8}\right)+83=45+83=128$

27. Let A(α, –2), B(α, 6) and $C\left(\frac{\alpha}{4},-2\right)$ be vertices of a ΔABC. If $\left(5,\frac{\alpha}{4}\right)$ is the circumcentre of ΔABC, then which of the following is NOT correct about ΔABC? **[JEE(Main)July2022]**

(A) area is 24 (B) perimeter is 25 (C) circumradius is 5 (D) inradius is 2

माना A(α, –2), B(α, 6) तथा $C\left(\frac{\alpha}{4},-2\right)$ एक त्रिभुज ABC के शीर्ष हैं। यदि ΔABC का परिकेन्द्र $\left(5,\frac{\alpha}{4}\right)$ है, तो इस त्रिभुज

d sfy, fuEu esa lsdkSl kl ghugha? **[JEE(Main)July2022]**

(A) क्षेत्रफल 24 है (B) परिमाप 25 है (C) परिवृत्त की त्रिज्या 5 है(D) अंतवृत्त की त्रिज्या 2

Ans. B

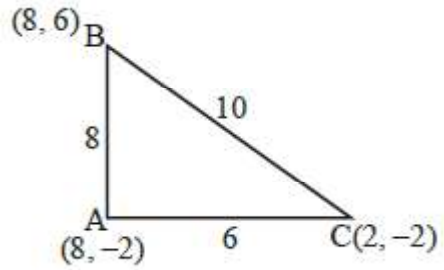
Sol. $A(\alpha,-2):B(\alpha,6):C\left(\frac{\alpha}{4},-2\right)$

since AC is perpendicular to AB.
So, ΔABC is right angled at A.

Circumcentre = mid point of BC. $=\left(\frac{5\alpha}{8},2\right)$

$\therefore \frac{5\alpha}{8}=5 \& \frac{\alpha}{4}=2$

$\alpha = 8$




Area $= \frac{1}{2}(6)(8) = 24$

Perimeter = 24

Circumradius = 5

Inradius $= \frac{\Delta}{s} = \frac{24}{12} = 2$

28. The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is (2, a), $-\frac{1}{2} < a < 2$, then p is equal to _____.

**[JEE(Main)July2022]**

एक त्रिभुज ABC की भुजाओं AB, BC तथा CA के समीकरण क्रमशः $2x + y = 0$, $x + py = 15a$ तथा $x - y = 3$ हैं। इसका लंब केन्द्र (2, a), $-\frac{1}{2} < a < 2$ है, तो p बराबर है_____ **[JEE(Main)July2022]**

Ans. 3

Sol. Coordinates of $A(1,-2), B\left(\frac{15a}{1-2p}, \frac{-30a}{1-2p}\right)$ and orthocentre H(2, a)

Slope of AH = p

$a + 2 = p$ _____(1)

Slope of BH = –1

$31a - 2ab = 15a + 4p - 2$ _____(2)

From (1) and (2)

$a = 1$ & $p = 3$

## Exercise J - A

1. For $a > b > c > 0$, the distance between (1, 1) and the point pf intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then : **[JEE(Adv.)2013]**

$a > b > c > 0$ के लिए, (1, 1) तथा रेखाओं $ax + by + c = 0$ व $bx + ay + c = 0$ के प्रतिच्छेद बिन्दु के बीच की दूरी $2\sqrt{2}$ से कम g§r c : **[JEE(Adv.)2013]**

(A) $a + b - c > 0$ (B) $a - b + c < 0$ (C) $a - b + c > 0$ (D) $a + b - c < 0$

**Ans.** AC

**Sol.** Point of intersection of both lines is

$$\left(-\frac{c}{(a+b)}, -\frac{c}{(a+b)}\right)$$

Distance between $\left(-\frac{c}{(a+b)}, -\frac{c}{(a+b)}\right)$ & (1,1) is

$$\text{Distance} = \sqrt{\frac{(a+b+c)^2}{(a+b)^2} \times 2} < 2\sqrt{2}$$

$a + b + c < 2(a + b)$

$a + b - c > 0$

According to given condition option (C) also correct.

2. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is _____. **[JEE(Adv.)2014]**

समतल में स्थित किसी बिन्दु P से रेखाओं $x - y = 0$ तथा $x + y = 0$ की दूरी क्रमशः $d_1(P)$ तथा $d_2(P)$ है। यदि क्षेत्र R उन सभी बिन्दुओं P से बना है जो प्रथम चतुर्थांश (quadrant) में स्थित हैं तथा $2 \le d_1(P) + d_2(P) \le 4$ को सन्तुष्ट करते हैं, तब क्षेत्र R का क्षेत्रफल है _____ **[JEE(Adv.)2014]**

**Ans.** 6

**Sol.** let p(h, k)

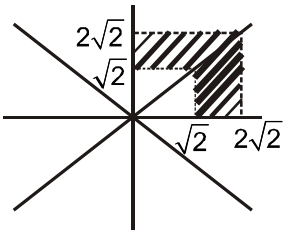
$$2 \le \left|\frac{h-k}{\sqrt{2}}\right| + \left|\frac{h+k}{\sqrt{2}}\right| \le 4$$

$\Rightarrow \quad 2\sqrt{2} \le |h - k| + |h + k| \le 4\sqrt{2}$

If $h \ge k$

$\Rightarrow \quad 2\sqrt{2} \le x - y + x + y \le 4\sqrt{2}$ or $\sqrt{2} \le x \le 2\sqrt{2}$

similarly when $k > h$



we have $\sqrt{2} \le y \le 2\sqrt{2}$

The required area = $(2\sqrt{2})^2 - (\sqrt{2})^2 = 6$.

3. Consider the lines $L_1$ and $L_2$

defined by $L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ let C be the locus of a point P such that the product of the distance of P from $L_1$ and the distance of P from $L_2$ is $\lambda^2$. The line $y = 2x + 1$ meets C at two points R and S. where the distance between R and S is $\sqrt{270}$. Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. The value of $\lambda^2$ is_____. **[JEE(Adv.)2021]**

माना रेखाऐं $L_1$ तथा $L_2$

$L_1 : x\sqrt{2} + y - 1 = 0$ व $L_2 : x\sqrt{2} - y + 1 = 0$ द्वारा परिभाषित हैं। एक निश्चित अचर λ के लिए, माना C एक बिन्दु P का बिन्दुपथ है जबकि P की $L_1$ से दूरी तथा P की $L_2$ से दूरी का गुणनफल $\lambda^2$ है। रेखा $y = 2x + 1$, C से दो बिन्दुओं R व S पर मिलती है, जहाँ R व S के मध्य दूरी $\sqrt{270}$ है। माना RS का लम्ब समद्विभाजक C से दो भिन्न बिन्दुओं R' व S' पर मिलता है। माना R' and S' के मध्य दूरी का वर्ग D है। $\lambda^2$ मा मान _____है। **[JEE(Adv.)2021]**

**Ans.** 9

**Sol.** Locus C,

$$\left|\frac{x\sqrt{2} + y - 1}{\sqrt{3}}\right|\left|\frac{x\sqrt{2} - y + 1}{\sqrt{3}}\right| = \lambda^2$$

$\Rightarrow |2x^2 - (y-1)^2| = 3\lambda^2$

C cuts $y - 1 = 2x$ at $R(x_1, y_1)$ and $S(x_2, y_2)$

So, $|2x^2 - (2x)^2| = 3\lambda^2$

$$\Rightarrow x = \pm\lambda\sqrt{\frac{3}{2}}$$

$$\therefore R\left(\lambda\sqrt{\frac{3}{2}}, 1 + \lambda\sqrt{6}\right) \text{ and } S\left(-\lambda\sqrt{\frac{3}{2}}, 1 - \lambda\sqrt{6}\right)$$

Distance between R and S is $\sqrt{270}$

$$\Rightarrow \sqrt{\left(\lambda\sqrt{6}\right)^2 + \left(2\lambda\sqrt{6}\right)^2} = \sqrt{270}$$

$$\therefore \lambda^2 = 9$$

4. Consider the lines $L_1$ and $L_2$ defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant $\lambda$ let C be the locus of a point P such that the product of the distance of P from $L_1$ and the distance of P from $L_2$ is $\lambda^2$. The line $y = 2x + 1$ meets C at two points R and S. where the distance between R and S is $\sqrt{270}$. Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. The value of D is_____. **[JEE(Adv.)2021]**

माना रेखाऐं $L_1$ तथा $L_2$

$L_1 : x\sqrt{2} + y - 1 = 0$ व $L_2 : x\sqrt{2} - y + 1 = 0$ द्वारा परिभाषित हैं।

एक निश्चित अचर $\lambda$ के लिए, माना C एक बिन्दु P का बिन्दुपथ है जबकि P की $L_1$ से दूरी तथा P की $L_2$ से दूरी का गुणनफल $\lambda^2$ है। रेखा $y = 2x + 1$, C से दो बिन्दुओं R व S पर मिलती है, जहाँ R व S के मध्य दूरी $\sqrt{270}$ है। माना RS का लम्ब समद्विभाजक C से दो भिन्न बिन्दुओं R' व S' पर मिलता है। माना R' तथा S' के मध्य दूरी का वर्ग D है। D का मान ____है। **[JEE(Adv.)2021]**

**Ans.** $\frac{540}{7}$

**Sol.** Locus C,

$$\left|\frac{x\sqrt{2}+y-1}{\sqrt{3}}\right|\left|\frac{x\sqrt{2}-y+1}{\sqrt{3}}\right| = \lambda^2$$

$\Rightarrow |2x^2 - (y-1)^2| = 3\lambda^2$

C cuts $y - 1 = 2x$ at $R(x_1, y_1)$ and $S(x_2, y_2)$

So, $|2x^2 - (2x)^2| = 3\lambda^2$

$\Rightarrow x = \pm\lambda\sqrt{\frac{3}{2}}$

$\therefore R\left(\lambda\sqrt{\frac{3}{2}}, 1+\lambda\sqrt{6}\right)$ and $S\left(-\lambda\sqrt{\frac{3}{2}}, 1-\lambda\sqrt{6}\right)$

Distance between R and S is $\sqrt{270}$

$$\Rightarrow \sqrt{\left(\lambda\sqrt{6}\right)^2 + \left(2\lambda\sqrt{6}\right)^2} = \sqrt{270}$$

$\therefore \lambda^2 = 9$

Now, the curve C is

$$\Rightarrow \left|2x^2-(y-1)^2\right|=27 \qquad ...(1)$$

Mid-point of RS is (0,1) and perpendicular bisector is

$$y=-\frac{1}{2}x+1 \qquad ...(2)$$

From (1) and (2), we have

$$\frac{7x^2}{2}=27 \Rightarrow x=\pm 6\sqrt{\frac{3}{7}}$$

$$\therefore R\left(6\sqrt{\frac{3}{7}},1-3\sqrt{\frac{3}{7}}\right) \text{ and } S\left(-6\sqrt{\frac{3}{7}},1+3\sqrt{\frac{3}{7}}\right)$$

$$D=(R'S')^2=\left(12\sqrt{\frac{3}{7}}\right)^2+\left(6\sqrt{\frac{3}{7}}\right)^2$$

$$\Rightarrow D=\frac{3}{7}\times 180=\frac{540}{7}$$

$$\therefore D\approx 77.14$$

# CHALLENGER EXERCISES

## Exercise C - I

### SINGLE CORRECT TYPE QUESTIONS

1. The orthocentre of the triangle ABC is 'B' and the circumcentre is 'S' (a, b). If A is the origin then the co–ordinates of C are :

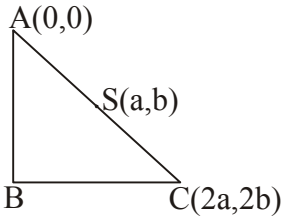
(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) none

त्रिभुज ABC का लम्बकेन्द्र 'B' है तथा परिकेन्द्र 'S' (a, b) है। यदि A मूल बिन्दु हो, तो शीर्ष C के निर्देशांक हैं :

(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**




B is right angle

2. A triangle ABC with vertices A $(-1, 0)$, B $(-2, 3/4)$ & C $(-3, -7/6)$ has its orthocentre H. Then the orthocentre of triangle BCH will be :

(A) $(-3, -2)$ (B) $(1, 3)$ (C) $(-1, 2)$ (D) none of these

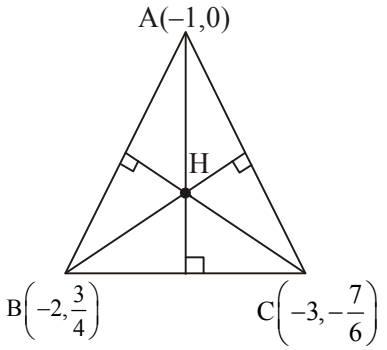
किसी त्रिभुज ABC के शीर्ष A $(-1, 0)$, B $(-2, 3/4)$ एवं C $(-3, -7/6)$ है तथा इसका लम्बकेन्द्र H है, तो त्रिभुज BCH का लम्बकेन्द्र gkxk:

(A) $(-3, -2)$ (B) $(1, 3)$ (C) $(-1, 2)$ (D) इनमें से कोई नहीं

**Ans.** D

**Sol.** **Alter :**

If H is orthocentre of triangle ABC, then orthocentre of triangle BCH is point A



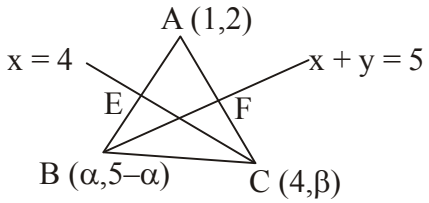
3. In a triangle ABC, co–ordinates of A are (1, 2) and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the co–ordinates of B and C will be :

त्रिभुज ABC में A के निर्देशांक (1, 2) है तथा B तथा C से गुजरने वाली माध्यिकाओं की समीकरण क्रमशः $x + y = 5$ एंव $x = 4$ है, तो B एवं C के निर्देशांक क्रमशः होगें :

(A) (–2, 7), (4, 3) (B) (7, –2), (4, 3) (C) (2, 7), (–4, 3) (D) (2, –7), (3, –4)

**Ans.** B

**Sol.**




$F \equiv F\left(\frac{5}{2}, \frac{\beta+2}{2}\right)$ $E \equiv \left(\frac{\alpha+1}{2}, \frac{5-\alpha+2}{2}\right)$

$\frac{5}{2} + \frac{\beta+2}{2} = 5$ $\frac{\alpha+1}{2} = 4$

$\beta + 7 = 10$ $\alpha = 7$

$\beta = 3$ $\Rightarrow E\left(\frac{5}{2}, \frac{5}{2}\right)$ & C (4,3) B(7, –2)

**Alter :**

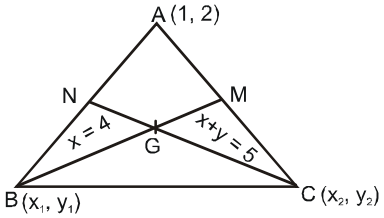
$x_1 + y_1 = 5$ _____(i)

$x_2 = 4$ _____(ii)

co - ordinates of G are ≡ (4, 1)

$\Rightarrow \quad \frac{1 + x_1 + x_2}{3} = 4$ _____(iii)

and $\frac{y_1 + y_2 + 2}{3} = 1$ _____(iv)



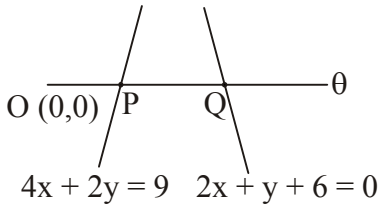

4. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio :

मूलबिन्दु O से जाने वाली एक सरल रेखा, दो समान्तर रेखाओं $4x + 2y = 9$ और $2x + y + 6 = 0$ से क्रमशः P और Q पर मिलती है, r ksfcUnq O, रेखाखण्ड PQ को किस अनुपात में विभाजित करेगा :

(A) 1 : 2 (B) 3 : 4 (C) 2 : 1 (D) 4 : 3

**Ans.** B

**Sol.**




Let $\frac{OP}{OQ} = \frac{\lambda}{1}$

$OP = \lambda k$

$OQ = k$

$P(\lambda k \cos\theta, \lambda k \sin\theta) \Rightarrow 4x + 2y = 9$

$Q(k\cos\theta, k\sin\theta) \Rightarrow 2x + y + 6 = 0$

$$2\lambda k = \frac{9}{2\cos\theta + \sin\theta}$$

$$k = \frac{-6}{2\cos\theta + \sin\theta}$$

$$2\lambda = -\frac{9}{6} = -\frac{3}{2}$$

$\lambda = -\frac{3}{4}$ O divides PQ in 3 : 4, internally

5. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is :

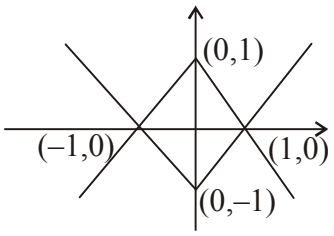
वक्रो $y = |x| - 1$ एवं $y = -|x| + 1$ से परिबद्ध क्षेत्रफल है :

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

**Ans.** B

**Sol.** $y = |x| - 1$ & $y = -|x| + 1$

$$y = |x| - 1 = \begin{cases} x - 1 & x \geq 0 \\ -x - 1 & x < 0 \end{cases}$$



$$\text{Area} = \frac{1}{2}d_1d_2 = \frac{1}{2}\times 2\times 2 = 2$$

6. The set of values of 'b' for which the origin and the point (1, 1) lie on the same side of the straight line, $a^2x + aby + 1 = 0 \ \forall\ a \in R, b > 0$ are :

मूल बिन्दु एवं बिन्दु (1,1) सरल रेखा $a^2x + aby + 1 = 0 \ \forall\ a \in R, b > 0$ के एक ही ओर स्थित होने के लिए b के मानों का समुच्चय gS

(A) $b \in (2, 4)$ (B) $b \in (0, 2)$ (C) $b \in [0, 2]$ (D) $(2, \infty)$

**Ans.** B

**Sol.** $a^2x + aby + 1 = 0$ origin and (1, 1) lies on same side. $a^2 + ab + 1 > 0 \ \forall\ a \in R$

$D < 0 \Rightarrow b^2 - 4 < 0 \Rightarrow b \in (-2, 2)$ but $b > 0 \Rightarrow b \in (0, 2)$

7. The point $(a^2, a + 1)$ is a point in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin if:

(A) $a \geq 1$ or $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$

(C) $a \in (0, 1)$ (D) none of these

बिन्दु $(a^2, a + 1)$ रेखाओं $3x - y + 1 = 0$ एवं $x + 2y - 5 = 0$ के मध्य उस कोण में स्थित होगा जिसमें मूल बिन्दु स्थित है यदि :

(A) $a \geq 1$ या $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$

(C) $a \in (0, 1)$ (D) इनमें से कोई नहीं

**Ans.** B

**Sol.** Origin & $(a^2, a + 1)$ lies same side w.r.t. to given lines

$a^2 + 2a + 2 - 5 < 0 \Rightarrow a^2 + 2a - 3 < 0$

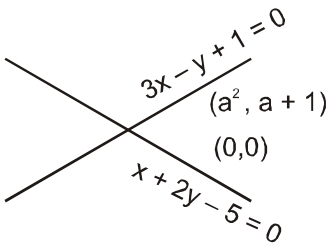
$\Rightarrow (a + 3)(a - 1) < 0$

$\Rightarrow a \in (-3, 1)$

$3a^2 - (a + 1) + 1 > 0 \Rightarrow 3a^2 - a > 0$

$\Rightarrow a(3a - 1) > 0$

$\Rightarrow a \in (\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$




take intersection we get $a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

8. The image of the point A (1, 2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point $(\alpha, \beta)$ then :

(A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) none of these

रेखा दर्पण $y = x$ में बिन्दु A (1, 2) का प्रतिबिम्ब B है तथा रेखा दर्पण $y = 0$ में B के प्रतिबिम्ब के निर्देशांक $(\alpha, \beta)$ है, तो :

(A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) इनमें से कोई नहीं

**Ans.** C

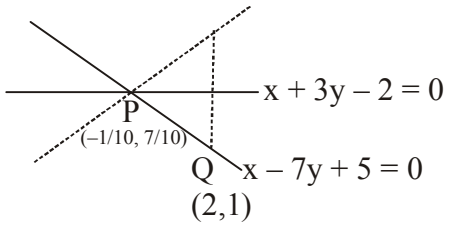
**Sol.** Image of A(1, 2) in line mirror $y = x$ is (2, 1); Image of B(2, 1) in $y = 0$ (x – axes) is 2, – 1)
Hence, $\alpha = 2, \beta = -1$

9. The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation $x - 7y + 5 = 0$. The equation of the other line is :
(A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) none

रेखा $x + 3y - 2 = 0$ एक सरल रेखा युग्म के मध्य कोण को समद्विभाजित करती हैं। यदि रेखा युग्म की एक समीकरण $x - 7y + 5 = 0$ हो, तो अन्य रेखा की समीकरण है :
(A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) इनमें से कोई नहीं

**Ans.** C

**Sol.**




$$P \equiv \left(\frac{-1}{10}, \frac{7}{10}\right)$$

$$Q' \to \frac{x-2}{1} = \frac{y-1}{3} = -2\left(\frac{2+3-2}{1+9}\right) = -\frac{3}{5} \qquad (Q' \equiv \text{image of } Q)$$

$$x = \frac{7}{5}, y = -\frac{4}{5}$$

Equation of line PQ′

$$\left(y - \frac{7}{10}\right) = \left(\frac{\frac{-4}{5} - \frac{7}{10}}{\frac{7}{5} + \frac{1}{10}}\right)\left(x + \frac{1}{10}\right)$$

$5x + 5y - 3 = 0$

10. On the portion of the straight line, $x + 2y = 4$ intercepted between the axes, a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has co–ordinates :
(A) (2, 3) (B) (3, 2) (C) (3, 3) (D) none

सरल रेखा $x + 2y = 4$ द्वारा अक्षों के मध्य काटे गये अन्तःखण्ड को एक भुजा लेकर एक वर्ग, मूल बिन्दु के विपरीत ओर, बनाया जाता है,
r ksbl d sfod . kksa ds chp esa fLFkr PNksa fcUnqvksa ds funsZ'kkad gksxs

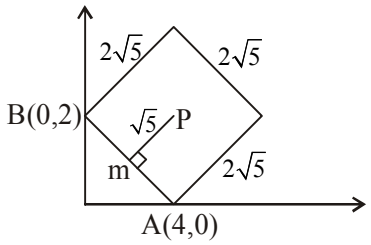
(A) (2, 3) (B) (3, 2) (C) (3, 3) (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $AB = \sqrt{16+4} = 2\sqrt{5}$

M(2,1)

Slope of PM = $\tan\theta = 2$




$\sin\theta = \frac{2}{\sqrt{5}}$

$\cos\theta = \frac{1}{\sqrt{5}}$

$r = \sqrt{5}$

$P\left(2 + r\cos\theta, 1 + r\sin\theta\right)$

= P(3,3)

11. AB is a variable line sliding between the coordinate axes is such a way that a lies on x–axis and B lies on y–axis. If P is a variable point on AB such that PA = b, PB = a and AB = a + b, then equation of locus of P is :

(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) none of these

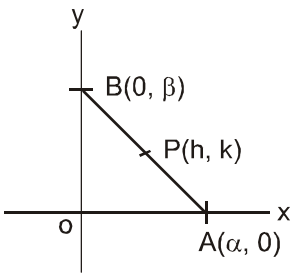
AB एक चर रेखा है जो निर्देशी अक्षों के बीच इस प्रकार फिसल रही है कि A, x–अक्ष पर स्थित है और B, y–अक्ष पर स्थित है। यदि AB पर एक चर बिन्दु P इस प्रकार है कि PA = b, PB = a और AB = a + b, तो P के बिन्दुपथ का समीकरण है :

(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) इनमें से कोई नही

**Ans.** A

**Sol.** M : 1 :-By geometry

$\alpha^2 + \beta^2 = (a + b)^2$ ....(i)




By section formula

$h = \frac{\alpha a}{a+b} \Rightarrow a = \frac{h(a+b)}{a}$

$k = \frac{\beta b}{a+b} \Rightarrow b = \frac{k(a+b)}{b}$

Put value of $\alpha$ and $\beta$ in (i)

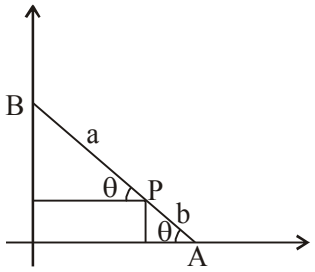
$$\frac{h^2(a+b)^2}{a^2}+\frac{k^2(a+b)^2}{b^2}=(a+b)^2 \quad \Rightarrow \quad \frac{h^2}{a^2}+\frac{k^2}{b^2}=1$$

Locus of P is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

**M : 2 :-** $P(h,k)=(a\cos\theta, b\sin\theta)$

$$\frac{h}{a}=\cos\theta$$

$$\frac{k}{b}=\sin\theta$$




$$\frac{h^2}{a^2}+\frac{k^2}{b^2}=1$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

12. Equations of the line pair through the origin and perpendicular to the line pair $xy-3y^2+y-2x+10=0$ is :

रेखा युग्म $xy-3y^2+y-2x+10=0$ के लम्बवत् एवं मूल बिन्दु से गुजरने वाले रेखा युग्म का समीकरण है :

(A) $xy-3y^2=0$ (B) $xy+3x^2=0$ (C) $xy+3y^2=0$ (D) $x^2-y^2=0$

**Ans.** B

**Sol.** Let equations of lines represented by the line pair $xy-3y^2+y-2x+10=0$ are

$y+c_1=0$, $x-3y+c_2=0$

lines $\perp$ to these lines and passing through origin are

$x=0$, $y=-3x$

Joint equation

$x(3x+y)=0 \quad \Rightarrow \quad xy+3x^2=0$

13. If the straight lines joining the origin and the points of intersection of the curve $5x^2+12xy-6y^2+4x-2y+3=0$ and $x+ky-1=0$ are equally inclined to the x–axis then the value of k is equal to :

वक्र $5x^2+12xy-6y^2+4x-2y+3=0$ एवं रेखा $x+ky-1=0$ के प्रतिच्छेद बिन्दुओं को मूल बिन्दु से मिलाने वाली सरल रेखाएँ x–अक्ष से समान कोण पर झुकी हुई हो, तो k का मान है :

(A) 1 (B) $-1$ (C) 2 (D) 3

**Ans.** B

**Sol.** Homogenize $5x^2+12xy-6y^2+4x-2y+3=0$ by $x+ky=1$

$$5x^2 + 12xy - 6y^2 + 4x(x + ky) - 2y(x + ky) + 3(x + ky)^2 = 0$$

it is equally indined with x-axes hence coeff. $xy = 0 \Rightarrow 12 + 4k - 2 + 6k = 0 \Rightarrow k = -1$

14. Area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals :

रेखाओं $y = mx$, $y = mx + 1$, $y = nx$ और $y = nx + 1$ से बनने वाले समान्तर चतुर्भुज का क्षेत्रफल है :

(A) $\frac{|m+n|}{(m-n)^2}$ (B) $\frac{2}{|m+n|}$ (C) $\frac{1}{|m+n|}$ (D) $\frac{1}{|m-n|}$

**Ans.** D

**Sol.** Area of ||gram $= \left|\frac{1}{m-n}\right|$

15. The area bounded by the angle bisectors of the lines $x^2 - y^2 + 2y = 1$ and the line $x + y = 3$, is :

रेखा युग्म $x^2 - y^2 + 2y = 1$ के कोण समद्विभाजको एवं रेखा $x + y = 3$ से निर्मित त्रिभुज का क्षेत्रफल है :

(A) 2 (B) 3 (C) 4 (D) 6

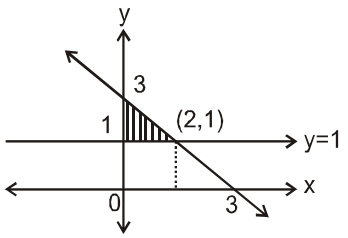
**Ans.** A

**Sol.** $x^2 - y^2 + 2y = 1 \quad \Rightarrow \quad x = \pm(y - 1)$

Bisector of above lines are $x = 0$, $y = 1$

so Area between $x = 0$, $y = 1$ and $x + y = 3$

$= \frac{1}{2} \times 2 \times 2 = 2$ sq. units




16. If L is the line whose equation is ax + by = c. Let M be the reflection of L through the y–axis, and let N be the reflection of L through the x–axis. Which of the following must be true about M and N for all choices of a, b and c :

(A) The x–intercept of M and N are equal (B) The y–intercepts of M and N are equal

(C) The slopes of M and N are equal (D) The slopes of M and N are reciprocal

यदि रेखा L जिसका समीकरण $ax + by = c$ है। माना L का y–अक्ष में प्रतिबिम्ब M तथा माना L का x–अक्ष में प्रतिबिम्ब N है, सभी a, b तथा c के लिए M तथा N के लिए निम्न में से कौनसा सत्य है?

(A) ड तथा N का x–काट बराबर है (B) ड तथा N का y–काट बराबर है

(C) ड तथा N की प्रवणता बराबर है (D) ड तथा N की प्रवणता व्युत्क्रमानुपाती है

**Ans.** C

**Sol.** L : $ax + by = c$

M : $-ax + by = c$

N : $ax - by = c$

M is parallel to N.

17. The complete set of real values of 'a' such that the point P(a, sin a) lies inside the triangle formed by the line $x - 2y + 2 = 0$, $x + y = 0$ and $x - y - \pi = 0$, is :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$ (C) $(0, \pi)$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

'a' के सभी वास्तविक मानों का सम्पूर्ण समुच्चय जिसके लिए बिन्दु P(a, sin a), रेखाओं $x - 2y + 2 = 0$, $x + y = 0$ तथा $x - y - \pi = 0$ से बनने वाले त्रिभुज के अन्दर स्थित है, होगा :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$

(C) $(0, \pi)$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

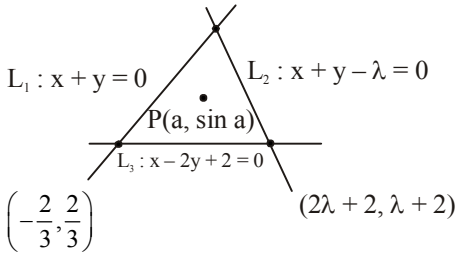
**Ans.** C

**Sol.** P(a, sin a)

$L_1 : a + \sin a > 0$ ...(1)

$L_2 : a - \sin a - \pi < 0$ ...(2)

$L_3 : a - 2\sin a + 2 > 0$ ...(3)




checking by option

Correct answer is C.

18. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, –2) and is running up the line $y = -5x + 18$. at the point (a, b) the mouse starts getting farther from the cheese rather than closer to it. The value of (a + b) is :

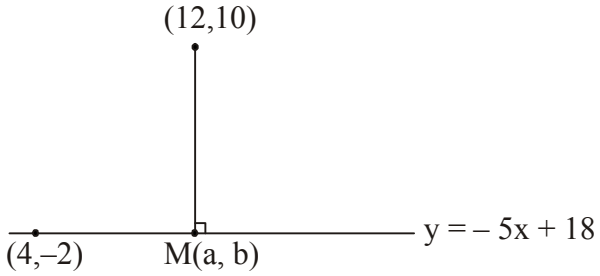
(A) 6 (B) 10 (C) 18 (D) 14

एक पनीर का टुकड़ा निर्देशांक समतल में बिन्दु (12, 10) पर स्थित है। एक चुहा बिन्दु (4, –2) पर है तथा रेखा $y = -5x + 18$ के vuq n' knkSj gkgSA tc (a, b) पर चुहा पनीर के नजदीक जाने की अपेक्षा दूर जाना शुरू हो जाता है, तो (a + b) का मान होगा :

(A) 6 (B) 10 (C) 18 (D) 14

**Ans.** B

**Sol.**




$$\frac{a-12}{5}=\frac{b-10}{1}=-\frac{(60+10-18)}{26}=-\frac{52}{26}=-2$$

$a = 2, b = 8 \Rightarrow a + b = 10$

19. If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to :

(A) $\frac{3}{8}$ (B) $\frac{-3}{4}$ (C) $\frac{1}{4}$ (D) –2

यदि समीकरण $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ तीन सरल रेखाओं को निरूपित करता है, जिसमें दो लम्बवत् रेखायें है, तो a के सभी l Hkkfor eku ksad k; ksx gksxk:

(A) $\frac{3}{8}$ (B) $\frac{-3}{4}$ (C) $\frac{1}{4}$ (D) –2

**Ans.** B

**Sol.** $4\left(\frac{y}{x}\right)^3 - a^2\left(\frac{y}{x}\right) - 3a\left(\frac{y}{a}\right)^2 + 8 = 0$

$4m^3 - 3am^2 - 8a^2m + 8 = 0$

$m_1m_2m_3 = -\frac{8}{4} = -2$ $\qquad$ $m_1m_2 = -1$

$m_3 = 2$

$\Rightarrow 32 - 12a - 16a^2 + 8 = 0$

$\Rightarrow 8 - 3a - 4a^2 + 2 = 0$

$\Rightarrow 4a^2 + 3a + 10 = 0$

$\text{sum of roots} = -\frac{3}{4}$

20. The graph of the function, $y = \cos x \cos(x+2) - \cos^2(x+1)$ is :

(A) a straight line passing through $(0, -\sin^2 1)$ with slope 2

(B) a straight line passing through (0 , 0)

(C) a parabola with vertex $(1, -\sin^2 1)$

(D) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to the x–axis

फलन $y = \cos x \cos(x+2) - \cos^2(x+1)$ का आरेख है :

(A) एक सरल रेखा है जो $(0, -\sin^2 1)$ से गुजरती है जिसकी प्रवणता 2 है

(B) एक सरल रेखा जो (0 , 0) से गुजरती है

(C) एक परवलय जिसका शीर्ष $(1, -\sin^2 1)$ है

(D) एक सरल रेखा जो $\left(\frac{\pi}{2}, -\sin^2 1\right)$ से गुजरती है तथा x–अक्ष के समान्तर है

**Ans.** D

**Sol.** $y = \cos x \cdot \cos(x+2) - \cos^2(x+1)$

$$y = \cos x \cdot \cos(x+2) - \left(\frac{1+\cos(2x+2)}{2}\right)$$

$2y = 2\cos x \cdot \cos(x+2) - 1 - \cos(2x+2)$

$2y = \cos(2x+2) + \cos 2 - 1 - \cos(2x+2)$

$2y = \cos 2 - 1$

$$y = \frac{\cos 2 - 1}{2} = -\sin^2(1)$$

Slope = $0 \Rightarrow$ parallel to x-axis

21. P is a point inside the triangle ABC. Lines are drawn through P, parallel to the sides of the triangle. The three resulting triangles with the vertex at P have areas 4, 9 and 49 sq. units. The area of the triangle ABC is :

(A) 121 (B) 100 (C) 81 (D) 144

त्रिभुज ABC के अन्दर एक बिन्दु P है, बिन्दु P से तीनों भुजाओं के समान्तर रेखा खींची जाती है इस प्रकार l s, d mHk, fu"V' kkZP से तीन त्रिभुज बनती है। जिनका क्षेत्रफल 4, 9 एवं 49 है, तो ABC का क्षेत्रफल होगा :

(A) 121 (B) 100 (C) 81 (D) 144

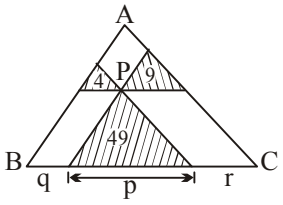
**Ans.** D

**Sol.** $$\frac{\text{ar}(PQR)}{QR^2} = \frac{\text{ar}(ABC)}{BC^2}$$

$\frac{49}{\Delta}=\frac{p^2}{a^2}$ or $\frac{7}{\sqrt{\Delta}}=\frac{p}{a}$ _____(1)

|||ly $\frac{2}{\sqrt{\Delta}}=\frac{q}{a}$ _____(2)

and $\frac{3}{\sqrt{\Delta}}=\frac{r}{a}$ _____(3)




(1) + (2) + (3) gives $\frac{12}{\sqrt{\Delta}}=\frac{p+q+r}{a}=1 \Rightarrow D=144$

22. The complete set of values of the parameter $\alpha$ so that the point $P\left(\alpha,(1+\alpha^2)^{-1}\right)$ does not lie outside the triangle formed by the lines $L_1: 15y=x+1$, $L_2: 78y=118-23x$ and $L_3: y+2=0$, is :

यदि बिन्दु $P\left(\alpha,(1+\alpha^2)^{-1}\right)$ भुजाओं $L_1: 15y=x+1$, $L_2: 78y=118-23x$ एवं $L_3: y+2=0$ से बनने वाली त्रिभुज के भीतर ; kf=kHkq i j fLFkr gS$r ksi kpy $\alpha$ का मान होगा :

(A) (0, 5) (B) [2, 5] (C) [1, 5] (D) [0, 2] (E) (2, 5]

**Ans.** B

**Sol.** As $P(\alpha, (1+\alpha^2)^{-1})$ lie on $y=\frac{1}{1+x^2}$

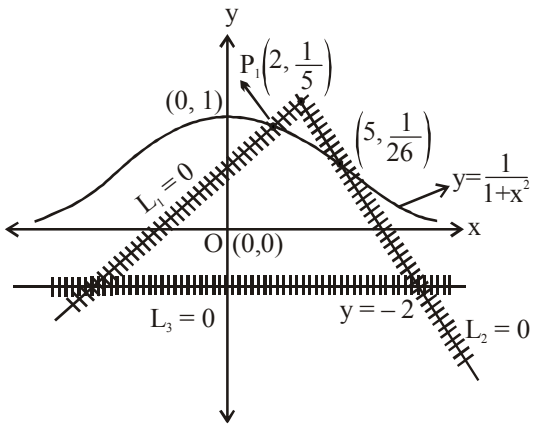
$\therefore$ On solving $y=\frac{1}{1+x^2}$ with $L_1$,

$\frac{15}{1+x^2}=x+1 \Rightarrow x=2,\ y=\frac{1}{5}$

we get $P_1\left(2,\frac{1}{5}\right)$ _____(1)

and with $L_2$ $\frac{78}{1+x^2}=118-23x$, we get $P_2\left(5,\frac{1}{26}\right)$ _____(2)

$\therefore$ From (1) and (2), we get $2 \le \alpha \le 5$



## ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Λινε $\frac{x}{a}+\frac{y}{b}=1$ cuts the co–ordinate axes at A(a, 0) and B (0, b) and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at A′ (–a′, 0) and B′ (0, –b′) . If the points A, B, A′, B′ are concyclic then the orthocentre of the triangle ABA′ is :

रेखा $\frac{x}{a}+\frac{y}{b}=1$ तथा $\frac{x}{a'}+\frac{y}{b'}=-1$ निर्देशांक अक्ष को क्रमशः A(a, 0), B (0, b) तथा A′ (–a′, 0) एवं B′ (0, –b′) पर काटती है यदि बिन्दु A, B, A′, B′ चक्रीय हो, तो त्रिभुज ABA′ का लम्ब केन्द्र होगा :

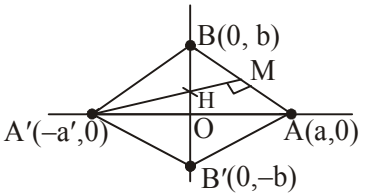
(A) (0, 0) (B) (0, b') (C) $\left(0, \frac{aa'}{b}\right)$ (D) $\left(0, \frac{bb'}{a}\right)$

**Ans.** BC

**Sol.** $\frac{x}{a}+\frac{y}{b}=1, \quad \frac{x}{a'}+\frac{y}{b'}=1$

$m_{AB}=-\frac{b}{a}$

$m_{A'M}=\frac{a}{b}$




$\frac{\beta-0}{0+a'}=\frac{a}{b}\Rightarrow\beta=\frac{aa'}{b}$

$H\left(0,\frac{aa'}{b}\right)\Rightarrow$ option (C)

ABA′B′ is concyclic then

Intersecting chord theorem

$a\times(a')=b\times(+b')$

$\Rightarrow b'=\frac{aa'}{b}$

$H\left(0,\frac{aa'}{b}\right)\equiv H(0,b')$

**Aliter** :

$m_{AB}=\frac{-b}{a}$

$$m_{A'H} = \frac{a}{b}$$

$$y - 0 = \frac{a}{b}(x + a')$$

$$y = \frac{a}{b}x + a\frac{a'}{b}$$

$$y - 0 = \frac{a'}{b}(x - a)$$

$$y = \frac{a'}{b}x - a\frac{a'}{b}$$

$$\text{P.O.I.} \left(0, \frac{aa'}{b}\right)$$

24. Consider the equation $y - y_1 = m(x - x_1)$. If m and $x_1$ are fixed and different lines are drawn for different values of $y_1$, then :

(A) the lines will pass through a fixed point (B) there will be a set of parallel lines

(C) all the lines intersect the line $x = x_1$ (D) all the lines will be parallel to the line $y = x_1$

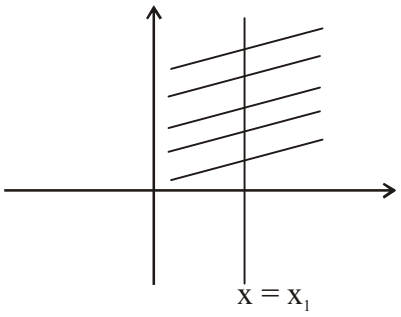
दिया हुआ रेखा का समीकरण $y - y_1 = m(x - x_1)$ है। यदि m एवं $x_1$ स्थिर है। तथा $y_1$ के अलग–अलग मान के लिये अलग–अलग j{kk[ kapht kr hgksxh ks:

(A) सारी रेखाऐं स्थिर बिन्दु से जायेगी (B) सारी रेखाऐं समान्तर होगी

(C) सारी रेखाऐं रेखा $x = x_1$ को काटेगी (D) सारी रेखाऐं $y = x_1$ के समान्तर होगी

**Ans.** BC

**Sol.** $(y - y_1) = m(x - x_1)$




(B) All lines are parallel.

(C) All the lines intersect the line $x = x_1$.

25. If $\frac{x}{c}+\frac{y}{d}=1$ is a line through the intersection of $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ and the lengths of the perpendiculars drawn from the origin to these lines are equal in lengths then :

(A) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2}$ (B) $\frac{1}{a^2}-\frac{1}{b^2}=\frac{1}{c^2}-\frac{1}{d^2}$

(C) $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}+\frac{1}{d}$ (D) none

यदि रेखाओं $\frac{x}{a}+\frac{y}{b}=1$ एवं $\frac{x}{b}+\frac{y}{a}=1$ के प्रतिच्छेद बिन्दु से गुजरने वाली रेखा का समीकरण $\frac{x}{c}+\frac{y}{d}=1$ है तथा (0, 0) से इन j[kkvksa ij Mkys x; sy Ec d hy Ecnqvksa ds ekiksa dh yEckbZ ekuk gks]r ks:

(A) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2}$ (B) $\frac{1}{a^2}-\frac{1}{b^2}=\frac{1}{c^2}-\frac{1}{d^2}$

(C) $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}+\frac{1}{d}$ (D) इनमें से कोई नहीं

**Ans.** AC

**Sol.** $\left.\begin{matrix} bx+ay=ab \\ ax+by=ab \end{matrix}\right] \Rightarrow y=\frac{ab}{a+b}$

$P\left(\frac{ab}{a+b},\frac{ab}{a+b}\right)$

$x=\frac{ab}{a+b}$

$\frac{x}{c}+\frac{y}{a}=1 \Rightarrow \frac{ab}{c(a+b)}+\frac{ab}{d(a+b)}=1$

$\left(\frac{1}{c}+\frac{1}{d}\right)=\left(\frac{1}{a}+\frac{1}{b}\right) \rightarrow \text{option (C)}$

$P=\frac{1}{\sqrt{\frac{1}{c^2}+\frac{1}{d^2}}}=\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}=\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2} \rightarrow \text{option (A)}$

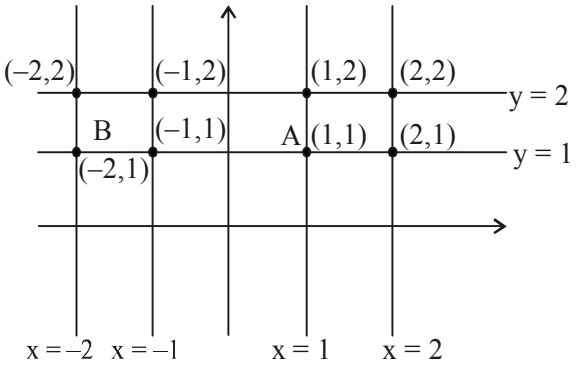
26. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2-3|x|+2=0$ and the y-coordinates of the vertices are the roots of the equation $y^2-3y+2=0$ then the possible vertices of the square is/are :

इकाई क्षेत्रफल वाले वर्ग के शीर्षो का भुज तथा कोटि समीकरण $x^2 - 3|x| + 2 = 0$ तथा $y^2 - 3y + 2 = 0$ का मूल है तो वर्ग के शीर्षो के सम्भावित निर्देशांक होगे।

(A) (1, 1), (2, 1), (2, 2), (1, 2)
(B) (–1, 1), (–2, 1), (–2, 2), (–1, 2)
(C) (2, 1), (1, –1), (1, 2), (2, 2)
(D) (–2, 1), (–1, –1), (–1, 2), (–2, 2)

**Ans.** AB

**Sol.** $x = \pm 1, \pm 2,\ y = 1, 2$




27. P(x, y) moves such that the area of the triangle formed by P, Q (a , 2a) and R (–a, – 2a) is equal to the area of the triangle formed by P, S (a, 2a) and T (2a, 3a). The locus of 'P' is a straight line given by :

यदि P (x, y) इस प्रकार से गति करता है कि ΔPQR का क्षेत्रफल ΔPST के क्षेत्रफल के बराबर है जहां Q (a , 2a) तथा R (– a, – 2a) एवं S (a, 2a) तथा T (2a, 3a) है तो P का बिन्दुपथ सरल रेखा द्वारा निरूपित होगा।

(A) $3x - y = a$ (B) $5x - 3y + a = 0$ (C) $5x - 5y + a = 0$ (D) $2y = ax$

**Ans.** AB

**Sol.** P (x, y) Q (a, 2a) & R (–a, –2a)

S (a, 2a) T (2a, 3a)

$$\begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ -a & -2a & 1 \end{vmatrix} = \pm \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ 2a & 3a & 1 \end{vmatrix}$$

–ve +ve

$$\Rightarrow \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ 3a & 5a & 0 \end{vmatrix} = 0 \qquad \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ a & a & 2 \end{vmatrix} = 2$$

$\Rightarrow -5ax + 3ay - a^2 = 0$ $3ax - ay - a^2 = 0$

$\Rightarrow 5x - 3y + a = 0$ $3x - y - a = 0$

28. Let $u \equiv ax + by + a\sqrt[3]{b} = 0$ $v \equiv bx - ay + b\sqrt[3]{a} = 0$ $a, b \in R$ be two straight lines. The equation of the bisectors of the angle formed by $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ for non zero real $k_1$ and $k_2$ are :

माना कि $u \equiv ax + by + a\sqrt[3]{b} = 0$, $v \equiv bx - ay + b\sqrt[3]{a} = 0$ जहाँ $a, b \in R$ रेखाओ $k_1u - k_2v = 0$ एवं $k_1u + k_2v = 0$ के कोणों की अर्द्धक रेखा का समीकरण होगा (जहाँ $k_1, k_2 \neq 0$) :

(A) $u = 0$ (B) $k_2u + k_1v = 0$ (C) $k_2u - k_1v = 0$ (D) $v = 0$

**Ans.** AD

**Sol.** Note that the lines are perpendicular.

Assume the co-ordinate axes to be directed along $u = 0$ and $v = 0$.

Now the lines $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ are equally inclined with $uv$ axes.

Hence the bisectors are $u = 0$ and $v = 0$. i.e. $\dfrac{k_1u - k_2u}{\sqrt{k_1^2 + k_2^2}} = \pm \dfrac{k_1u + k_2u}{\sqrt{k_1^2 + k_2^2}}$

29. The bisectors of angle between the st. lines, $y - b = \dfrac{2m}{1-m^2}(x-a)$ and $y - b = \dfrac{2m'}{1-m'^2}(x-a)$ are :

रेखाओं $y - b = \dfrac{2m}{1-m^2}(x-a)$ तथा $y - b = \dfrac{2m'}{1-m'^2}(x-a)$ के कोणों की अर्द्धक रेखा होगी।

(A) $(y-b)(m+m') + (x-a)(1-mm') = 0$ (B) $(y-b)(m+m') - (x-a)(1-mm') = 0$

(C) $(y-b)(1-mm') + (x-a)(m+m') = 0$ (D) $(y-b)(1-mm') - (x-a)(m+m') = 0$

**Ans.** AD

**Sol.** $M_1 = \tan 2\theta_1$ $M_2 = \tan 2\theta_2$ $\tan\theta_1 = M_1 \tan\theta_2 = M$

$\phi_1 = \dfrac{2\theta_1 + 2\theta_2}{2} = \theta_1 + \theta_2$ $\phi_2 = \dfrac{\pi}{2} + \theta_1 + \theta_2$

$\tan\phi_1 = \dfrac{m+m'}{1-mm'}$ $\tan\phi_2 = -\cos(\theta_1 + \theta_2)$

$= -\left(\dfrac{mm'-1}{m+m'}\right)$

$(y-b) = \dfrac{m+m'}{1-mm'}(x-a) \rightarrow (D)$

$(y-b) = \left(\dfrac{mm'-1}{m+m'}\right)(x-a) \rightarrow (A)$

30. If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s)?

(A) centriod (B) incentre (C) circumcentre (D) orthocentre

यदि $\Delta PQR$ के शीर्ष यदि परिमेय है, तो इनमें से कौन सा निर्देशांक हमेशा परिमेय होगा

(A) केन्द्रक (B) अन्तः केन्द्र (C) परिकेन्द्र (D) लम्बकेन्द्र

**Ans.** ACD

**Sol.** put $m = \tan\alpha$ and $m' = \tan\beta$

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$ all are rational.

Hence

(A) Centroid $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$ is rational.

(B) In centre $\left(\frac{\Sigma a x_1}{\Sigma a}, \frac{\Sigma a y_1}{\Sigma a}\right)$ can be irrational as side length can be irrational.

(C) Circumcentre will be rational as coffiecient in equations of $\perp$ bisector will be rational.

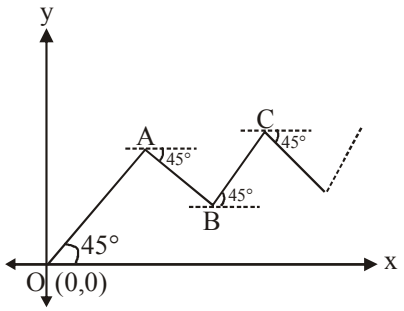
Hence coordinate of there Point of Intersection will also be rational.

(D) Similarly, Orthocentre is also rational.

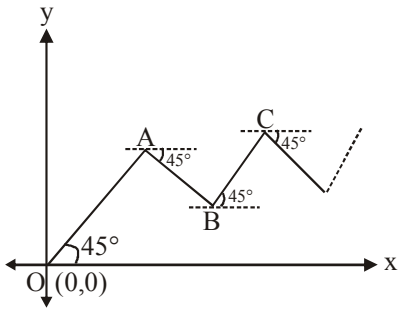
## Exercise C - II

### SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Points O, A, B, C.....are shown in figure where OA = 2AB = 4BC = ..... so on. Let A is the centroid of a triangle whose orthocentre and circumcenter are (2, 4) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point O(0, 0) along the straight line in zig–zag fashion and terminates ultimately at point P($\alpha$, $\beta$) then find the value of ($\alpha + \beta$)




आरेख में प्रदर्शित बिन्दुओं O, A, B, C ..... जहाँ OA = 2AB = 4BC = ..... यदि बिन्दु A, किसी त्रिभुज का केन्द्रक है एवं जिसका लम्बकेन्द्र (2, 4) एवं परिकेन्द्र $\left(\frac{7}{2}, \frac{5}{2}\right)$ पर स्थित है, यदि कोई कीट मूल बिन्दु O(0, 0) से रेखा के साथ उतार–चढ़ाव (zig–zag) में बिन्दु P तक चलता है। यदि बिन्दु P के निर्देशांक ($\alpha$, $\beta$) हो तो ($\alpha + \beta$) का मान ज्ञात करो।




**Ans.** 8

**Sol.** $H(2,4) \bullet \underset{G}{\overset{2 \quad 1}{\longrightarrow}} \bullet 0\left(\frac{7}{2}, \frac{5}{2}\right)$

$$\left(\frac{7+2}{+3}, \frac{5+4}{3}\right) \equiv G(3,3)$$

$$\alpha = \frac{3\sqrt{2}}{\sqrt{2}} + \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} + .....\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{1-\frac{1}{2}} = 6$$

$$\beta = \frac{3\sqrt{2}}{\sqrt{2}} - \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} .....\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{\frac{3}{2}} = 2$$

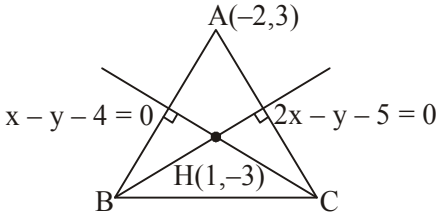
$\alpha + \beta = 6 + 2 = 8$

2. The equations of perpendiculars of the sides AB & AC of triangle ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and point of intersection of perpendiculars bisectors is $\left(\frac{3}{2}, \frac{5}{2}\right)$, find the equation of medians to the sides AB & AC respectively.

त्रिभुज ABC समें भुजाओं AB एवं AC के लम्बो का समीकरण क्रमशः $x - y - 4 = 0$ एवं $2x - y - 5 = 0$ हो, यदि शीर्ष बिन्दु A के निर्देशांक $(-2, 3)$ हो, एवं $\left(\frac{3}{2}, \frac{5}{2}\right)$ भुजाओं के लम्ब अर्द्धको का प्रतिच्छेद बिन्दु हो, तो क्रमशः AB एवं AC की मध्यिकाओं के समीकरण ज्ञात करो।

**Ans.** $x + 4y = 4$ ; $5x + 2y = 8$

**Sol.**




$$\begin{array}{rrrrrr} x & - & y & - & 4 & = & 0 \\ 2x & - & y & - & 5 & = & 0 \\ - & & + & & + & & \\ \hline & - & x & + & 1 & = & 0 \\ \hline \end{array}$$

$x = 1 \qquad y = -3$

Eqn of $\perp$ bisector of AB and AC $\rightarrow$ AB $\perp^r$ Bisector $\rightarrow x - y + 1 = 0$

$x - y + 1 = 0$

AC $\perp^r$ Bisector

$$2x - y - \frac{1}{2} = 0$$

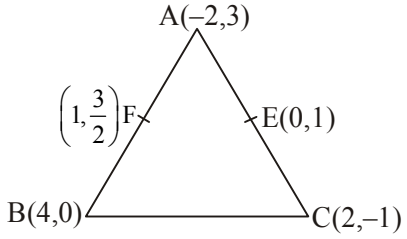
$$4x - 2y - 1 = 0$$

$B \rightarrow$ Image of A(–2, 3) in $4x - 2y - 1 = 0$

$$\frac{x+2}{4} = \frac{y-3}{-2} = -2\frac{(-8-6-1)}{20} = \frac{3}{2} \Rightarrow B(4,0)$$

$C \rightarrow$ Image of A(–2, 3) in $x - y + 1 = 0$

$$\frac{x+2}{1} = \frac{y-3}{(-1)} = -2\frac{(-2-3+1)}{2} = 4 \Rightarrow C(2,-1)$$




CF : $\rightarrow$ $\quad (y+1) = \left(\frac{\frac{3}{2}+1}{1-2}\right)(x-2)$

$$2y + 2 = -5x + 10$$

$$5x + 2y = 8 \text{ Ans.}$$

BE : $\rightarrow$ $\quad (y-0) = \frac{1}{-4}(x-4)$

$$-4y = x - 4$$

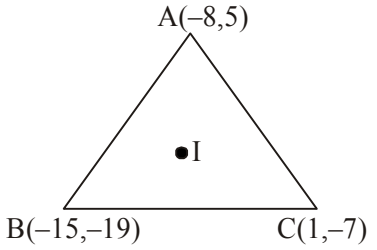
$$x + 4y = 4 \text{ Ans.}$$

3. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are A(–8, 5); B(–15, –19) and C(1, – 7) has the equation $ax + 2y + c = 0$. Find 'a' and 'c'.

यदि त्रिभुज ABC के शीर्ष बिन्दु के निर्देशांक A(–8, 5); B(–15, –19) एवं C(1, – 7) है, एवं कोण के आंतरिक कोण अर्द्धक का समीकरण $ax + 2y + c = 0$ है तो 'a' एवं 'c' का मान ज्ञात करो।

**Ans.** $a = 11$ , $c = 78$

**Sol.**




$I \equiv$ Incentre

$$a = BC = \sqrt{16^2 + 12^2}$$

$$a = 20$$

$$b = AC = \sqrt{9^2 + 12^2}$$

$$b = 15$$

$$C = AB = \sqrt{7^2 + 24^2}$$

$$C = 25$$

$$I = \left( \frac{-160 - 225 + 25}{60}, \frac{100 - 285 - 175}{60} \right) \qquad \left( \because I \equiv \frac{ax_1 + by_1 + cz_1}{a+b+c}, \frac{ax_2 + by_2 + cz_2}{a+b+c} \right)$$

$$I \equiv (-6, -6)$$

AI εθν. : $\rightarrow$ $\quad (y+6) = \frac{11}{-2}(x+6)$

$$-2y - 12 = 11x + 66$$

$$11x + 2y + 78 = 0$$

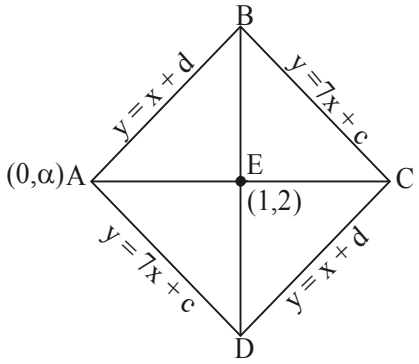
$$a = 11,\ c = 78$$

4. Two sides of a rhombous ABCD are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombous intersect at the point (1, 2) & the vertex A is on the y-axis, find the possible coordinates of A.

किसी समचतुर्भुज ABCD की दो भुजाऐं, रेखाओं $y = x + 2$ एवं $y = 7x + 3$ के समान्तर हो, एवं समचतुर्भुज के विकर्णो का प्रतिच्छेद बिन्दु (1, 2) हो तथा बिन्दु A, y-अक्ष पर स्थित हो, तो बिन्दु A के सम्भावित निर्देशांक ज्ञात करो।

**Ans.** $(0\,,\,0)$ or $\left(0, \frac{5}{2}\right)$

**Sol.**




Diagonals will be parallel to angle bisectors of line

$y = x + 2$

$y = 7x + 3$

$$\frac{x - y + 2}{\sqrt{2}} = \pm\frac{(7x - y + 3)}{5\sqrt{2}}$$

(+) (–)

$5x - 5y + 10 = 7x - y + 3$ $5x - 5y + 10 = -7x + y - 3$

$B_1 : 2x + 4y - 7 = 0$ $B_2 : 12x - 6y + 13 = 0$

$$x + 2y - \frac{7}{2} = 0 \qquad 2x - y + \frac{13}{6} = 0$$

$$m_{AE} = \frac{-2}{4} \text{ or } \frac{12}{6}$$

$$\Rightarrow \frac{2-\alpha}{1-0} = -\frac{1}{2} \text{ or } \frac{2-\alpha}{1-0} = 2$$

$$\alpha = \frac{5}{2} \text{ or } \alpha = 0$$

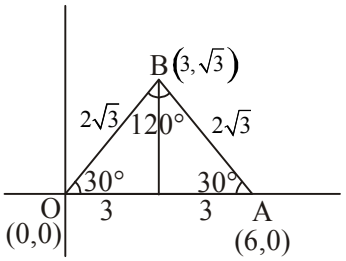
Coordinates $\left(0, \frac{5}{2}\right)$ or (0,0)

5. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of $\Delta OAB$. Let R be the region consisting of all those points P inside $\Delta OAB$ which satisfy d(P, OA) ≤ minimum $\left(d(P, OB), d(P, AB)\right)$ where d(P, OA), d(P, OB) and d(P, AB) represent the distance of P from the sides OA, OB and AB respectively. If the area of region R is 9 ($a - \sqrt{b}$) where a and b are coprime, then find the value of (a + b).

यदि O(0, 0), A (6, 0) एवं B(3, $\sqrt{3}$) त्रिभुज OAB के शीर्ष बिन्दु है तथा R त्रिभुज OAB के अंदर स्थित सभी बिन्दुओं के क्षेत्र को प्रदर्शित करता है जो कि संबंध d(P, OA) ≤ न्यूनतम मान $\left(d(P,OB), d(P,AB)\right)$ को संतुष्ट करता है जहाँ d(P, OA), d(P, OB) तथा d(P, AB) बिन्दु क्रमशः भुजाओं OA, OB एवं AB दूरी को दर्शाता है। यदि क्षेत्र R का क्षेत्रफल 9 ( a – $\sqrt{b}$ ) है जहाँ a एवं b परस्पर अभाज्य (coprime) है, तो (a + b) का मान ज्ञात कीजिये।

**Ans.** 5

**Sol.**




Let P(h, k)

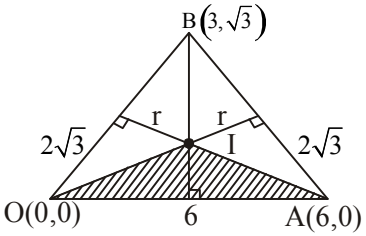
$OB \equiv \sqrt{3}y = x$

$AB = \sqrt{3}y = -x + 6$

$d(P,OA) \le \min\{d(P,OB), d(P,AB)\}$

$d(P,OA) \le d(P,OB)$

$d(P,OA) \le d(P,AB)$




$I\left(\frac{18+0+12\sqrt{3}}{6+4\sqrt{3}}, \frac{6\sqrt{3}+0+0}{6+4\sqrt{3}}\right)$

$\text{Area} = R = \frac{1}{2}\times(6)\times\left(\frac{6\sqrt{3}}{6+4\sqrt{3}}\right)$

$$= 9\left(\frac{\sqrt{3}}{3+2\sqrt{3}}\right) = 9\left(\frac{\sqrt{3}\left(2\sqrt{3}-3\right)}{3}\right)$$

$$= 9\left(2-\sqrt{3}\right)$$

a = 2, b = 3, a + b = 5

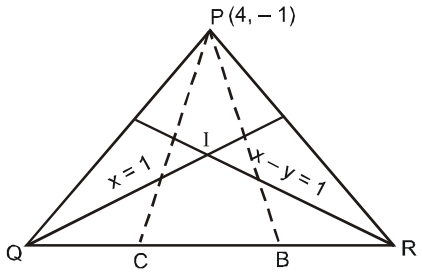
6. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines x – 1 = 0 and x – y –1 = 0 are the equations of two internal bisectors of its angles.

त्रिभुज की भुजाओं के समीकरण ज्ञात करो यदि एक शीर्ष बिन्दु के निर्देशांक (4, –1) एवं दो आन्तरिक कोर्णाद्धको के समीकरण x – 1 = 0 एवं x – y –1 = 0 है।

**Ans.** 2x – y + 3 = 0, 2x + y – 7 = 0, x – 2y – 6 = 0

**Sol.** Method I :

**Alter :**




Mirror image of a vertex of ΔABC w.r.t. it's angle bisector always lie on the side opposite to the vertex.
Reflection of (4, – 1) about x = 1 is B and about x – y = 1 is C
$B \equiv (x_1, y_1)$
$C \equiv (x_2, y_2)$

$\left(\frac{x_1+4}{2}, \frac{y_1-1}{2}\right)$ will lie on x = 1

$\Rightarrow \quad \frac{x_1+4}{2} = 1 \quad \Rightarrow \quad x_1 = -2 \qquad$ and $\qquad \frac{y_1+1}{x_1-4} = 0 \quad \Rightarrow \quad y_1 = -1$

$\therefore \quad B \equiv (-2, -1) \Rightarrow \left(\frac{x_2+4}{2}, \frac{y_2-1}{2}\right)$ will lie on x – y – 1 = 0

$x_2 + 4 - y_2 + 1 - 2 = 0 \quad \Rightarrow \quad x_2 - y_2 + 3 = 0 \quad$ _____(i)

and $\left(\frac{y_2+1}{x_2-4}\right).1 = -1 \quad \Rightarrow \quad x_2 + y_2 - 3 = 0 \quad$ _____(ii)

from (i) & (ii) $x_2 = 0, y_2 = 3$

$\therefore$ C(0, 3) $\therefore$ Equation of BC

$y + 1 = \frac{4}{2}(x + 2) \quad \Rightarrow \quad 2x - y + 3 = 0$

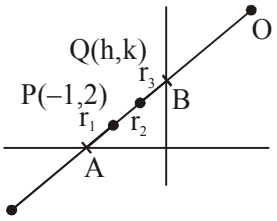
To find PQ & PR we have to find Q and R ;
solving QI & BC, we get Q (1, 5)
equation of PQ is $2x + y - 7 = 0$
For R, we solve RI & BC
$\therefore$ R (–4, – 5)
equation of PR is $x - 2y - 6 = 0$

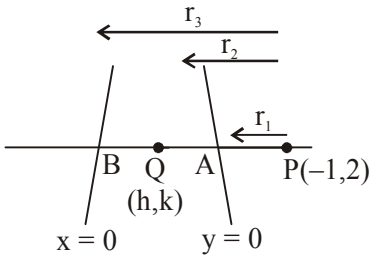
7. P is the point (–1, 2), a variable line through P cuts the x & y axes at A & B respectively Q is the point on AB such that PA, PQ, PB are H.P. Find the locus of Q.

यदि P के निर्देशांक (–1, 2) हो, एवं एक चर रेखा जो बिन्दु P से गुजरती है, तथा x एवं y अक्षों को क्रमशः A एवं B पर काटती है, एवं Q, AB पर स्थित कोई बिन्दु है, तथा PA, PQ, PB हरात्मक श्रेणी में हो तो, बिन्दु Q का बिन्दुपथ ज्ञात करो।

**Ans.** $y = 2x$

**Sol.**




$$Q(h,k) = Q(-1 + r_2\cos\theta, 2 + r_2\sin\theta)$$

$$A(-1 + r_1\cos\theta, 2 + r_1\sin\theta) \Rightarrow x - \text{axis}$$

$$r_1\sin\theta = -2$$

$$r_1 = \frac{-2}{\sin\theta}$$

$$B(-1 + r_3\cos\theta, 2 + r_3\sin\theta) \Rightarrow y - \text{axis}$$

$$-1 + r_3\cos\theta = 0$$

$$r_3 = \frac{1}{\cos\theta}$$

$$\frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3} = \frac{-\sin\theta}{2} + \frac{\cos\theta}{1}$$

$$2 = \frac{-r_2 \sin\theta}{2} - r_2 \cos\theta$$

$$4 = -(k-2) + 2(h+1)$$

$2h - k = 0$

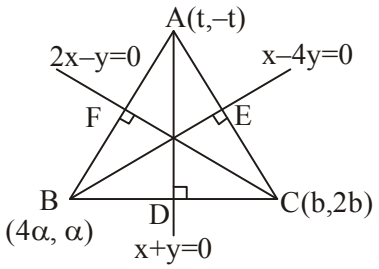
$y = 2x$

8. The equations of the altitudes AD, BE, CF of a triangle ABC are $x+y=0$, $x-4y=0$ and $2x-y=0$ respectively. The coordinates of A are $(t, -t)$. Find coordinates of B & C. Prove that if t varies the locus of the centroid of the triangle ABC is $x+5y=0$.

त्रिभुज ABC के लम्बो AD, BE, CF के समीकरण के क्रमशः $x+y=0$, $x-4y=0$ एवं $2x-y=0$ है एवं बिन्दु A के निर्देशांक $(t, -t)$ है। तो बिन्दु B एवं C के निर्देशांक ज्ञात कीजिये। दर्शाओं यदि t प्राचल हो तो त्रिभुज ABC के केन्द्रक का बिन्दुपथ $x+5y=0$ होगा।

**Ans.** $B\left(-\frac{2t}{3}, -\frac{t}{6}\right)$, $C\left(\frac{t}{2}, t\right)$

**Sol.**




$$\frac{-t-\alpha}{t-4\alpha} = \frac{-1}{2}$$

$2t + 2\alpha = t - 4\alpha$

$t = -6\alpha$

$$\alpha = \frac{t}{-6}$$

$$\frac{2b+t}{b-t} = -4$$

$2b + t = 4t - 4b$

$6b = 3t$

$t = 2b$

$$b = \frac{t}{2}$$

$$B\left(\frac{-2t}{3}, \frac{-t}{6}\right) \qquad C\left(\frac{t}{2}, t\right)$$

$$G\left(\frac{t - \frac{2t}{3} + \frac{t}{2}}{3}, \frac{-t - \frac{t}{6} + t}{3}\right) \equiv (h, k)$$

$$\left.\begin{array}{l} \frac{5t}{18} = h \Rightarrow t = \frac{18h}{5} \\ t = -18k \end{array}\right\rangle \quad h + 5k = 0$$

$$\Rightarrow x + 5y = 0$$

hence, locus of G is st. line $x + 5y = 0$

9. The distance of a point $(x_1, y_1)$ from each of two straight lines which passes through the origin of co-ordinates is $\delta$; find the combined equation of these straight lines.

बिन्दु P $(x_1, y_1)$ की मूल बिन्दु से गुजरने वाली दो रेखाओं में प्रत्येक दूरी $\delta$ है। तो दोनो रेखाओं का सम्मिलित समीकरण ज्ञात करो।

**Ans.** $(y_1^2 - \delta^2)\, x^2 - 2\, x_1 y_1\, xy + (x_1^2 - \delta^2)\, y^2 = 0$

**Sol.** $$P_1\, P_2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$\delta^2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$y = m_1x, \qquad y = m_2x$

$$\delta = \left|\frac{m_1x_1 - y_1}{\sqrt{1 + m_1^2}}\right| \qquad \delta = \left|\frac{m_2x_2 - y_2}{\sqrt{1 + m_2^2}}\right|$$

$$\delta^2 = \frac{(mx_1 - y_1)^2}{1 + m_1^2} \Rightarrow \delta^2 = \frac{m^2x_1^2 - 2x_1y_1m + y_1^2}{1 + m_1^2}$$

$$\left(x_1^2 - \delta^2\right)m^2 - 2x_1y_1m + \left(y_1^2 - \delta^2\right) = 0$$

Put $m = \frac{y}{x}$

$\left(y_1^2 - \delta^2\right)x^2 - 2x_1y_1xy + \left(x_1^2 - \delta^2\right)xy^2 = 0$

10. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point (–5, –1). If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b) then find $\left(a + \frac{1}{b^2}\right)$.

त्रिभुज की दो भुजाओं का संयुक्त रूप का समीकरण $x^2 - 3y^2 - 2xy + 8y - 4 = 0$ तथा तीसरी चर भुजा जो कि हमेशा बिन्दु (–5, –1) से गुजरती है। तीसरी भुजा की प्रवणता का अन्तराल (a, b) इस तरह है, कि मूल बिन्दु त्रिभुज का आंतरिक बिन्दु है तो $\left(a + \frac{1}{b^2}\right)$ का मान ज्ञात करो।

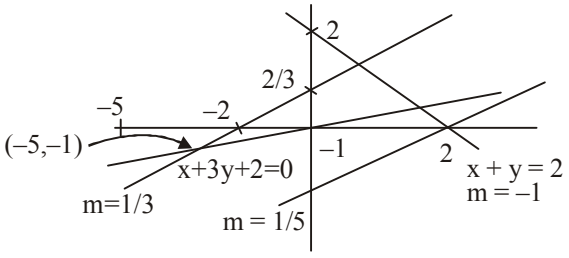
**Ans.** 24

**Sol.** $x^2 - 2xy - 3y^2 + 8y - 4 = 0$

$x^2 - 2xy - 3y^2 = x^2 - 3xy + xy - 3y^2 = (x + y)(x - 3y) = 0$

$(x + y + c_1)(x - 3y + c_2) = 0$

$$\begin{array}{rrrrr} & C_1 & + & C_2 & = 0 \\ - & 3C_1 & + & C_2 & = 8 \\ + & & - & & - \\ \hline & & 4C_1 & = & -8 \\ \hline \end{array}$$




$C_1\, C_2 = -4$

$C_1 = -2 \qquad C_2 = 2$

$L_1 : x + y - 2 = 0 \qquad L_2 : x - 3y + 2 = 0$

Let equation of variable line be

$y + 1 = m(x + 5)$

For origin to lie inside the required $\Delta$, y intercept for the variable line should be –ve.

$\therefore\ 5m - 1 = 0 \Rightarrow m < \frac{1}{5}$

$(-5, -1)$ also lie on $x - 3y + 2 = 0$

Now, $\Delta$ can form when $m > -1$.

So, $m \in \left(-1, \frac{1}{5}\right)$, $a + b = 4$

11. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A, B and C.

(a) Find the sum of the contangents of the interior angles of the triangle ABC.

(b) Find the area of triangle ABC

(c) Find the radius of the circle touching all the 3 sides of the triangle.

एक रेखा युग्म $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ तथा रेखा L जिसकी प्रवणता 3 एवं मूलबिन्दु से गुजरती है। तो रेखा युग्म एवं रेखा L से बनने वाले त्रिभुज के शीर्ष बिन्दु A, B एवं C है।

(a) ड वज ड वज ड वज का मान ज्ञात कीजिये। (जहाँ A,B,C त्रिभुज ABC के आन्तरिक कोण हैं)

(b) त्रिभुज ABC का क्षेत्रफल ज्ञात करो

(c) त्रिभुज की तीनो भुजाओं को स्पर्श करने वाले वृत की त्रिज्या ज्ञात करे।

**Ans.** (a) $\frac{50}{7}$; (b) $\frac{63}{10}$; (c) $\frac{3}{10}\left(8\sqrt{5} - 5\sqrt{10}\right)$

**Sol.** $2x^2 + 3xy - 2y^2 - 2y^2 - 10x + 15y - 28 = 0$

$y = 3x$

$2x^2 + 3xy - 2y^2 = 0$

$2 + 3m - 2m^2 = 0 \qquad \left(\because \frac{y}{x} = m\right)$

$2m^2 - 3m - 2 = 0$

$2m^2 - 4m + m - 2 = 0$

$(2m + 1)(m - 2) = 0$

$m = -\frac{1}{2}, \qquad m = 2$

$m_1 = 3$

$m_2 = 2$

$$m_3 = -\frac{1}{2}$$

$$\tan I = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{3-2}{1+6} = \frac{1}{7}$$

$$\tan II = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{2 + \frac{1}{2}}{1-1} = \infty$$

$$\tan III = \frac{-\frac{1}{2} - 3}{1 - \frac{3}{2}} = \frac{-17}{1+1}$$

(a) $\sum \cot I = 7 + \frac{1}{7} + 0 = \frac{50}{7}$

(d) $2x^2 + 3xy - 2y^2 = 0$

$2x^2 + 4xy - xy - 2y^2 = 0$

$2x(2x + 2y) - y(x + 2y) = 0$

$(x + 2y)(2x - y) = 0$

$2C_1 + C_2 = -10$ $\qquad C_1 C_2 = -28$

$-C_1 + 2C_2 = 15$

$C_1 = -7 \qquad C_2 = 4$

$x + 2y + C_1 = 0 \qquad \Rightarrow x + 2y - 7 = 0$ and $y = 3x$

$2x - y + C_2 = 0 \qquad \Rightarrow 2x - y + 4 = 0$

P.O.I $\quad A\left(-\frac{1}{5}, \frac{18}{5}\right)$

B (4, 12)

C (1, 3)

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} -\frac{1}{5} & \frac{18}{5} & 1 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix} = \frac{1}{10}\begin{vmatrix} -1 & 18 & 5 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{10}\begin{vmatrix} -6 & 3 & 5 \\ 3 & 9 & 1 \\ 0 & 0 & 1 \end{vmatrix} \Rightarrow \frac{1}{10}[-54-9] = \frac{63}{10}$$

12. The two line pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon find
(i) area of the polygon (ii) length of its diagonals.

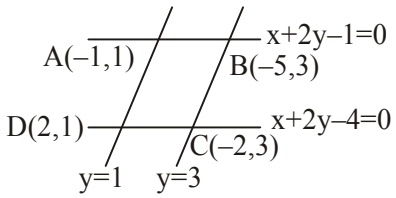
दो रेखा युग्मों $y^2 - 4y + 3 = 0$ एवं $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ से चार भुजाओं वाला उत्तल बहुभुज निर्मित होता है, तो
(i) बहुभुज का क्षेत्रफल (ii) विकर्णो की लम्बाई ज्ञात करो

**Ans.** (i) area = 6 sq. units, (ii) diagonals are $\sqrt{5}$ & $\sqrt{53}$

**Sol.** $y^2 - 4y + 3 = 0$ $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$

$y = 3, 1$ $(x + 2y)^2 - 5(x + 2y) + 4 = 0$

$(x + 2y - 4)(x + 2y - 1) = 0$




(i) Area of ploygon

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -1 & 2 & -2 & -5 & -1 \\ 1 & 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}[-1 - 2 + 6 + 2 - 6 + 15 - 5 + 3]$$

$= 6$

(ii) Length of diagonals

$AC = \sqrt{1+4} = \sqrt{5}$

$BD = \sqrt{49+4} = \sqrt{53}$

13. Find the equation of the two straight lines which together with those given by the equation $6x^2 - xy - y^2 + x + 12y - 35 = 0$ will make a parallelogram whose diagonals intersect in the origin.

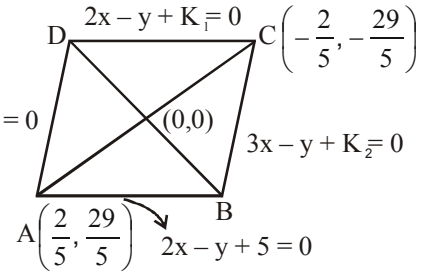
उन दोनों रेखाओं का समीकरण ज्ञात करो जो कि समीकरण $6x^2 - xy - y^2 + x + 12y - 35 = 0$ से प्रदर्शित होने वाली अन्य रेखाओं के साथ मिलकर एक सामान्तर चतुर्भुज इस प्रकार निर्मित करती है कि समान्तर चतुर्भुज के विकर्ण मूलबिन्दु पर प्रतिच्छेद करते है।

**Ans.** $6x^2 - xy - y^2 - x - 12y - 35 = 0$

**Sol.** $6x^2 - xy - y^2 + x + 12y - 35 = 0 \quad \Rightarrow \quad (2x - y + 5)(3x + y - 7) = 0$

Now point A is intersection point of $2x - y + 5 = 0$ and $3x + y - 7$

$\therefore \quad A\left(\frac{2}{5}, \frac{29}{5}\right)$




Now C $\left(-\frac{2}{5}, -\frac{29}{5}\right)$ as midpoint of diagonals is (0, 0)

$2x - y + K_1 = 0$ and $3x + y + K_2 = 0$ are passing through C

$\therefore \quad K_1 = -5$, and $K_2 = 7$

Now combined equation $(2x - y - 5)(3x + y + 7) = 0$

$6x^2 - xy - y^2 - x - 12y - 35 = 0$

14. A straight line passing through O(0, 0) cuts the lines $x = a$, $y = b$ and $x + y = 8$ at A, B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(a, b) \le 0$

where $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$.

(i) Find the point of intersection of lines $x = a$ and $y = b$.

(ii) Find the value of $(OA + OB + OC)$.

(iii) Find the equation of line OA.

एक सरल रेखा जो कि मूल बिन्दु O(0, 0) से गुजरती है, तथा रेखाओं $x = a$, $y = b$ एवं $x + y = 8$ को क्रमशः A, B एवं C पर इस प्रकार

i fr PNs d j r hg\$fd $OA \cdot OB \cdot OC = 48\sqrt{2}$ एवं $f(a, b) \le 0$

जहाँ $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$

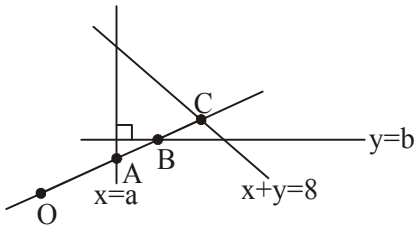
(i) रेखाओं $x = a$ एवं $y = b$ का प्रतिच्छेद बिन्दु ज्ञात करो

(ii) ;ट । ट ठ ट ड द्धका मान ज्ञात करो।

(iii) ट । का समीकरण ज्ञात करो।

**Ans.** (i) a = 2 and b = 3; (ii) $9\sqrt{2}$ ; (iii) x – y = 0

**Sol.**




(i) $f(a, b) \le 0$

$\Rightarrow f(a, b) = 0$ (only possible)

$\frac{b}{a} = \frac{3}{2}$

$2b = 3a$

$ea + 2b - ea - 6 = 0$

$(a, b) = (2, 3)$ $\quad b = 3$

$\quad a = 2$

(ii) $A(r_1 \cos\theta, r_1 \sin\theta)$ $\quad \Rightarrow r_1 \cos\theta = 2$

$B(r_2 \cos\theta, r_2 \sin\theta)$ $\quad \Rightarrow r_2 \sin\theta = 3$

$C(r_3 \cos\theta, r_3 \sin\theta)$ $\quad \Rightarrow r_3 = \frac{8}{\cos\theta + \sin\theta}$

$r_1 r_2 r_3 = 48\sqrt{2}$

$\frac{2 \times 3 \times 8}{\cos\theta.\sin\theta.(\cos\theta + \sin\theta)} = 48\sqrt{2}$ $\quad \Rightarrow \sin\theta\cos\theta(\sin\theta + \cos\theta) = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin 2\theta.(\sin\theta + \cos\theta) = \sqrt{2}$

$\sin 2\theta = 1,$ $\quad \sin\theta + \cos\theta = \sqrt{2}$

$\theta = 45°$

$r_1 = 2\sqrt{2}$ , $r_2 = 3\sqrt{2}$ , $r_3 = \frac{8}{2}\sqrt{2} = 4\sqrt{2}$

$r_1 + r_2 + r_3 = 9\sqrt{2}$

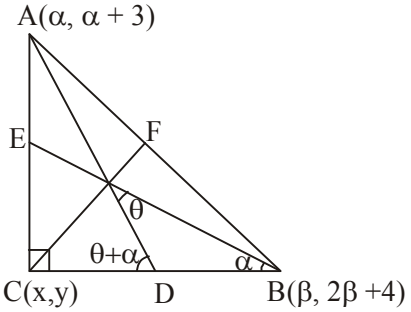
(iii) $y = x$ $\quad m = \tan 45° = 1,$ $\quad O(0,0)$

15. The triangle ABC, right angled at C, has median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60, find the area of the triangle ABC.

त्रिभुज ABC, जिसक कोण C समकोण है, तथा माध्यिकाये AD, BE एवं CF है जहाँ AD एवं BE क्रमशः रेखाओं $y = x + 3$ एवं $y = 2x + 4$ के अनुसरित है यदि कर्ण की लम्बाई 60 है तो त्रिभुज ABC का क्षेत्रफल ज्ञात करो।

**Ans.** 400 sq. units

**Sol.**




$AC^2 + BC^2 = 60^2$

$b^2 + a^2 = 60^2$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{1-2}{1+2}\right| = \frac{1}{3}$$

In $\Delta CBE$

$$\tan\theta = \frac{CE}{BC} = \frac{\frac{b}{2}}{a} = \frac{b}{2a}$$

In $\Delta ADC$

$$\tan(\theta + \alpha) = \frac{AC}{CD} = \frac{b}{\frac{a}{2}} = \frac{2b}{a}$$

$$\frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha} = \frac{2b}{a} \Rightarrow \frac{\frac{1}{3} + \frac{b}{2a}}{1 - \frac{1}{3}\cdot\frac{b}{2a}} = \frac{2b}{a}$$

$$\Rightarrow \frac{2a + 3b}{6a - b} = \frac{2b}{a}$$

$\Rightarrow ab = 800$

$ar.(\Delta ABC) = 400$

16. A straight line L through the origin meets the line $x+y=1$ and $x+y=3$ at P and Q respectively. Through P and Q two straight lines $L_1$ and $L_2$ are drawn, parallel to $2x-y=5$ and $3x+y=5$ respectively. Lines $L_1$ and $L_2$ intersect at R. Show that the locus of R, as L varies, is a straight line.

मूलबिन्दु से होकर जाने वाली एक सरल रेखा L रेखाओं $x+y=1$ और $x+y=3$ से क्रमशः P और Q पर मिलती है। P और Q से दो सरल रेखाऐं $L_1$ एवं $L_2$ क्रमशः $2x-y=5$ और $3x+y=5$ के समान्तर खींची जाती है। रेखाऐं $L_1$ व $L_2$ बिन्दु R पर प्रतिच्छेद करती है। दिखाइए कि बिन्दु R का बिन्दुपथ एक सरलरेखा है।

**Ans.** $x-3y+5=0$;

**Sol.** The line $y=mx$ meets the given lines in

$$P\left(\frac{1}{m+1},\frac{m}{m+1}\right) \text{ and } Q\left(\frac{3}{m+1},\frac{3m}{m+1}\right)$$

Hence equation of $L_1$ is

$$y-\frac{m}{m+1}=2\left(x-\frac{1}{m+1}\right)$$

$$\Rightarrow y-2x-1=-\frac{3}{m+1} \qquad \text{_____(1)}$$

and that of $L_2$ is

$$y-\frac{3m}{m+1}=-3\left(x-\frac{3}{m+1}\right)$$

$$\Rightarrow y+3x-3=\frac{6}{m+1} \qquad \text{_____(2)}$$

From (1) and (2) $\frac{y-2x-1}{y+3x-3}=-\frac{1}{2}$

$\Rightarrow x-3y+5=0$ which is a straight line

17. A straight line L with negative slope passes through the point (8,2) and cuts the positive coordinates axes at points P and Q. Find the absolute minimum value of OP + OQ, as L varies, where O is the origin.

ऋणात्मक प्रवणता वाली सरलरेखा L बिन्दु (8,2) से गुजरती है तथा निर्देश अक्षों का धनात्मक दिशा में P तथा Q बिन्दुओं पर काटती है। L के परिवर्तित होने के अनुसार OP + OQ का न्यूनतम निरपेक्ष मान ज्ञात कीजिये, जहाँ O मूल बिन्दु है।

**Ans.** 18

**Sol.** Equation of line

(A) $y-2=m(x-8)$ and $m<0 \quad \Rightarrow \quad P\left(8-\frac{2}{m},\ 0\right)$ and $Q(0,\ 2-8m)$

$$OP+OQ=\left|8-\frac{2}{m}\right|+|2-8m|=10+\frac{2}{(-m)}+8(-m) \geq 10+2\sqrt{\frac{2}{(-m)}\times 8(-m)} \geq 18.$$

Area of $\Delta OPQ$ is minimum when (8,2) is midpoint of line. So, P(16, 0), Q(0,4) $\Delta OPQ = \frac{1}{2}(16)(4) = 32$.

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question Nos. 18 to 20**

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE = AF. Let P be a point inside the square ABCD.

माना ABCD एक वर्ग है जिसकी भुजाओं की लम्बाई एक है। बिन्दु E तथा F क्रमशः भुजाओं AB तथा AD पर इस प्रकार लिये गये है कि AE = AF है। माना बिन्दु P वर्ग ABCD के अन्दर है।

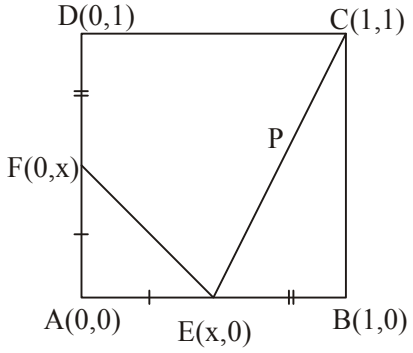
18. The maximum possible area of quadrilateral CDFE is :

चतुर्भुज CDFE का अधिकतम क्षेत्रफल है :

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

**Ans.** C

**Sol.**



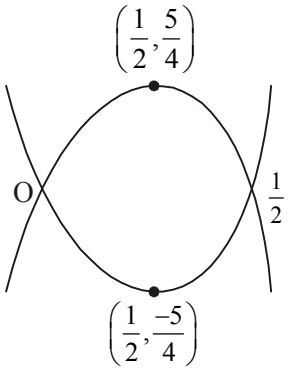

$$\text{Area} = \frac{1}{2}\begin{vmatrix} x & 0 & 0 & 1 & x \\ 0 & x & 1 & 1 & 0 \end{vmatrix}$$

$$A = \frac{1}{2}\left|x^2 - 1 - x\right|$$

$$A = \frac{1}{2}\left|x^2 - x - 1\right|$$

$$x \in (0,1)$$

$$\frac{-b}{2a} = \frac{1}{2}$$



$$\frac{-D}{4a}=\frac{-(1+4)}{4}=\frac{-5}{4}$$

$$A_{max}=\frac{5}{8}$$

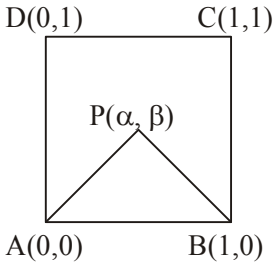
19. The value of $(PA)^2-(PB)^2+(PC)^2-(PD)^2$ is equal to :

$(PA)^2-(PB)^2+(PC)^2-(PD)^2$ का मान है :

(A) 3 (B) 2 (C) 1 (D) 0

**Ans.** D

**Sol.**




$$Z=\alpha^2+\beta^2-(\alpha-1)^2-\beta^2+(\alpha-1)^2+(\beta+2)^2-\alpha^2-(\beta\alpha)^2$$

$Z=0$

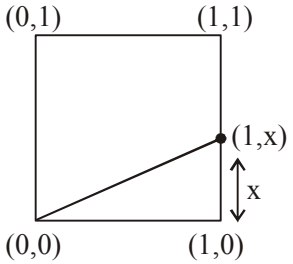
20. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is :

माना एक रेखा बिन्दु A से गुजरती है जो वर्ग ABCD को दो भागों में विभाजित करती है जिनमें एक का क्षेत्रफल अन्य से दोगुना है तो रेखा d soxZd sHkr j fgLl sd hy Ecbl Zgs

(A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

**Ans.** B

**Sol.**




$3A=1$

$$A = \frac{1}{3}$$

$$\frac{1}{2} \times 1 \times x = \frac{1}{3}$$

$$x = \frac{2}{3}$$

$$\text{Length} = \sqrt{1 + x^2}$$

$$= \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3}$$

**Paragraph for Question Nos. 21 to 23**

The base of an isosceles triangle is equal to 4, the base angle is equal to 45°. A straight line cuts the extension of the base at a point M at the angle θ and bisects the lateral side of the triangle which is nearest to M.

एक समद्विबाहु त्रिभुज का आधार 4 है तथा आधार कोण 45° है एक रेखा इस बढ़े हुये आधार को बिन्दु M पर θ कोण पर काटती है तथा M से करीब भुजा को समद्विभाजित करती है।

21. The area 'A' of the quadrilateral which the straight line cuts off from given triangle is :

यह रेखा त्रिभुज के साथ जो चतुर्भुज बनाती है। उसका क्षेत्रफल 'A' होगा :

(A) $\frac{3 + \tan\theta}{1 + \tan\theta}$ (B) $\frac{3 + 2\tan\theta}{1 + \tan\theta}$ (C) $\frac{3 + \tan\theta}{1 - \tan\theta}$ (D) $\frac{3 + 5\tan\theta}{1 + \tan\theta}$

**Ans.** D

22. The range of values of 'A' for different values of θ, lie in the interval :

θ के भिन्न–भिन्न मान के लिय 'A' का परिसर होगा :

(A) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (B) (4, 5) (C) $\left(4, \frac{9}{2}\right)$ (D) (3, 4)

**Ans.** D

23. The length of portion of straight line inside the triangle may lie in the range :
इस रेखा का वह हिस्सा जो कि त्रिभुज के भीतर है, उसकी लम्बाई का परिसर होगा :

(A) (2, 4) (B) $\left(\frac{3}{2}, \sqrt{3}\right)$ (C) $(\sqrt{2}, 2)$ (D) $(\sqrt{2}, \sqrt{3})$

**Ans.** C

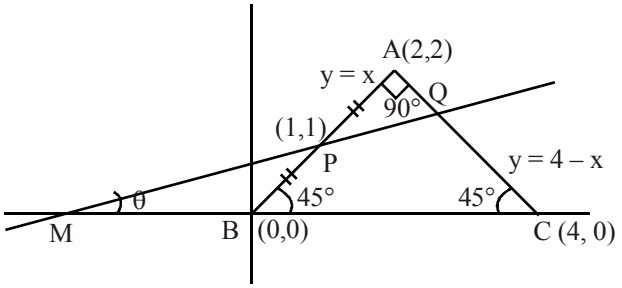
**Sol.** Equation of line PM :

$y - 1 = \tan\theta\,(x - 1)$

Intersection point 'Q' of AC and MP

$4 - x - 1 = \tan\theta\,(x - 1)$

$Q \equiv \left(\frac{3+\tan\theta}{1+\tan\theta}, \frac{1+3\tan\theta}{1+\tan\theta}\right)$




$$\text{Area of } \Delta APQ = \text{modulus of } \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ \frac{3+\tan\theta}{1+\tan\theta} & \frac{1+3\tan\theta}{1+\tan\theta} & 1 \end{vmatrix} = \frac{1-\tan\theta}{1+\tan\theta}$$

## MATCH THE COLUMN TYPE QUESTION(S)

24. Match the following column :

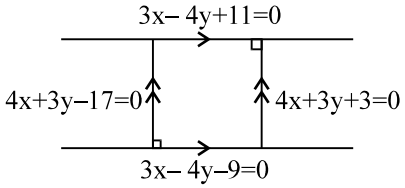
| **Column–I** | **Column–II** |
|---|---|
| (A) The four lines $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ and $4x + 3y - 17 = 0$ enclose a figure which is | (P) A quadrilateral which is neither a parallelogram nor a trapezium nor a kite. |
| (B) The lines $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ and $x + 2y = 3$ from a figure which is | (Q) A parallelogram which is neither a rectangle nor a rhombus |
| (C) If 'O' is the origin, P is the intersection of the lines $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$, A and B are the points in which these lines are cut by the line $x + 2y - 5 = 0$, then the points O, A, P, B (in some order) are the vertices of | (R) A rhombus which is not a square |
| | (S) A square |

निम्न कॉलम को सुमेलित कीजिए :

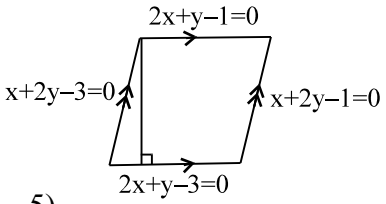
| **Column–I** | **Column–II** |
|---|---|
| (A) चार भुजाऐं $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ एवं $4x + 3y - 17 = 0$ किस आकृति को बनाएगी। | (P) एक चतुर्भुज जो न तो समान्तर चतुर्भुज एवं नही काइट है |
| (B) रेखाऐं $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ तथा $x + 2y = 3$ किस आकृति को बनाएगी | (Q) एक समान्तर चतुर्भुज जो न तो आयत एवं न ही समचतुर्भुज है |
| (C) यदि रेखा $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$ का प्रतिच्छेद बिन्दु P हो तथा इस रेखा युग्म को दूसरी रेखा $x + 2y - 5 = 0$ यदि A एवं B पर काटती है तथा O मूलबिन्दु है। तो O, A, P, B किसके शीर्ष होगे | (R) एक समचतुर्भुज जो वर्ग नहीं है |
| | (S) वर्ग |

**Ans.** (A) S; (B) R; (C) Q

**Sol.** (A) $\left.\begin{array}{l} d_1 = \dfrac{20}{5} = 4 \\ d_2 = \dfrac{20}{5} = 4 \end{array}\right] \Rightarrow$ square

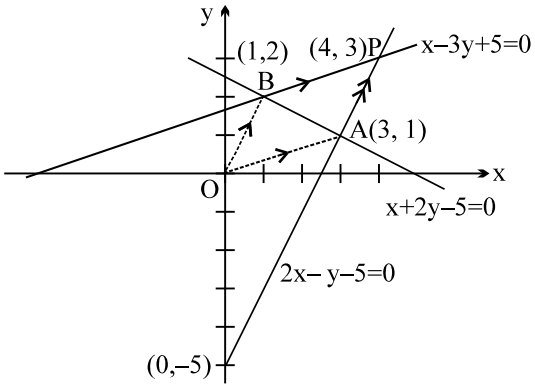


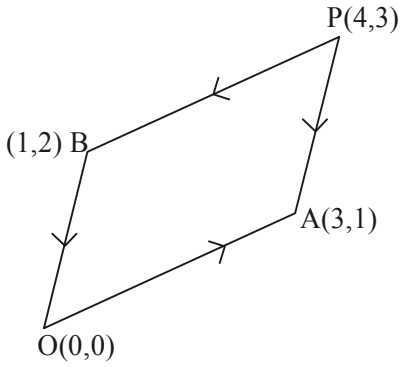

(B) $\left.\begin{array}{l} d_1 = \dfrac{2}{\sqrt{5}} \\ d_2 = \dfrac{2}{\sqrt{5}} \end{array}\right] \Rightarrow$ interior not 90° $\Rightarrow$ rhombus




(C) $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0 \equiv (x - 3y + 5)(2x - y - 5)$

the point of intersection is (4, 3)




$m_{OA} = \dfrac{1}{3}$ $\qquad$ $m_{BP} = \dfrac{1}{3}$

$m_{AP} = \dfrac{2}{1}$ $\qquad$ $m_{OB} = \dfrac{2}{1}$

$OA \parallel AP$ & $AP \parallel OB$

$OA \not\perp AP$

Hence, parallelogram.

25. Consider the 3 linear equations

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

where $a, b, c \in R$.

| **Column–I** | **Column–II** |
|---|---|
| (A) If $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then | (P) Entire xy plane |
| (B) If $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then | (Q) The lines are concurrent |
| (C) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then | (R) Line are concident |
| (D) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then | (S) Lines are neither coincident nor concurrent |

तीन रेखाओं का समीकरण दिया है

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

जहाँ $a, b, c \in R$.

Match the following column :

| **Column – I** | **Column – II** |
|---|---|
| (A) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो | (P) xy तल होगा |
| (B) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो | (Q) रेखाऐं संगामी होगी |
| (C) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो | (R) रेखाऐं सम्पाती होगी |
| (D) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो | (S) न तो रेखाऐं सम्पाती होगी एवं न ही संगामी होगी |

**Ans.** (A) Q; (B) P; (C) S; (D) R

**Sol.** $D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ and $D = 0$ is condition of concurrency

$D = -(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

(A) if $a + b + c = 0$ but $\sum a^2 \neq \sum ab$ i.e. a, b, c are not all equal, then $D = 0$

hence lines are concurrent $\Rightarrow$ (Q)

(B) if $a + b + c = 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

$\therefore a = 0; \quad b = 0; \quad c = 0$

$\Rightarrow$ lines becomes identical and of the form $0x + 0y + 0 = 0$

any ordered pair (x, y) will satisfy $\Rightarrow$ complete xy plane $\Rightarrow$ (P)

(C) if $a + b + c \neq 0$ and $\sum a^2 \neq \sum ab$ $\Rightarrow$ a, b, c are not all equal $\Rightarrow D \neq 0$

In this case equations represents set of lines which are neither concident nor concurrent $\Rightarrow$ (S)

(D) if $a + b + c \neq 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

hence lines becomes identical or concident $\Rightarrow$ (R)

26. Match the following column :

| **Column–I** | **Column–II** |
|---|---|
| (A) The lines $y = 0$; $y = 1$; $x - 6y + 4 = 0$ and $x + 6y - 9 = 0$ constitute a figure which is | (P) a cyclic quadraliteral |
| (B) The points A(a, 0), B(0, b), C(c, 0) and D(0, d) are such that $ac = bd$ and a, b, c, d are all non-zero. The points A, B, C and D always constitute | (Q) a rhombus |
| (C) The figure formed by the four lines $ax \pm by \pm c = 0$ $(a \neq b)$, is | (R) a square |
| (D) The line pairs $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$ constitute a figure which is | (S) a trapezium |

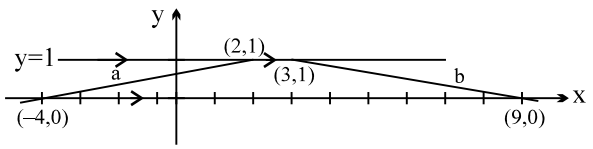
निम्न कॉलम को सुमेलित कीजिए :

| **Column–I** | **Column–II** |
|---|---|
| (A) रेखा $y = 0$; $y = 1$; $x - 6y + 4 = 0$ एवं $x + 6y - 9 = 0$ से बनने वाली आकृति होगी | (P) चक्रीय चतुर्भुज |
| (B) यदि $a, b, c, d \neq 0$ तथा A(a, 0), B(0, b), C(c, 0) एवं D(0, d) इस प्रकार से है कि $ac = bd$ है तो आकृति A B C D इनमें से क्या होगा | (Q) समचतुर्भुज |
| (C) चार रेखाओं $ax \pm by \pm c = 0$ $(a \neq b)$ से बनने वाली आकृति होगी | (R) वर्ग |
| (D) रेखा युग्म $x^2 - 8x + 12 = 0$ एवं $y^2 - 14y + 45 = 0$ कौनसी आकृति बनाएगें। | (S) समलम्ब चतुर्भुज |

**Ans.** (A) P, S; (B) P; (C) Q; (D) P, Q, R

**Sol.** (A) Obviously trapezium

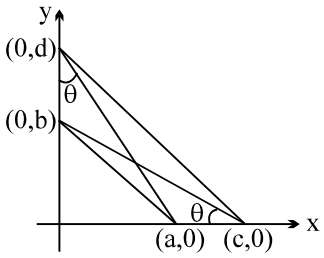
$$\left.\begin{matrix} a = \sqrt{37} \\ b = \sqrt{37} \end{matrix}\right] \quad \Rightarrow \quad a = b$$




hence isosceles trapezium

$\Rightarrow$ a cyclic quadrilateral also $\Rightarrow$ P, S

(B) $ac = bd$ $\Rightarrow$ $\dfrac{b}{c} = \dfrac{a}{d}$




$$\left.\begin{matrix} \tan\theta = \dfrac{b}{c} \\ \tan\phi = \dfrac{a}{d} \end{matrix}\right] \quad \Rightarrow \quad \theta = \phi$$

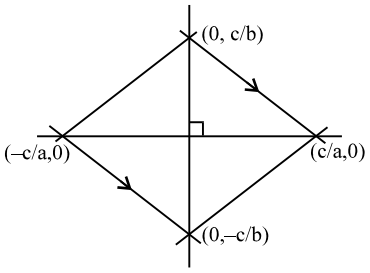
hence cyclic quadrilateral $\Rightarrow$ P

(C) $ax \pm by \pm c = 0$

if $y = 0$, $x = \pm \frac{c}{a}$

if $x = 0$, $y = \pm \frac{c}{b}$

$\Rightarrow$ rhombus $\Rightarrow$ Q




(D) $(x - 6)(x - 2) = 0$

$x = 6$ and $x = 2$

$y^2 - 14y + 45 = 0$

$(y - 9)(y - 5) = 0$

$\Rightarrow$ a square $\Rightarrow$ P, Q, R

